

Multinational Cloud Networks and Market Fragmentation*

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Abstract

We study how cloud regulation reshapes multinational infrastructure networks and transmits price effects across borders. Providers choose locations under private costs and variable markups, accounting for rivals' portfolios. We develop an exact method for combinatorial portfolio choice given rivals' strategies, without requiring ordered marginal location returns, and derive sufficient conditions for equilibrium uniqueness under dispersed private costs. Domestic sourcing requirements need not attract more infrastructure. Relative to unrestricted cross-border sourcing, allowing eligible supply across the EU expands modeled providers' EU presence, whereas restricting supply to individual countries reduces it. Higher costs of serving China from abroad increase Chinese providers' domestic presence while shrinking their global networks and reducing their operating profits. Direct restrictions drive most of the regulated-market price increase; network adjustment generates smaller price increases elsewhere. Competitor responses substantially moderate global network contraction.

Keywords: multinational networks, digital trade, regulatory fragmentation, portfolio equilibrium, variable markups, combinatorial choice

JEL Codes: F12, F23, L86, C72, C62, L13

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1. INTRODUCTION

Multinational cloud providers serve customers across borders from networks of facilities. A location can serve several markets, while its return depends on other locations in the same network and on competing providers' networks. Regulation that restricts cross-border supply can therefore change location incentives beyond the regulated market. How do such restrictions reshape multinational cloud networks, and how are their effects transmitted to customers in other countries?

A cloud region is a group of data centers providing a range of computing services. Regions can serve customers abroad, while smaller local facilities can extend service into countries without a supporting region of their own. For example, AWS's Copenhagen Local Zone is linked to a parent region in Stockholm. A provider can therefore offer local services in a country without locating a region there. Procurement rules can restrict where services purchased by public agencies are supplied from, while leaving private customers' access unchanged.

We develop a quantitative model of multinational cloud production and cross-border service supply. Eight providers observe private location costs and choose which of 42 countries host their cloud regions, the core infrastructure of the model. We group each provider's regions within a country into one presence and hold the locations of smaller local facilities fixed. Each complete portfolio determines the provider's delivered costs and service accessibility across customer markets, including local facilities supported by a core abroad. Prices follow the Atkeson–Burstein (AB) share-dependent markup schedule, with other suppliers represented in aggregate.

Geographic restrictions affect customers both directly, through the supply available at existing networks, and indirectly, through providers' location choices. A rule that narrows the markets a location can serve changes its value relative to other locations and alters competition in those markets. Providers may then expand, contract, or relocate their networks, changing costs and competition in markets the rule does not cover. Their choices also alter the returns to competing networks.

These interdependencies make the model hard to solve. If AWS includes France in its core network, the return depends on how Microsoft and Google adjust their own networks; their responses change prices and market shares and feed back, in equilibrium,

into AWS’s location incentive. Even with rivals’ networks fixed, adding locations can first raise and then lower the return to another location as prices and market shares adjust. Marginal location returns therefore need not move monotonically as a network expands. Restrictions exploited by existing methods, including complementarity (Jia, 2008), single-crossing (Arkolakis et al., 2026), and diminishing marginal profits (Sabal, 2026), need not hold in this environment. The challenge is to solve complete-portfolio choices without imposing those restrictions and to determine when the resulting strategic responses form a unique equilibrium; unique prices at a given network resolve neither problem.

We develop an exact method for each provider’s combinatorial location problem, given rivals’ strategies, and derive sufficient conditions for uniqueness of the portfolio equilibrium. For the optimization problem, our completion bounds combine limits on market benefits with the costs of the locations generating them. Conditioning on the number of additions and tying benefits to costs rules out groups of networks that cannot improve profit, without requiring ordered marginal location returns. This supports global portfolio search while allowing both the number and composition of locations to change.

For the equilibrium problem, we bound how changes in rivals’ networks move the probabilities of whole-portfolio choices. The bounds exploit the shared location-cost shocks across overlapping portfolios, accounting jointly for choices that change several locations at once. They yield sufficient conditions under which private-cost dispersion dominates strategic feedback, giving a unique portfolio equilibrium and convergence of distributional best responses. Together, these results provide a way to analyze geographic restrictions when independent location decisions and stable rankings of location returns are unavailable.

Two policy experiments change the reach of cross-border service supply. The EU experiment compares two rules for the same public-cloud customers: *EU-wide* eligibility permits supply from eligible service origins throughout the participating EU countries, whereas *National* eligibility restricts supply to domestic origins. A baseline without geographic eligibility restrictions allows the same customers to use the ordinary cross-border supply network. The comparison corresponds to a choice now under discussion

in the EU.¹ The China experiment lowers or raises the cost of supplying mainland customers from abroad, around existing restrictions on directly controlled mainland cores and partner-operated services. The China comparison thus starts from a market that is already segmented.

Domestic sourcing requirements need not attract more infrastructure. Relative to the baseline, EU-wide eligibility expands the modeled providers' core presence within the EU, whereas National eligibility reduces it. For these providers, EU-wide eligibility directs regulated demand toward EU origins while allowing each eligible origin to serve that demand across participating countries. National eligibility favors domestic service origins over foreign ones, but limits each eligible origin to domestic regulated demand. Serving that demand need not justify a domestic core. All eight providers have fewer EU core presences under National than under EU-wide eligibility, and their worldwide core presence is about 2.5 percent lower. For National relative to EU-wide eligibility, most of the regulated-market price increase arises directly from the loss of eligible supply at existing networks. Network adjustment raises prices for EU private customers and customers outside the EU. Competitor responses leave the regulated-price effect nearly unchanged but substantially moderate the global contraction and its international price effects. Other suppliers serve most regulated demand under National but do not choose networks in the model; the contraction therefore concerns the eight modeled providers.

Greater domestic presence need not translate into higher profits. In the China experiment, Chinese providers become more likely to operate a mainland core, yet their worldwide networks contract and their operating profits fall. These providers also serve mainland customers from offshore facilities, so the shock raises the cost of supply from part of their own networks. Competitor responses substantially moderate this contraction: under the larger tightening, the global decline is about 47 percent smaller than when providers respond to unchanged rival networks. Partial easing reverses the aggregate pattern, expanding worldwide networks and lowering prices in China and abroad.

The two applications distinguish domestic presence from the global networks sup-

¹The Commission's 2026 proposal for a Cloud and AI Development Act includes an EU-wide framework for assessing cloud sovereignty (European Commission, 2026b), while member states currently apply their own public-sector qualifications, such as France's SecNumCloud (Agence nationale de la sécurité des systèmes d'information, 2022). Our experiment isolates the geographic scope of eligible service supply, rather than reproducing the proposal or the qualification and ownership requirements of existing national schemes.

porting it: national eligibility can reduce even domestic presence, while higher cross-border costs can increase domestic presence alongside a global contraction. Competitor responses matter for this international adjustment even when they change the regulated-market price effect little. The resulting price effects on customers whose own access rules are unchanged depend on which providers and locations adjust, not simply on the total change in core presence.

The calibration uses worldwide core-hosting country counts and global-share targets to discipline provider-level net location costs, relative competitiveness, and the aggregate contribution of other suppliers. It closely matches worldwide scale and shares while predicting more EU core presences than observed. We assess the implied geography by comparing regional presence and host-country characteristics with the data. The policy comparisons hold calibrated parameters fixed. The EU results' magnitude depends on starting geography and the eligibility and supply of providers outside the modeled eight.

The paper builds on quantitative trade models in which a firm's choices across countries are interdependent. In canonical export models a firm serves a destination when its profit there covers the fixed cost, a market-by-market rule that is exact under separable payoffs (Chaney, 2008; Dickstein and Morales, 2018). Interdependence arises when earlier destinations change later entry costs (Morales et al., 2019; Alfaro-Ureña et al., 2025), when sourcing locations jointly determine input costs (Antràs et al., 2017), and when production locations jointly determine market access (Tintelnot, 2017; Castro-Vincenzi, 2024; Antràs et al., 2026). These models develop exact methods for a single firm's problem. We build on this view of interdependent locations in a game where geographic restrictions on service supply are transmitted through rivals' responses to networks and prices elsewhere.

The two experiments connect to work on digital trade barriers: data-localization and sovereignty rules that raise the cost of operating or using foreign infrastructure (Ferracane et al., 2020; Demirer et al., 2024; Manganelli and Schnurr, 2024), and China's restrictions on cross-border digital services (Sun and Trefler, 2023; Sun, 2024). Those studies measure the restrictions and their effects on firms; we ask how they reshape the multinational networks that supply the services.

Order-based methods solve complete combinations of locations under restrictions on marginal returns. Arkolakis et al. (2026) exploit single-crossing, Jia (2008) complementarity in strategic chain entry, and Castro-Vincenzi et al. (2025) known signs of comple-

mentarity and substitutability in a multiproduct firm’s problem; single-firm models of fulfillment and plant networks impose analogous structure (Houde et al., 2023; Oberfield et al., 2024). Our completion bounds apply when no such ordering can be established, which repricing under variable markups can produce; when the ordering holds, those procedures remain available.

Most closely related, Sabal (2026) develops choice-probability bounds for estimation and counterfactual analysis in a strategic product-entry game. His equilibrium algorithm exploits decreasing marginal profits as own and rival product sets expand to bound the distributions consistent with equilibrium, allowing for multiplicity. Our approach addresses portfolio games in which this monotonicity need not hold. We construct completion bounds for globally optimal responses to given rival strategies and separately bound how portfolio-choice distributions change when those strategies change. The latter bounds yield sufficient conditions for equilibrium uniqueness under location-cost heterogeneity. The distinction concerns both the restrictions on payoffs and the objects being bounded: equilibrium distributions in Sabal, and remaining profit gains and response sensitivity here.²

Strategic spillovers across markets are central to multimarket oligopoly (Bulow et al., 1985) and to models of product and network choice (Holmes, 2011; Wollmann, 2018; Fan and Yang, 2020; Nishida, 2015). Here the connection is a multinational service network whose geographic reach is restricted by policy. Our uniqueness results also relate to contraction arguments comparing heterogeneity with strategic interaction (Mason and Valentinyi, 2007; Melo, 2022); we exploit the shared location-cost shocks across overlapping portfolios to bound changes in whole-network choice probabilities.

Section 2 describes cloud networks and regulatory access. Sections 3 and 4 present the model and theory; Sections 5 and 6 report calibration and policy results. The appendix contains proofs and supporting evidence.

²Sabal’s equilibrium algorithm also extends to the supermodular case, and his estimation bounds require weaker conditions than submodularity. Our example in Section 4.3 has marginal location returns that both rise and fall as the network expands, so neither submodularity nor supermodularity holds.

2. CLOUD NETWORKS AND REGULATORY ACCESS

Cloud networks connect core regions, local computing facilities, and customer markets across national borders. These connections determine which parts of a network can support service in a country. Regulatory eligibility then determines which customers may use that service.

The country in which computing services are available locally need not contain the supporting core region. Figure 1 gives four documented AWS examples. The Copenhagen and Helsinki Local Zones have the Stockholm region as their parent, and the Warsaw Local Zone has Frankfurt as its parent. Hamburg–Frankfurt provides a within-country comparison. In the 42-country sample, 33 AWS Local Zone records identify their parent regions: 24 are within-country and nine are cross-country. These are counts of documented parent connections, not capacity or traffic shares, and the available parent documentation is much more complete for AWS than for the other providers. They therefore establish the existence of cross-border topology, not its prevalence across the industry.

Regulation can restrict the services available to particular buyers without eliminating a provider’s physical presence. France’s SecNumCloud scheme assesses cloud services against security requirements; qualification applies to a specified service and legal entity (Agence nationale de la sécurité des systèmes d’information, 2022). In mainland China, local operators supply AWS and Microsoft services through separate contracts and accounts. Korean certification similarly determines which services public agencies can purchase from existing cloud regions. Appendix Table 11 documents these arrangements and their institutional sources.

Providers also expand across several countries within the same year. In the five-provider launch panel, covering AWS, Microsoft, Google, Oracle, and Alibaba, 115 first country entries over 2007–2024 occur in 52 provider-years with at least one entry. Twenty-eight include at least two countries, accounting for 91 entries. This clustering motivates studying multicountry networks; it does not identify joint decision timing or strategic responses. Appendix Table 10 reports the eight-provider extension and its measurement limits. The five-provider series has more consistently documented historical launch dates than the additional Huawei, Tencent, and OVHcloud records.

The records distinguish infrastructure presence from customer-specific eligibility.

They guide the counterfactual supply mappings, holding ownership, partnerships, and certification effort fixed. Location and price responses are model predictions; these records do not identify their causal effects.

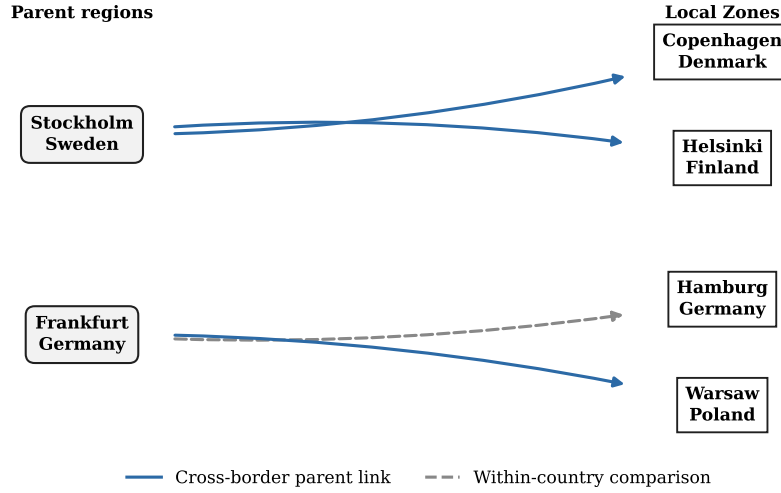


Figure 1: Selected Local Zone–Parent Region Connections

Notes: The nodes show four AWS Local Zones and their parent regions. Blue solid links cross a national border; the gray dashed link remains within Germany. The diagram records product topology, not traffic, capacity, or the legal route followed by a particular workload. AWS describes a Local Zone as an extension of a parent region but also documents direct internet and Direct Connect paths that need not traverse that parent. Sources: the AWS Local Zone roster and architecture documentation, archived September 11, 2026.

3. A MODEL OF MULTINATIONAL CLOUD SUPPLY

A provider chooses the countries in which to locate core infrastructure that can serve several customer markets. A core’s value depends on other facilities in the same network, the rules governing service from each origin, and competitors’ networks. This section specifies those links and the downstream pricing payoffs. Section 4 develops the portfolio game and solution method for a general finite set of links.

3.1 Core location, service paths, and regulation

Firm f chooses a set S_f of countries hosting its core infrastructure. Each provider’s core regions within a country are grouped into one location alternative, so $K_f = |S_f|$ counts its core-hosting countries. The model captures the extensive margin of core presence across countries; it does not separately choose the number or capacity of regions within a host country. Aggregate core counts sum these provider-country presences across firms; expected counts average over the private-cost realizations described in Section 3.3 and the portfolio choices they induce.

Installed local facilities can provide service from a country without a core there. Their locations are held fixed. A local service with a documented parent operates when that parent core is in the provider's chosen portfolio, including when the parent is abroad. When the parent is unconfirmed, we maintain observed local availability independently of the chosen cores.³ Several local facilities in the same country provide one service origin.

A customer segment n is a destination country and customer group. The EU application distinguishes regulated public demand from private demand within each participating country; other countries have one customer segment. A regime r determines which supply paths are eligible for each segment. EU public-cloud rules restrict the geographic origin of eligible supply; the China application changes offshore delivery costs while maintaining specified local operating modes. The latter restrictions are detailed in Section 6.2. The policies change which customers a network can reach or the cost of reaching them, and hence the return to its core locations.

The service origin is the country in which the modeled core or local service is located. Geographic eligibility applies to that origin. A local service can therefore serve domestic regulated demand when its supporting core is abroad, provided that the required parent core is in the provider's portfolio. Parent presence and service-origin eligibility are separate conditions: National does not require the supporting parent to be domestic. For unconfirmed parent links, the maintained local availability is subject to the same service-origin rule.

The service mapping distinguishes the contribution of a core from that of a local service alone. Let $u_{fin}^r(S_f)$ indicate that firm f has a core in country i eligible to serve segment n under regime r , and let $o_{fin}^r(S_f)$ indicate coverage through either an eligible core or an operating local path. Thus $o_{fin}^r(S_f) \geq u_{fin}^r(S_f)$. Effective supply is

$$t_{fin}^r(S_f) = u_{fin}^r(S_f) + \gamma_{loc} [o_{fin}^r(S_f) - u_{fin}^r(S_f)], \quad \gamma_{loc} = 0.5.$$

³This is an assumption about unobserved support. Among 60 screened local records, 33 have confirmed parents, all for AWS; 27 have unconfirmed parents, including 25 for OVHcloud. These counts describe documentation coverage rather than industry frequencies. Appendix G.2 describes the sources.

A core contributes one unit, a local-only path one half, and an unavailable path zero. Core and local supply in the same country are not added together. The lower local weight represents narrower service capability.⁴

Each eligible origin can lower the cost of serving a customer. Let w_i denote its operating input-cost index, τ_{in} the cost of delivering service from i to segment n , and a_f the provider's relative competitiveness. Delivered marginal cost aggregates effective supply as

$$\Sigma_{fn}^r(S_f) = \sum_i t_{fin}^r(S_f)(w_i \tau_{in})^{-\theta}, \quad c_{nf}^r(S_f) = a_f^{-1}[\Sigma_{fn}^r(S_f)]^{-1/\theta}. \quad (3.1)$$

This is the Fréchet cost aggregation used in models of multinational production, with its common scale factor absorbed into a_f (Tintelnot, 2017). The parameter θ governs how additional supply reduces cost. A core can improve supply in a country already served by a local facility without creating a new service origin. For scale, start with a single core origin and hold delivered input costs $w_i \tau_{in}$ equal across origins. At $\theta = 3$, adding a second core origin multiplies marginal cost by $2^{-1/3}$, a reduction of about 21 percent; adding a local-only origin instead multiplies it by $1.5^{-1/3}$, a reduction of about 13 percent. The weight t determines each origin's contribution to this cost reduction.

Customers also value access to a provider's network. Let q_{in} measure accessibility from origin i to segment n . Additional eligible origins can improve access even when the most accessible origin is unchanged:

$$A_{fn}^r(S_f) = \lambda_A^{-1} \log \left(1 + \sum_i o_{fin}^r(S_f) e^{\lambda_A q_{in}} \right), \quad \psi_{fn}^r(S_f) = e^{\chi[A_{fn}^r(S_f) - A_n^{\text{frontier}}] + d_{nf}}. \quad (3.2)$$

Here A_n^{frontier} is access with all origins available, held fixed across regimes; λ_A controls aggregation across origins, χ scales the demand advantage from access, and d_{nf} permits a home-market advantage. The accessibility score is a proxy, not a direct measure of latency or willingness to pay; Section 5.1 describes its construction. Neither a_f nor d_{nf} is separately identified as physical productivity or a causal preference effect. Both cost and access depend on the complete portfolio because local supply can require a parent core elsewhere. A provider with no eligible supply has zero share in that segment.

⁴The weight is a modeling assumption rather than a measured capacity ratio. It affects the return to a core in countries already served by local facilities.

At a given eligible origin, a core and a local-only service each enter access through $o = 1$, while their cost contributions use $t = 1$ and $t = \gamma_{\text{loc}}$, respectively. Access reflects service geography, whereas cost also reflects supply capability.

3.2 Market shares, variable markups, and pricing

Customer expenditure R_n in each segment is fixed across policy regimes. The real quantity of the service composite, R_n/P_n , therefore falls when its price index rises. The model holds total cloud expenditure and its allocation across customer segments constant. Operating input prices are also fixed: this is a partial-equilibrium model in which core locations and cloud-service prices adjust.⁵ For each portfolio profile and regime, price-index contributions and revenue shares satisfy

$$y_{nf} = \psi_{fn} p_{nf}^{1-\sigma}, \quad s_{nf} = \frac{y_{nf}}{b_n + \sum_g y_{ng}}, \quad b_n \geq 0. \quad (3.3)$$

The positive term b_n is the aggregate contribution of other suppliers (the residual) to the price index. In the baseline it is common across customer segments, $b_n = b$, with its level calibrated to global shares. It summarizes their available services and prices, rather than physical capacity or their expenditure share. The latter is $b_n/(b_n + \sum_g y_{ng})$ and changes when the eight modeled providers' prices or eligibility change. In the central comparison, these suppliers remain eligible in every segment and their aggregate contribution is fixed. Sensitivity exercises vary their contribution and exposure to restrictions. Residual suppliers do not choose portfolios. A provider without eligible supply contributes zero to the sum.

For a portfolio profile S and regime r , the corresponding price index is

$$P_n^r(S) = \left[b_n^r + \sum_f \psi_{fn}^r(S_f) p_{nf}^r(S)^{1-\sigma} \right]^{1/(1-\sigma)}. \quad (3.4)$$

This is the cost of the segment's CES service composite under the maintained quality

⁵The demand system abstracts from customer-specific switching histories and explicit fees for switching or using multiple clouds. These differ from the geographic service costs studied here. Competition and Markets Authority (2025, Chapter 6) documents technical barriers and egress fees affecting switching and multi-cloud use.

weights. It can rise because suppliers charge more, because a network loses accessibility, or because a previously available service becomes ineligible.⁶

We take the share-dependent markup schedule of Atkeson and Burstein (2008), used in quantitative trade models by Edmond et al. (2015), as a conduct specification, holding customer-segment expenditure fixed:⁷

$$p_{nf} = \mu(s_{nf})c_{nf}, \quad \mu(s) = \frac{\mathcal{E}(s)}{\mathcal{E}(s) - 1}, \quad \frac{1}{\mathcal{E}(s)} = \frac{1-s}{\sigma} + \frac{s}{\eta}, \quad \sigma > \eta > 1. \quad (3.5)$$

A larger supplier faces a lower perceived elasticity and charges a higher markup, bounded between $\sigma/(\sigma - 1)$ and $\eta/(\eta - 1)$. With $\eta = 2$, the markup approaches an upper bound of two as the provider's share approaches one. Here η governs pricing conduct; it does not estimate the response of total segment expenditure, which is fixed. The CES comparison in Appendix D.3 instead holds markups constant at $\sigma/(\sigma - 1)$.

Operating profit is

$$VP_{nf} = \frac{R_n s_{nf}}{\mathcal{E}(s_{nf})} = R_n \left[\frac{s_{nf}}{\sigma} + \left(\frac{1}{\eta} - \frac{1}{\sigma} \right) s_{nf}^2 \right], \quad \Pi_f = \sum_n VP_{nf}. \quad (3.6)$$

At $(\sigma, \eta) = (5, 2)$, the expression in brackets is $0.2s_{nf} + 0.3s_{nf}^2$. A location changes delivered costs and accessibility, but its return also depends on the share and markup obtained after every supplier reprices. Even when two locations are technological substitutes, these market-share effects can alter the incentive to include both in a portfolio.

Proposition 1 (Pricing uniqueness). *For $\sigma > \eta > 1$, every fixed portfolio profile has a unique AB pricing equilibrium. In markets with at least two active firms, the update Φ of log contributions $\log y$ is a global sup-norm contraction with modulus $\bar{\lambda}_{AB}$ in (A.15). Equilibrium prices vary smoothly with strictly positive marginal costs on each fixed active set.*

Appendix A.6 defines the scalar pricing update and proves the result. Its uniform modulus is 0.375 at the maintained elasticities. Adding a fixed residual contribution preserves

⁶The index allows service availability and accessibility to vary, so it need not move with posted prices for continuing services. Feenstra (1994) studies the related role of changing varieties in CES price measurement. Security benefits and network choices by unmodeled suppliers are outside the comparison; these price effects do not measure the full welfare consequences of regulation.

⁷Atkeson and Burstein derive this schedule under Cournot competition with nested-CES demand. We retain its share-markup relationship while suppressing expenditure responses across customer segments. Downstream payoffs follow the stated pricing system; the general portfolio results also apply to other continuation payoffs satisfying their assumptions.

the bound because it reduces the weight of responsive strategic rivals in each supplier's competitor aggregate. Unique prices make each portfolio profile's operating payoff well defined. They do not establish uniqueness of the portfolio equilibrium, which requires a separate bound on portfolio responses.

3.3 Private costs and sequence of choices

A core portfolio incurs the additive net cost

$$C_f(S_f, Z_f) = \ell_f^{\text{ref}} \sum_{i \in S_f} [\alpha_f + \delta M_i + \kappa D_{fi} + \sigma_\varepsilon Z_{fi}] , \quad Z_{fi} \stackrel{\text{iid}}{\sim} N(0, 1). \quad (3.7)$$

The firm-specific intercept α_f governs worldwide portfolio size: a higher common per-core charge makes larger portfolios less attractive. The intercept is allowed to be signed and is interpreted as a reduced-form net cost of core presence, not an estimate of cash construction expenditure. An intercept below zero represents a net benefit before distance and missing-data penalties. It can absorb returns omitted from modeled IaaS profits, such as other services, but the calibration does not identify the source of that benefit. Negative private-cost realizations are also retained.

Here D_{fi} is distance from the firm's home country (the nearest model-EU country for OVHcloud), and M_i indicates a missing original construction-cost observation. Costs are expressed in operating-profit units using a fixed provider-specific loading $\ell_f^{\text{ref}} = G_f^{\text{ref}}/K_f^{\text{ref}}$, the ratio of reference operating profit to a reference core count.⁸ Section 5 states the maintained values and the calibration of α_f .

The sequence is static: firms observe their own cost vectors, choose core portfolios simultaneously, and then compete in each customer segment. They know rivals' cost distributions but not their realizations. Portfolio size is endogenous.⁹ A policy changes the eligibility or delivered cost of service paths; the resulting portfolio response can change

⁸The reference values are fixed inputs, not the profit and core count of the portfolio being evaluated. Holding the other coefficients fixed, changing ℓ_f^{ref} changes both deterministic costs and private-cost dispersion, and can therefore change location choices. It is a maintained scale assumption, not merely a relabeling of units. Appendix F reports the reference values.

⁹A lower expected core count denotes a different network configuration, not a prediction of immediate facility closure. Installed local-service topology and operating arrangements remain fixed. Construction delays and transitions between network configurations would require a dynamic framework, such as Ericson and Pakes (1995).

both the number and location of core-hosting countries. The counterfactuals compare static equilibria under different policy regimes, without tracing the transition between them.

4. EQUILIBRIUM IN COMPLETE PORTFOLIOS

The cloud model poses two linked questions. Given rivals' portfolio distributions, which complete network should a firm choose? When do the distributions induced by these choices form a unique equilibrium? Inclusion cutoffs and completion bounds address the first question; bounds on changes in choice probabilities address the second. Completion bounds support the portfolio searches in our applications, while the uniqueness results state additional conditions on strategic responses. The results apply to any finite feasible portfolio family, including endogenous portfolio sizes.

4.1 Portfolio game and equilibrium

There are F firms. Firm f has a finite set of candidate locations, called links in the general portfolio notation, $\mathcal{A}_f$ and a nonempty feasible family $\mathcal{X}_f \subseteq 2^{\mathcal{A}_f}$. For every pure portfolio profile $S = (S_1, \dots, S_F)$, first solve the downstream game and record the resulting operating profits. If that game has multiple equilibria, a fixed rule $p^\dagger(S)$ selects one continuation outcome. The selected payoff, net of deterministic link costs, is

$$r_f(S) = \Pi_f(S; p^\dagger(S)) - \sum_{j \in S_f} C_{fj}^0. \quad (4.1)$$

Here C_{fj}^0 is the deterministic net cost of link j . It is a separate primitive from the positive private-shock loading ℓ_{fj} . The cloud specification is obtained by setting

$$C_{fj}^0 = \ell_f^{\text{ref}}(\alpha_f + \delta M_j + \kappa D_{fj}), \quad \ell_{fj} = \ell_f^{\text{ref}}, \quad \varepsilon_{fj} = Z_{fj}.$$

These substitutions recover (3.7).

Firm f observes the vector of private link costs $\varepsilon_f = (\varepsilon_{fj})_{j \in \mathcal{A}_f}$. Its payoff is

$$u_f(S; \varepsilon_f) = r_f(S) - \sigma_\varepsilon \sum_{j \in S_f} \ell_{fj} \varepsilon_{fj}, \quad \ell_{fj} > 0. \quad (4.2)$$

Throughout, $\sigma_\varepsilon > 0$. The loading ℓ_{fj} converts the standardized private-cost shock into payoff units.

Assumption 1 (Private link costs). Private types are independent across firms, and each firm's type vector has an absolutely continuous joint distribution. Within a firm, the coordinates may be dependent. For every j , the conditional density of ε_{fj} given $\varepsilon_{f,-j}$ is bounded by $\bar{g}_{fj} < \infty$ almost surely. Independent normal or logistic link shocks satisfy this condition.

Let $Q_f \in \Delta(\mathcal{X}_f)$ be firm f 's portfolio distribution before its private costs are observed, and write $Q_{-f} = \bigotimes_{g \neq f} Q_g$. In equilibrium, each type chooses an optimal portfolio, and the type distribution induces Q_f .¹⁰ Its interim portfolio value, averaging over rivals' choices, is

$$V_f(S_f; Q_{-f}) = \sum_{S_{-f}} r_f(S_f, S_{-f}) Q_{-f}(S_{-f}). \quad (4.3)$$

Independence across firms gives the product Q_{-f} .

The type-specific global best response and the distribution it induces are

$$b_f(\varepsilon_f; Q_{-f}) \in \arg \max_{S_f \in \mathcal{X}_f} \left\{ V_f(S_f; Q_{-f}) - \sigma_\varepsilon \sum_{j \in S_f} \ell_{fj} \varepsilon_{fj} \right\}, \quad (4.4)$$

$$\mathcal{B}_f(Q_{-f})(S_f) = \Pr\{b_f(\varepsilon_f; Q_{-f}) = S_f\}. \quad (4.5)$$

Definition 1 (Portfolio Bayesian Nash equilibrium). A portfolio Bayesian Nash equilibrium is a profile $Q^* = (Q_1^*, \dots, Q_F^*)$ satisfying

$$Q_f^* = \mathcal{B}_f(Q_{-f}^*), \quad f = 1, \dots, F.$$

¹⁰Independence of link-cost shocks does not imply independent location choices. Shared links induce dependence among portfolio costs, while nonadditive operating profits couple optimal inclusion decisions. The application uses independent Gaussian link shocks; the general theory also permits dependence within a firm's cost vector.

Each type chooses a globally optimal portfolio in $\mathcal{X}_f$, with ties resolved by a fixed rule. The term *exact* below refers to optimization over this entire feasible family.

Proposition 2 (Existence). *Suppose the continuation selection in (4.1) is fixed independently of entry-cost types and beliefs, all selected payoffs are finite, types are independent across firms, and each within-firm type vector has an absolutely continuous joint distribution. Then an exact portfolio BNE exists.*

Appendix A.1 gives the proof. With finite action sets and absolute continuity, the global best response is unique for almost every type. The resulting distributional best-response map is continuous on $\prod_f \Delta(\mathcal{X}_f)$. Brouwer's theorem then applies.

4.2 Why an inclusion cutoff still requires portfolio optimization

Holding the other link costs fixed, compare the best network containing a country with the best network excluding it. The former value falls with that country's own cost; the latter does not. Their crossing gives an inclusion cutoff, but finding either value can require changing several other countries at once.

Fix rival beliefs Q_{-f} and write $v_f(S_f) = V_f(S_f; Q_{-f})$ for the resulting deterministic interim portfolio value. For a link j whose inclusion varies across the feasible family, define the best values among portfolios that include and exclude j :

$$V_{fj}^{\text{in}} = \max_{\substack{S_f \in \mathcal{X}_f \\ j \in S_f}} \left\{ v_f(S_f) - \sigma_\varepsilon \sum_{k \in S_f \setminus \{j\}} \ell_{fk} \varepsilon_{fk} \right\},$$

$$V_{fj}^{\text{out}} = \max_{\substack{S_f \in \mathcal{X}_f \\ j \notin S_f}} \left\{ v_f(S_f) - \sigma_\varepsilon \sum_{k \in S_f} \ell_{fk} \varepsilon_{fk} \right\}.$$

The comparison portfolios may contain different other links, so the cutoff is contingent rather than a separate optimization for link j .

Lemma 1 (Contingent inclusion cutoff). *Except on a probability-zero tie set, the globally optimal portfolio contains link j if and only if*

$$\varepsilon_{fj} < \bar{\varepsilon}_{fj}(\varepsilon_{f,-j}; v_f) \equiv \frac{V_{fj}^{\text{in}} - V_{fj}^{\text{out}}}{\sigma_\varepsilon \ell_{fj}}. \quad (4.6)$$

Example 1: A two-link portfolio. Suppose all four portfolios $\emptyset$, $\{a\}$, $\{b\}$, and $\{a, b\}$ are feasible, and write their deterministic values as $v_\emptyset$, v_a , v_b , and v_{ab} . Conditional on ε_b , the cutoff for link a is

$$\begin{aligned} \sigma_\varepsilon \ell_a \bar{\varepsilon}_a(\varepsilon_b) = & \underbrace{\max\{v_a, v_{ab} - \sigma_\varepsilon \ell_b \varepsilon_b\}}_{\text{best value with } a} \\ & - \underbrace{\max\{v_\emptyset, v_b - \sigma_\varepsilon \ell_b \varepsilon_b\}}_{\text{best value without } a}. \end{aligned} \quad (4.7)$$

The cutoff is a reservation cost for including a after reoptimizing the rest of the network. Both values are net of deterministic link costs and private costs on other included links; only a 's own private-cost component is excluded. The best portfolio containing a can switch between $\{a\}$ and $\{a, b\}$, while the best portfolio excluding a can switch between $\emptyset$ and $\{b\}$. The cutoff therefore depends on the other link's realized cost even though the firm still chooses one globally optimal portfolio. If $v_{ab} - v_a = v_b - v_\emptyset$, the two maximization terms move together and the cutoff reduces to the separable value $(v_a - v_\emptyset)/(\sigma_\varepsilon \ell_a)$. Outside that special case, the four joint portfolio probabilities contain information that the two marginal entry rates do not.

When do independent country rules recover the optimum?

Remark 1 (Independent thresholds as a special case). If every subset is feasible and costs have rich support, fixed country cutoffs reproduce the complete-portfolio response for almost every cost realization if and only if interim operating value is additive across links. Subtracting additive link costs does not change this condition. When value is additive, the include and exclude problems share the same optimization over other links, so their difference is a constant country value. With independent own shocks the resulting distribution also factors into independent link choices.

Complementarity instead creates choices between no links and both links; substitution creates choices between one link and another. These boundaries depend on sums or differences of realized costs and cannot all be represented by fixed country cutoffs. Appendix D proves the additive characterization and approximation results. An additive approximation with uniform payoff error ϱ_f loses at most $2\varrho_f$. The sharper bound uses the range of the residual across portfolios, since a common shift in all values cannot affect choices. The exact portfolio is preserved when its advantage over the second-best exceeds

that range. These payoff bounds also apply to constrained portfolios if the approximate objective is optimized subject to the same constraints.

This distinction matters for the economic comparison. Two networks may earn nearly equal profits while serving customers from different countries. A small profit error can then change exposure to a geographic eligibility rule and the costs of serving other markets. Nonadditivity alone does not establish a large policy error, and a small profit error alone does not establish accurate location or price effects.

4.3 Completion bounds for global portfolio responses

Branch and bound groups unsearched portfolios, bounds the best payoff in each group, and discards groups unable to improve on the best portfolio already found. Repricing every evaluated network alone does not justify link elimination: the required ordering can fail even with unique prices.¹¹

Example 2: Unique prices without ordered entry incentives. Consider two firms in one market, with AB parameters $(\sigma, \eta, \theta) = (5, 2, 3)$, unit quality weights, and no residual supply. Hold firm 2's marginal cost at one. Firm 1 has cost $[0.001 + \sum_{j \in T} d_j]^{-1/3}$, where $(d_1, d_2, d_3) = (0.02, 0.02, 0.099)$. Link 1 costs 0.007; the other links have zero cost. Let $\Delta_1(T)$ be the net gain from adding link 1 to T , with both firms repricing before and after the addition. Normalizing market expenditure to one gives

T	$\emptyset$	$\{2\}$	$\{2, 3\}$
$\Delta_1(T)$	-0.000337	+0.000897	-0.001047.

Link 1 becomes profitable after link 2 is added, then unprofitable after link 3 is added. Neither expansion nor contraction preserves its positive marginal return. The rise and subsequent fall also violate submodularity and supermodularity, respectively; subtracting an additive link cost does not change either comparison. Thus the monotonicity underlying the equilibrium bounds in Sabal (2026) is not guaranteed in this environment. The changes in market share and the perceived demand elasticity occur alongside the techno-

¹¹ Arkolakis et al. (2026) solve for a complete combination under single-crossing restrictions on marginal values. Independent country cutoffs impose a different restriction. The example below shows that unique downstream prices alone do not guarantee the ordering needed by their procedure.

logical cost reduction; unique pricing does not force their combined effect to be ordered. This is a constructed counterexample, not a measured policy loss in the eight-provider economy. Appendix E.2 supplies the calculation and the violated AES conditions. Completion bounds do not require those conditions: they discard a group only when its entire remaining profit potential is too small.

For a type ε_f , write the objective as

$$\widetilde{V}_f(T_f; \varepsilon_f, Q_{-f}) = V_f(T_f; Q_{-f}) - \sigma_\varepsilon \sum_{j \in T_f} \ell_{fj} \varepsilon_{fj}. \quad (4.8)$$

The first term is the type-independent interim value; only included links incur their realized cost shocks. A partial-portfolio subproblem (S_f, U_f) records links already selected in S_f and links still undecided in U_f . Its feasible completions form the set $\mathcal{X}_f(S_f, U_f)$, and a completion upper bound is any value $B_f(S_f, U_f)$ that dominates the payoff of every such completion:

$$\mathcal{X}_f(S_f, U_f) = \{T_f \in \mathcal{X}_f : S_f \subseteq T_f \subseteq S_f \cup U_f\}, \quad B_f(S_f, U_f) \geq \max_{T_f \in \mathcal{X}_f(S_f, U_f)} \widetilde{V}_f(T_f; \varepsilon_f, Q_{-f}). \quad (4.9)$$

If $B_f(S_f, U_f)$ does not exceed the best payoff already found, no portfolio represented by that subproblem can improve on the best choice already found.

The useful bound retains one feature of the common portfolio: its number of additional links. Write the objective (4.8) equivalently as $\sum_\varsigma q_\varsigma \pi_{f\varsigma}(T_f) - \sum_{j \in T_f} C_{fj}$, separating operating profit from total link costs: ς indexes a finite rival-portfolio state, $q_\varsigma \geq 0$ is the probability the rival portfolio distribution assigns to state ς (a point mass on the realized rival profile in a complete-information benchmark), and $C_{fj} = C_{fj}^0 + \sigma_\varepsilon \ell_{fj} \varepsilon_{fj}$ combines deterministic and realized private costs. Suppose the continuation payoff in state ς is bounded by a coordinatewise increasing function of nonnegative additive statistics, indexed by $m = 1, \dots, L$:

$$z_{\varsigma m}(T_f) = z_{\varsigma m}^0 + \sum_{j \in T_f} a_{\varsigma m j}, \quad a_{\varsigma m j} \geq 0, \quad \pi_{f\varsigma}(T_f) \leq \bar{\pi}_{f\varsigma}(z_{\varsigma 1}(T_f), \dots, z_{\varsigma L}(T_f)).$$

For k additional links, the bound combines the k largest contributions to each statistic with the k cheapest costs, even when these come from different sets of locations. Let $a_{\varsigma m, (h)}^\downarrow(U_f)$

be the h th largest contribution to statistic (ς, m) among undecided links, and let $C_{f,(h)}^\uparrow(U_f)$ be the h th smallest link cost. Define

$$\begin{aligned}\bar{z}_{\varsigma m}(S_f, U_f, k) &= z_{\varsigma m}(S_f) + \sum_{h=1}^k a_{\varsigma m, (h)}^\downarrow(U_f), \\ \underline{C}_f(S_f, U_f, k) &= \sum_{j \in S_f} C_{fj} + \sum_{h=1}^k C_{f,(h)}^\uparrow(U_f).\end{aligned}$$

Proposition 3 (Cardinality-conditioned completion bound). *Under the conditions above,*

$$B_f^{\text{card}}(S_f, U_f) = \max_{0 \leq k \leq |U_f|} \left\{ \underbrace{\sum_{\varsigma} q_{\varsigma} \bar{\pi}_{f\varsigma}(\bar{z}_{\varsigma}(S_f, U_f, k))}_{\text{operating-profit bound with } k \text{ additions}} - \underbrace{\underline{C}_f(S_f, U_f, k)}_{\text{total cost lower bound}} \right\} \quad (4.10)$$

is an upper bound on every feasible completion represented by the subproblem. The result applies to any finite feasible portfolio family; including cardinalities with no feasible completion only weakens the bound.

The relaxation may credit one country's best contribution in one market and another country's best contribution elsewhere, while subtracting the cheaper cost. That combination need not be feasible; precisely for that reason it can overstate, but not understate, the best completion. A common number k of additions prevents crediting all remaining locations while paying for only a few. Maximizing over k permits endogenous portfolio size, including negative realized link costs. Appendix C gives the proof and the CES and AB envelopes. Where parent-supported paths are not additive in selected cores, validity requires a dominating envelope for that complete service mapping, not treating each path as an independent core.

A tighter bound connects each location's benefits to its own cost. Suppose one undecided country offers the largest reduction in European delivery costs but is expensive, while another is cheap and mainly serves Asia. Separate market and cost bounds can credit the European benefit and charge the Asian country's cost when only one addition is allowed. The joint-affine refinement instead ranks countries by a common affine bound on their total benefit net of cost. It cannot combine attributes from different countries into that single addition. Proposition C.1 in Appendix C.4 gives the construction and proves

that its bound is no larger than the cardinality bound built from the same component envelopes. A tighter bound need not take less total computation time because evaluating it is itself costly. The completion bounds combine standard branch-and-bound and top- k tools (Feldman and Topaloglu, 2015; Leitner et al., 2024; Zhang et al., 2025) with restrictions linking market benefits and location costs across rival states. Appendix C.5 describes their use in the cloud policy searches.

Optimizing each type's portfolio against given rivals is the inner problem. An equilibrium also requires the resulting portfolio distributions to be mutually consistent. The next subsection gives conditions for uniqueness of that fixed point; Appendix C.7 separates search, integration, and equilibrium approximation errors.

4.4 Probability flow and uniqueness

A rival's portfolio change perturbs the value of every own portfolio. Only differences in that perturbation affect choices: adding the same amount to all values leaves the optimizer unchanged. Private-cost dispersion limits the mass of types close enough to a moving choice boundary to switch. For a single optional link with deterministic net benefit a , inclusion has probability $F_\varepsilon(a/(\sigma_\varepsilon \ell))$, where F_ε is the cost-shock distribution function. If its density is bounded by $\bar{g}$, a change Δa alters this probability by at most $\bar{g}|\Delta a|/(\sigma_\varepsilon \ell)$. With several links, one switch can change several locations at once. The following bound accounts jointly for those changes, using conditional densities when costs within a firm are dependent.

Let $K_{fg}^\# \geq 0$ bound the influence of rival g on firm f , with $K_{ff}^\# = 0$. These coefficients form an interaction matrix, distinct from the core counts K_f . Appendix A.3 constructs these coefficients from changes in relative portfolio payoffs and conditional cost densities. Its probability-flow argument accounts for types that switch several locations together without charging the same movement separately at every location. The resulting bound is

$$\underbrace{d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f}))}_{\text{own response}} \leq \sum_{g \neq f} \underbrace{\frac{K_{fg}^\#}{\sigma_\varepsilon}}_{\text{sensitivity bound}} \underbrace{d_{\text{TV}}(Q_g, Q'_g)}_{\text{rival change}}. \quad (4.11)$$

Total variation bounds the change in the probability of any collection of portfolios, such as those containing an EU location. The coefficients bound response magnitudes; they allow

a rival to raise or lower the relative return to an own portfolio.

Theorem 1 (Uniqueness under sufficiently dispersed private costs). *Under the assumptions in Section 4.1, each firm’s distributional best response is Lipschitz in rivals’ portfolio distributions with coefficient matrix $K^\#/\sigma_\varepsilon$ (inequality (4.11)). Consequently, if*

$$\sigma_\varepsilon > \rho(K^\#), \quad (4.12)$$

the exact portfolio BNE is unique, and simultaneous distributional best-response iteration converges to it from every initial portfolio distribution. The same conclusion holds with any verified componentwise upper bound on $K^\#$, including the include-exclude matrix K defined in Appendix A.2.

The spectral radius measures feedback around cycles. To see why it governs uniqueness, consider two equilibria in a two-firm game. Their component distances satisfy $d_1 \leq M_{12}d_2$ and $d_2 \leq M_{21}d_1$, where $M_{fg} = K_{fg}^\#/\sigma_\varepsilon$. Thus $d_1 \leq M_{12}M_{21}d_1$: if $M_{12}M_{21} < 1$, both distances must be zero. A large response bound in one direction can therefore coexist with uniqueness if the reverse bound is small. The spectral-radius condition extends this argument to feedback among many firms. Appendix A.5 proves the result; Proposition A.2 shows that reweighting firms cannot improve this test for the same interaction matrix.

A uniform bound may be conservative. A localized test first establishes that all equilibria lie in a smaller region, then bounds responses within it. A one-sided test can retain signed cancellations without requiring simple response iteration to converge. Appendix B gives these refinements and their assumptions, including Gaussian containment for arbitrary finite feasible families. None is verified merely by numerical convergence from several initial values. Nor does the general theory require fixing portfolio size. The calibration does not impose this condition. In the China application, response iteration from three different initial distributions converges to portfolio distributions that agree to within 0.26 percentage points in any country-inclusion probability and 0.11 in expected world-wide core counts; for the EU regimes, re-solving at higher integration accuracy changes worldwide policy core-count differences by less than 0.10 presences and the regulated price effect by less than 0.02 percentage points (Appendix H.4).

5. QUANTIFYING MULTINATIONAL CLOUD NETWORKS

We calibrate provider-specific cost intercepts and relative competitiveness to worldwide core counts and market shares. Spatial cost coefficients and private-cost dispersion are held fixed. Regional composition and the characteristics of hosting countries are used to assess the implied geography.

5.1 Data and assigned inputs

Our 2026 eight-provider sample (F8) covers Amazon Web Services, Microsoft Azure, Google Cloud, Oracle Cloud, Alibaba Cloud, Huawei Cloud, Tencent Cloud, and OVHcloud. It records 169 core-country presences in a 42-country universe of 336 provider-country choices. As in Section 3.1, multiple core regions of the same provider in one country count as a single presence. Every country is available as a core location in the EU specification, including unobserved and unannounced locations. The China application adds explicit operating-mode restrictions. Current cores and the historical eight-provider extension in Appendix G, with 170 pairs having some documented presence, have different dates and supply definitions.

Expenditure is proportional to $(\text{GDP} \times \text{internet-use rate})^{0.7}$, normalized to one in the United States. Appendix F.2 gives its formula and source conventions. The model EU contains Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, the Netherlands, Poland, Spain, and Sweden. Splitting each into regulated and private demand gives 54 customer segments. The regulated fraction $\omega = 0.18$ sets the assumed exposure of EU expenditure to the procurement rule. Both segments share supply rules in the baseline.¹²

The operating input-cost index is $w_i = 0.75 + 0.20(E_i/E_{US}) + 0.05(L_i/L_{US})$. Electricity prices use large industrial-user tariffs, including Eurostat band IF where available, and documented fallbacks. Labor costs use a human-capital-adjusted compensation proxy. These weights describe a maintained operating-cost index, not estimated production shares.

¹²The exposure share assigns 18 percent of total modeled EU cloud expenditure to the regulated segment as a scenario assumption, rather than an estimate from procurement spending. The Commission's figures for the scope of SecNumCloud concern shares of public-sector use cases, not shares of total cloud expenditure (European Commission, 2026a, Part 1, p. 43). They therefore do not directly calibrate this parameter or establish its order of magnitude.

Delivered-service costs use a three-level geographic rule:

$$\tau_{in} = \begin{cases} 1, & i = n, \\ 1.15, & i \neq n, \text{ same continent or within 5,000 km,} \\ 1.50, & \text{otherwise.} \end{cases} \quad (5.1)$$

At $\theta = 3$, these foreign delivery wedges give an origin approximately 66 or 30 percent of an otherwise identical domestic origin's contribution to the cost aggregator, since $1.15^{-3} \simeq 0.66$ and $1.50^{-3} \simeq 0.30$. The delivery-distance rule uses the input file's city reference coordinates, rather than individual facilities. Thus all foreign EU-to-EU origins have the same baseline cost wedge. These levels are maintained assumptions; measured latency is not used to estimate the resource cost of serving a customer abroad.

Latency instead helps order accessibility within the EU. For each EU destination, we rank the eleven other EU origins by measured round-trip time and assign the highest existing accessibility score to the fastest origin. The procedure preserves that destination's distribution of scores, while changing which origins receive them. Domestic scores and pairs outside this EU block retain the geographic specification. The measurements are RIPE anchor-to-anchor ping statistics; they are not provider-specific private-network performance or customer-weighted service quality. The score formula, sample convention, and fixed frontier are specified in Appendix F.2. This use of latency disciplines relative location rankings, not a causal willingness to pay for one millisecond. Table 1 separates calibrated parameters from assigned parameters and inputs.

The demand shifter d_{nf} equals $\log 4$ for U.S. providers serving U.S. customers, Chinese providers serving China, and OVHcloud serving the model EU; otherwise it is zero. OVHcloud's preference is regional. At given costs and prices, home preference raises a provider's price-index contribution by a factor of four; equilibrium shares do not increase fourfold. The factor of four is assigned, rather than estimated from these global share targets; relative competitiveness is calibrated conditional on it.

In the net cost of core presence (3.7), distance is measured in 10,000 kilometers. We maintain $\delta = 0.614$, $\kappa = 0.5$, and $\sigma_\varepsilon = 0.5$. The level of the construction-cost index does not enter (3.7); its missing-data indicator does. An additional 10,000 kilometers raises deterministic cost by $0.5\ell_f^{\text{ref}}$ per core, equal to one standard deviation of that provider's

Table 1: Calibrated Parameters and Assigned Inputs

Parameter or input	Value	Role
<i>Panel A. Calibrated parameters</i>		
α_f (eight providers)	$(-.119, -.040, .365, .498, .675, .798, .746, 1.268)$	Match expected worldwide core-country counts through signed net entry costs.
a_f/a_{AWS}	$(1, .817, .628, .514, .692, .665, .528, .210)$	Seven relative competitiveness parameters match global-share ratios jointly with residual intensity; $a_{\text{AWS}} = 1$.
Residual intensity	Log multiplier .1396; multiplier 1.150	Match the residual share with a common price-index contribution across markets.
<i>Panel B. Assigned parameters and inputs</i>		
(σ, η, θ)	$(5, 2, 3)$	Substitution, perceived demand elasticity, and supply aggregation.
Missing-data loading	$\delta = .614$	Cost penalty where original construction-cost data are missing.
Home-country distance	$\kappa = .5$	Net location-cost gradient in distance from the provider's home.
σ_ε	.5	Private-cost dispersion in reference payoff units.
γ_{loc}	.5	Contribution of local-only service relative to a core.
Accessibility	$\lambda_A = 1.5, \chi = .4$	Accessibility mapping in the price-index contribution.
Home preference	$d_{nf} = \log 4$ at home; 0 elsewhere	Domestic demand advantage; the home market for OVHcloud is the model EU.

Notes: Firm vectors are ordered AWS, Microsoft, Google, Oracle, Alibaba, Huawei, Tencent, and OVHcloud. The parameters are held fixed across policy regimes in both applications. Values are rounded for display. The specification includes no regional cost intercept. Policy outcomes did not select these values.

private per-core cost shock. There is no regional cost intercept. The missing-data term is an assigned penalty.¹³ Fixed provider-specific loadings convert these standardized costs into operating-profit units; Appendix F reports their values and normalization. The loadings remain unchanged when the number of chosen locations changes. Private-cost dispersion is set independently of the uniqueness condition.

¹³Missing construction-cost data need not imply higher location costs. The term δM_i imposes that association; it is not an estimated construction-cost effect. Its economic implications depend on which locations lack data. Belgium is the only model-EU country affected; Appendix F.3 lists all affected countries.

5.2 Matching worldwide scale and shares

We match expected worldwide core counts using eight intercepts α_f . Seven relative competitiveness parameters, normalized by $a_{\text{AWS}} = 1$, and one residual-supply intensity match eight nonredundant global-share ratios. The residual represents unmodeled suppliers. Its price-index contribution is assumed equal across markets, and this common level is calibrated; its realized share is endogenous. The competitiveness parameters absorb relative supply and demand advantages conditional on the other inputs and are not separately identified physical productivities.

Table 2 lists the targets. Five providers' share targets come from public 2024 worldwide IaaS data. The targets for Oracle, Tencent, and OVHcloud are assumed because the public source used here does not report their shares; the residual is the remaining share. These targets are not a complete observed 2026 F8 distribution. The infrastructure and share data refer to different years, and detailed country-level revenue shares are not matching targets.

The two matching blocks are coupled through portfolio choice and pricing. We choose cost intercepts to match expected counts, solve the global-share equations for competitiveness and residual intensity, and update portfolio distributions and prices as the parameters change. Worldwide shares average expenditure shares over priced portfolio profiles; pricing an average network would give a different object. The resulting parameters and portfolio distribution must jointly satisfy the count and share targets and the firms' best-response conditions. This simulation-based calibration nests count and share matching; it does not treat the 336 entry decisions as independent observations in a likelihood. Conditional count matching does not by itself identify the counterfactual elasticity of core counts. That response also depends on the assigned private-cost dispersion, spatial costs, service technology, and the profit changes induced by pricing and rivals' choices. Calibrated parameters are held fixed across policy regimes.

5.3 Location moments and baseline fit

We use 40 composition moments to assess core geography. They also informed specification assessment and are correlated summaries of the eight observed portfolios. The reported calibration adjusts count and share parameters conditional on the assigned spatial-

Table 2: Worldwide Market Shares and Core-Country Counts

Provider	World share (%)		World core count	
	Target	Model	Target	Model
AWS	37.7	37.66	28	27.99
Microsoft	23.9	23.85	33	32.97
Google	9.0	8.94	29	28.91
Oracle	4.0	3.98	27	26.92
Alibaba	7.2	7.09	17	16.84
Huawei	4.1	4.05	12	11.85
Tencent	2.0	2.37	13	14.86
OVHcloud	0.1	0.10	10	9.57
Eight providers	88.0	88.03	169	169.92
Residual	12.0	11.97	—	—

Notes: Model values are evaluated at the calibrated parameters and the computed baseline portfolio distribution. Shares average two independently scrambled samples of 2,048 portfolio profiles; expected core counts are calculated directly from the distribution. The maximum share error is 0.37 percentage points and the core-count RMSE is 0.68 countries. Both worldwide counts and share ratios are calibration targets. Appendix H reports numerical accuracy and the remaining search gaps.

cost coefficients and private-cost dispersion. For each provider, the moments are the fractions of core presences in the model EU, Asia-Pacific, and the Americas, the mean construction-cost index with imputed missing values, and mean log expenditure in its hosting countries. The three regional fractions contribute 24 moments; construction costs and expenditure contribute eight each. With variable portfolio size, the moment for country feature h_{ij} is

$$m_{fj}(Q_f) = \frac{\mathbb{E}_{Q_f}[\sum_{i \in S_f} h_{ij}]}{\mathbb{E}_{Q_f}[|S_f|]}. \quad (5.2)$$

The ratio of expectations weights countries by their inclusion probabilities. It differs from averaging each portfolio's fraction and retains zero-core types in the unconditional expectations. Composition is undefined if the expected core count is zero.

These moments assess the geographic implications of the count and share calibration. We also summarize their discrepancies with a weighted score, defined in Appendix F.4; the reported calibration does not estimate additional spatial coefficients by minimizing that score. Missing construction data use a projection on public covariates and a median fallback. The fraction of chosen countries with missing data is reported separately from the 40 moments.

Table 2 reports worldwide share and core-count targets and their model counterparts. Aggregate fit does not ensure accurate geographic allocation: a provider may choose the right number of countries while allocating too many cores to the EU or to particular EU countries. Such discrepancies matter for policies that restrict where eligible supply can originate. Core counts and policy-eligible supplier counts also differ when local paths provide service.

The current baseline distribution matches the global-share targets with a maximum absolute error of 0.37 percentage points. Its firm-level worldwide core-count RMSE is 0.68 countries, and it predicts 169.92 worldwide provider-country links against the target of 169. Two discrepancies remain despite the close aggregate fit. Tencent has 14.86 predicted core countries against 13 in the data, and the model predicts 52.54 EU links against 40. The latter is a 31.3 percent overprediction. Microsoft, Google, and OVHcloud are comparatively close to their observed EU totals, while AWS, Oracle, Alibaba, Huawei, and Tencent account for most of the excess. Tencent’s discrepancy is a remaining calibration error: meeting the portfolio solver’s convergence criterion does not imply exact count matching.

The calibration disciplines worldwide scale and shares without forcing their regional allocation. We retain common spatial cost coefficients rather than fitting an EU-specific intercept. Such an intercept would change the returns to both EU and non-EU locations and require joint recalibration and new portfolio equilibria; it would not correct this discrepancy in isolation. The observed-network comparison in Table 6 instead shows how the fixed-network price effect depends on starting geography, without estimating such an intercept.

Figure 2 distinguishes core presence from service coverage. The model underpredicts core presence in Germany but overpredicts it strongly in Finland, Austria, and Denmark. Local-service paths also make coverage less responsive than core presence in some countries. These distinctions matter for the EU policy, which restricts eligible service origins rather than the number of core-hosting countries.

For the counterfactuals, we approximate the Bayesian portfolio equilibrium in Section 4 using finite candidate sets of complete networks and simulated private-cost vectors. Locations and core counts can both change. Each provider chooses its most profitable candidate for each simulated type against rivals’ current portfolio distributions. These choices update its distribution, preserving dependence across its country choices. Prices

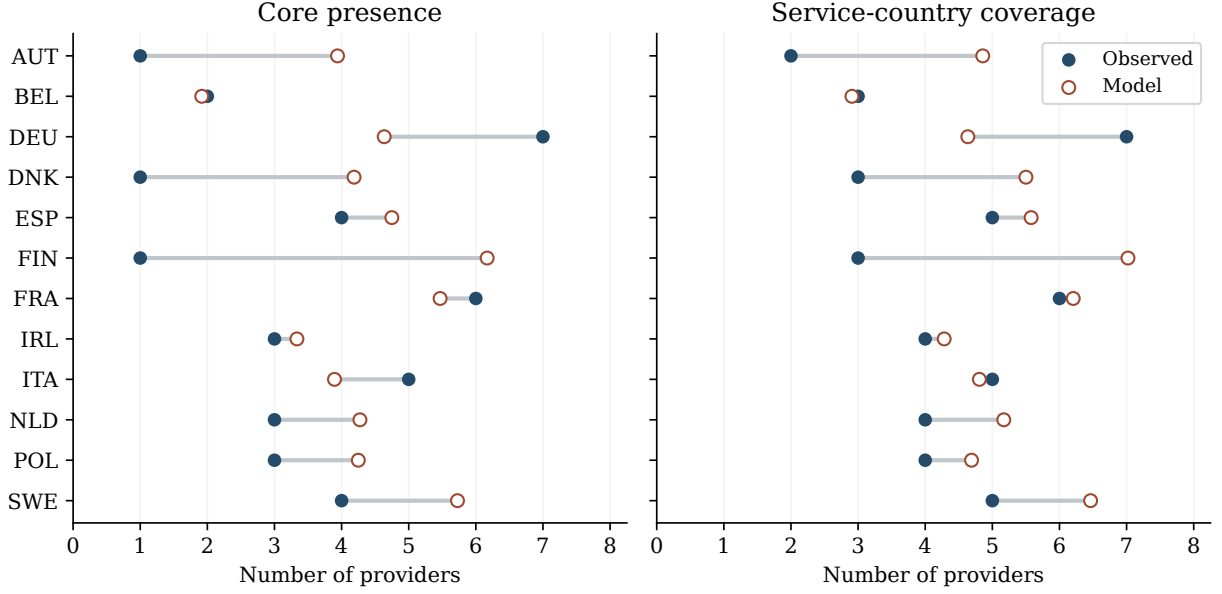


Figure 2: EU geography: core presence and service coverage

Notes: Observed provider counts and expectations under the current baseline portfolio distribution. Both panels use the same country universe and local topology. Model coverage evaluates whether the required parent cores are present for each complete portfolio before averaging. The panels are geographic diagnostics, not additional independent estimation targets.

are solved jointly for each portfolio profile before profits and market outcomes are averaged. This preserves the joint core-presence events that determine parent-supported local service. Branch-and-bound searches at selected cost types add candidates; all policy regimes within an application use the expanded sets in common.

Policy price effects compare expected log indices, using fixed expenditure weights to aggregate customer segments. We average log-index differences over portfolio profiles and report $100[\exp(\overline{\Delta \log P}) - 1]$.

We assess these approximations using payoff regret within the candidate sets and the undamped residual of the portfolio-distribution map. Appendix H reports convergence measures, remaining search gaps, and sensitivity to initialization and numerical integration. Each branch-and-bound search reports an upper bound on the remaining payoff improvement for the evaluated cost type and the supplied rival portfolio distribution.

6. TWO APPLICATIONS OF CLOUD-MARKET FRAGMENTATION

Both applications change the geographic reach of service supply and the returns to multinational locations, using the same calibrated parameter vector. Core-country counts and locations can adjust, while entry-cost parameters, competitiveness, home preference, and baseline residual intensity remain fixed across regimes. The EU experiment changes geographic eligibility for a segment of public demand. The China experiment lowers or raises the cost of supplying mainland customers from offshore facilities. These policies affect different parts of a network; their price effects need not be similar even at identical portfolios. The EU experiment allows cross-border supply within each provider. The China application conditions on existing operating modes and partner paths and uses its own baseline portfolio distribution, recalculated under those arrangements. The two applications share parameters but differ in their feasible operating arrangements.

6.1 EU public cloud: EU-wide eligibility and national fragmentation

The EU experiment changes the markets that a given service origin can reach. Under EU-wide eligibility, qualifying supply can serve regulated public customers throughout the model EU. Under national fragmentation, it can serve only domestic regulated demand. With the same regulated expenditure share $\omega = 0.18$, the eligibility rules are

$$e_{in}^E = \mathbf{1}\{i \in \text{EU12}\}, \quad e_{in}^N = \mathbf{1}\{i = n\}, \quad n \text{ a regulated EU segment.}$$

The baseline imposes no geographic eligibility restriction on these customers, allowing them to use the ordinary global supply network. Unregulated demand keeps its original eligibility in both policy regimes. National minus EU-wide therefore changes the geographic scope of eligibility while holding the regulated customer group and expenditure share fixed. National fragmentation is a scenario for sensitive public demand. It is not a general EU localization mandate: EU non-personal-data rules prohibit national localization subject to a proportionate public-security exception (European Parliament and Council of the European Union, 2018, Article 4). The experiment does not value security benefits or model ownership tests, sovereignty grades, and procurement auctions.

Table 3: Core Presence by Eligibility Regime

Core presences	Baseline	EU-wide	National
World	169.92	166.27	162.16
EU	52.54	53.34	50.55
Non-EU	117.38	112.93	111.61

Notes: Expected provider-country core presences of the eight modeled providers under the computed portfolio distribution of each regime (equilibrium responses). The baseline imposes no geographic eligibility restriction on regulated public customers; EU-wide and National are defined in Section 6.1. Table 4 reports the National minus EU-wide differences. Residual suppliers do not choose locations in the model.

Separating access restrictions from network responses. Let Q^O be the baseline portfolio distribution. *Fixed networks* retain Q^O under each policy. *Responses to unchanged rivals* combine each provider’s optimal response to its rivals’ baseline distributions, $Q^{r,u} = \bigotimes_f \widehat{\mathcal{B}}_f^r(Q_{-f}^O)$, and jointly reprice the resulting profiles. *Equilibrium responses* use the policy-specific distribution Q^r , approximating the portfolio equilibrium on the common candidate sets. All eight firms optimize in the latter two comparisons; the difference is which rival portfolio distributions they face.

For comparison h , define

$$G_h = Y_N(Q^{N,h}) - Y_E(Q^{E,h}).$$

Prices are solved in every market and portfolio profile before aggregation. For a group of customer markets, Y is the expenditure-weighted expected log price index, and the percentage effect is $100[\exp(G_h) - 1]$. Comparing equilibrium responses with fixed networks measures the total contribution of portfolio adjustment. Comparing equilibrium responses with responses to unchanged rivals measures the additional effect of portfolio feedback.¹⁴ Table 5 reports the individual provider responses; Appendix H gives the distributional definitions.

Domestic sourcing requirements need not attract more infrastructure. Table 3 reports expected core presences under the computed portfolio distribution of each regime. EU-wide eligibility expands the modeled providers’ core presence within the EU relative to

¹⁴Prices adjust jointly in both response comparisons. The difference isolates responses to changed competitor portfolios; it is neither a temporal adjustment path nor the error of a country-cutoff algorithm. Separate one-firm effects need not add to the combined effect.

Table 4: National Fragmentation: Core Networks, Prices, and Competitor Responses

Outcome	Fixed networks	Responses to unchanged rivals	Equilibrium responses
<i>Core networks</i>			
World core presences	0.00	-6.20	-4.11
EU core presences	0.00	-3.35	-2.79
Non-EU core presences	0.00	-2.84	-1.32
<i>Prices</i>			
Regulated price (%)	27.49	27.82	27.88
EU private price (%)	0.00	0.51	0.38
Non-EU price (%)	0.00	0.38	0.24
<i>Modeled providers and residual supply</i>			
Conditional provider HHI	1,133	1,238	1,218
Eligible modeled providers	-2.48	-2.68	-2.62
Residual share (pp)	55.95	56.39	56.43

Notes: National minus EU-wide under the network comparisons defined in Section 6.1. Prices are solved jointly for each portfolio profile before aggregation. Core counts are expected provider-country presences. HHI conditions on the eight providers' shares and is not a whole-market concentration measure; zero is assigned when none is eligible. Residual supply remains locally eligible with a fixed price-index contribution.

the baseline, whereas National eligibility reduces it: the expected EU core count is 52.54 under the baseline, 53.34 under EU-wide, and 50.55 under National. For the modeled providers, EU-wide eligibility directs regulated demand toward EU origins while preserving access to buyers across member countries. National eligibility favors domestic service origins over foreign ones, but limits each eligible origin to its domestic regulated market. Under EU-wide eligibility, an eligible origin can serve regulated customers throughout the model EU; under National, it can serve only domestic regulated demand. Access to that domestic demand need not justify maintaining a core when the location can no longer serve regulated buyers in other EU countries.

The contraction extends beyond the regulated region. Worldwide core presence falls from 169.92 under the baseline to 166.27 under EU-wide and 162.16 under National, and presence outside the EU from 117.38 to 112.93 and 111.61. Thus EU-wide shifts core presence toward the EU while reducing the global total, with the reduction occurring entirely outside the EU, whereas National reduces presence both within the EU and worldwide. Table 4 reports the National–EU-wide difference: 4.11 fewer worldwide core-country presences, comprising 2.79 fewer in the EU and 1.32 fewer outside it. The worldwide contrac-

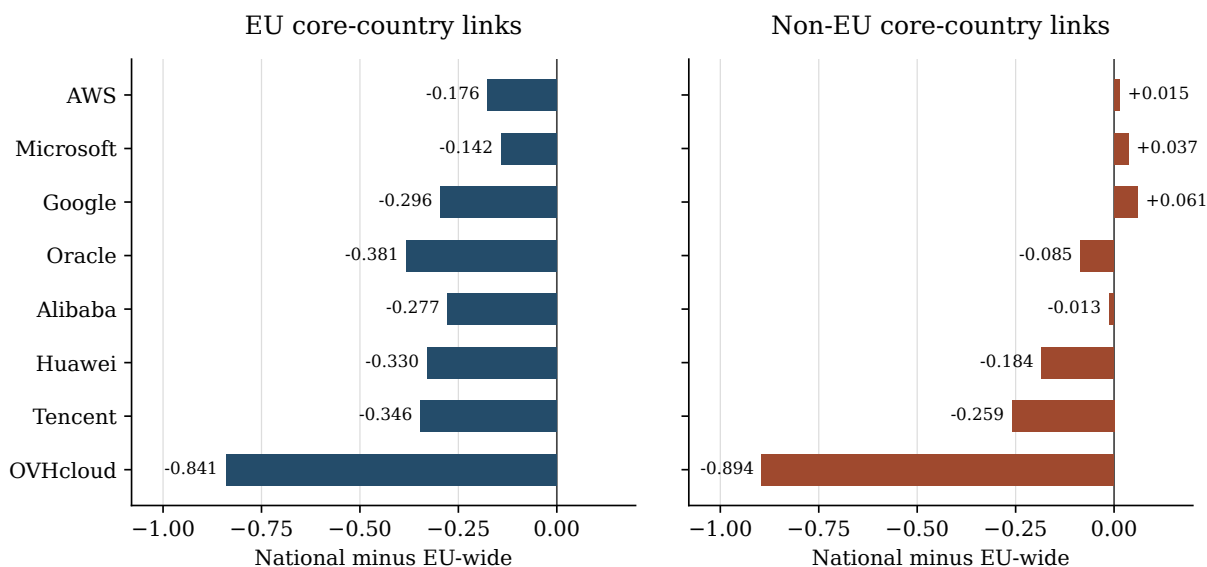


Figure 3: Where providers change their core networks

Notes: National minus EU-wide expected core-country links under the computed portfolio distributions. EU and non-EU changes add to each firm's worldwide change. These are provider-country core presences, not building counts or installed capacity. All eight providers have fewer EU core links; the non-EU response differs across firms.

tion is about 2.5 percent of the EU-wide network. These counts concern the eight modeled providers; residual suppliers do not choose locations in the model.

Figure 3 shows that all eight providers have fewer EU core presences under National than under EU-wide. Their non-EU responses differ: AWS, Microsoft, and Google add a small number of non-EU links, while the other providers contract there. OVHcloud accounts for 42 percent of the world difference, 30 percent of the EU difference, and 68 percent of the non-EU net contraction. These fractions describe core counts, not OVHcloud's contribution to price effects.

Network adjustment transmits the restriction to other markets. The regulated-market price increase arises mainly from the loss of eligible supply at existing networks. The price index rises by 27.49 percent with fixed networks and by 27.88 percent with equilibrium responses. Portfolio adjustment adds about 0.39 percentage points to this difference. Its small contribution to the regulated-price effect does not imply a small role in international transmission.

Network changes raise the EU-private price index by 0.38 percent and the non-EU in-

Table 5: EU Policy: Provider Responses and Prices Outside the EU

Responding provider	Worldwide core links	Non-EU price (%)
AWS	-0.182	0.037
Microsoft	-0.174	0.038
Google	-0.370	0.041
Oracle	-0.833	0.047
Alibaba	-0.351	0.050
Huawei	-0.602	0.075
Tencent	-1.382	0.082
OVHcloud	-2.305	0.002
Responses to unchanged rivals	-6.199	0.378
Equilibrium responses	-4.112	0.243

Notes: National minus EU-wide. In each provider row, only that provider adjusts its portfolio, taking rivals' baseline portfolios as given; all firms' prices are re-solved. Responses to unchanged rivals combine the eight individual responses and jointly reprice them. Equilibrium responses use the policy-specific portfolio distributions. Core changes are expected provider-country counts. Price effects are nonlinear and cannot be added across provider rows.

dex by 0.24 percent. At fixed networks both effects are zero: those markets' supply rules and expenditures are unchanged, and pricing has no cross-market capacity or input-price channel. Within the model, portfolio adjustment is therefore the channel through which the restriction reaches unregulated EU buyers and customers in other countries.

Competitor responses moderate international adjustment. Allowing providers to respond to changed rival networks substantially moderates the global contraction while leaving the regulated-price effect nearly unchanged. Combining responses to rivals at the baseline produces 6.20 fewer worldwide core presences under National than under EU-wide. Equilibrium responses reduce that gap to 4.11, an attenuation of about one third. The associated EU-private price effect falls from 0.51 to 0.38 percent, and the non-EU effect from 0.38 to 0.24 percent. The regulated-price increases are much closer: 27.82 and 27.88 percent, respectively. Evaluating providers against unchanged rivals would thus give a similar regulated-price effect while overstating the contraction of their global networks and its price effects in other markets.

Which providers transmit the policy to other markets? The number of locations a provider removes is not sufficient to predict its international price effect. Table 5 allows

one provider at a time to respond to each policy while keeping all rivals at the baseline. OVHcloud has the largest worldwide contraction in this comparison: 2.305 core-country links under National relative to EU-wide. Yet its one-firm response raises the non-EU price index by only 0.002 percent. Tencent removes fewer links, 1.382, but its response raises that index by 0.082 percent.

The price index in (3.4) weights service contributions, whereas core counts give each provider-country presence the same weight. Removing a link matters for customers according to the markets it serves, the provider's remaining network, and its share relative to competing supply. OVHcloud's small global-share target and regional demand advantage are therefore relevant to interpreting its large footprint response. Counts alone cannot establish a provider's contribution to aggregate price effects.

Residual supply and the magnitude of the access restriction. National reduces the expenditure-weighted number of eligible modeled providers by 2.48 at fixed networks and 2.62 with equilibrium responses. The residual accounts for 12.50 percent of regulated demand in the baseline. With portfolio responses, its share is 33.73 percent under EU-wide and 90.16 percent under National. Most regulated expenditure in the National scenario is consequently supplied outside the eight strategic firms. This large share does not mean that all eight are usually absent. Under National, the expenditure-weighted probability that a regulated segment has no eligible modeled provider is below 0.01 percent. Most residual sales occur while at least one modeled provider remains eligible but has little competitive supply. Consequently, HHI conditional on the eight providers' sales describes concentration within a small part of the regulated market. It is not a total-market HHI; the number and shares of residual suppliers remain unspecified.

The baseline assumes that residual supply remains locally eligible and has a fixed price-index contribution. It does not establish how many unmodeled suppliers meet a procurement standard. Halving or doubling this residual contribution in both regimes changes the fixed-network price effect to 39.97 and 17.86 percent, respectively. Residual supply is therefore important for the magnitude even though it does not change the direction in these scenarios.

We also allow part of residual supply to lose eligibility under National. For regulated EU customers, let $b_n^N = (1 - \xi)b_n^E$, where ξ is the fraction of the residual price-index contri-

bution that becomes unavailable. The parameter is a scenario, not an observed fraction of suppliers or revenue. Holding the baseline networks recomputed with greater integration accuracy fixed, the regulated-price effect rises from 27.50 percent at $\xi = 0$ to 35.97 percent at $\xi = 0.25$ and 48.39 percent at $\xi = 0.5$. The National residual share falls from 90.06 to 87.43 and 82.85 percent. Table 6 reports these comparisons separately from changes in residual supply in both regimes. They show the importance of the residual eligibility assumption for prices; they do not establish how portfolio responses would change.

The table also uses observed cores with the same parameters and installed local topology. The fixed-network price effect then falls from 27.49 to 22.02 percent, showing its sensitivity to starting geography. This comparison changes the entire network rather than isolating the excess number of EU locations. Starting geography and residual supply therefore matter for the magnitude. These comparisons reprice given networks; they do not establish robustness of the portfolio response under alternative residual assumptions. The residual's share also prevents interpreting the F8 network contraction as a contraction of the entire industry's networks: location and capacity choices of those suppliers are not modeled.

Provider profits and costs of core presence. Appendix Figure 5 compares worldwide operating profit with net portfolio payoff after modeled link costs. National lowers both measures for each of the eight providers relative to EU-wide, despite the higher price index paid by regulated customers. That price increase is accompanied by a shift toward residual supply; it does not imply higher profits for the named firms. OVHcloud experiences the largest proportional decline in both profit measures. Its operating profit falls by 24.69 percent and its net portfolio payoff by 21.78 percent; the corresponding operating-profit losses for AWS, Microsoft, and Google are 2.49, 2.17, and 2.73 percent.

Each firm's worldwide operating margin changes by less than 0.07 percentage points. These margins are ratios of expected operating profit to expected revenue before costs of core presence. A small change in the ratio can accompany a substantial loss of sales and profit; it does not contradict the contraction in networks. Net portfolio payoff additionally subtracts modeled signed link costs. Appendix H gives the profit accounting.

Appendix H reports convergence and the remaining search gaps. Residual eligibility and the baseline's excess EU core presence are the main economic qualifications to these

Table 6: EU Price Magnitudes Depend on Residual Supply and Starting Geography

Assumption	Regulated price	Residual share (%)	
	National–EU-wide (%)	EU-wide	National
<i>A. Residual supply and starting geography</i>			
Residual term $\times 0.5$	39.97	21.57	82.85
Baseline residual term	27.49	34.11	90.06
Residual term $\times 2$	17.86	49.04	94.56
Observed cores, baseline residual	22.02	40.92	90.80
<i>B. National eligibility of residual supply</i>			
National exposure $\xi = 0$	27.50	34.10	90.06
National exposure $\xi = 0.25$	35.97	34.10	87.43
National exposure $\xi = 0.5$	48.39	34.10	82.85

Notes: Panel A’s first three rows hold the baseline portfolio distribution fixed and multiply residual price-index contribution in both policy regimes by 0.5, 1, or 2. Its final row uses observed cores with the same installed local topology and parameters. Panel B reduces the residual contribution only in National regulated markets, $b_n^N = (1 - \xi)b_n^E$, leaving EU-wide supply unchanged. It uses the baseline distribution recomputed with greater integration precision; the unexposed price effect differs from Panel A’s baseline by less than 0.01 percentage points. All rows re-solve prices at fixed networks. These are conditional repricing comparisons, not recalibrations or portfolio equilibria under alternative residual supply. They assess magnitudes and starting geography without establishing the robustness of the portfolio response itself.

magnitudes.

6.2 China: cross-border supply costs and global infrastructure

The second application asks how a change in the cost of serving mainland customers from offshore infrastructure changes global core networks. A higher cross-border cost can favor a mainland location while making a provider’s offshore network less valuable. The relevant network response is therefore the change in its entire portfolio, including facilities outside China. We examine partial easing and two tightening scenarios to assess the same channel in both directions.

Policy experiment and maintained operating arrangements. The policy parameter d changes only

$$\tau_{i,\text{CHN}}(d) = \tau_{i,\text{CHN}}(0) \exp\{d \mathbf{1}(i \neq \text{CHN})\}, \quad d \in \{-0.10, 0, 0.10, 0.25\}.$$

Thus easing reduces the offshore-to-China trade-cost factor by $1 - e^{-0.10} = 9.52$ percent, approximately 10 percent. The two tightening scenarios raise it by 10.52 and 28.40 percent. The paired ± 0.10 changes are symmetric in log costs rather than their levels. These are changes around an existing segmented environment, not estimates of removing or introducing the firewall. China-to-China supply, other destinations' trade costs, accessibility q , expenditure, costs of core presence, and demand preferences do not change.

The calculation holds current operating arrangements fixed. AWS and Microsoft retain fixed partner-operated branded paths serving mainland customers. Those paths do not count as direct core choices and do not receive private location-cost draws. The five non-Chinese providers cannot add a directly controlled mainland core in this scenario; Alibaba, Huawei, and Tencent can choose mainland and offshore cores. Offshore service to China remains available to foreign providers. This restriction represents the maintained operating arrangements; it does not imply that all foreign participation in cloud services is legally prohibited. New partnership formation and changes in profit-sharing contracts are outside its scope. The central profit accounting assigns all modeled branded operating profit to the provider, including on the two partner paths; it does not estimate the division of rents between partners.

The calibrated parameter vector is retained and the baseline portfolio distribution is re-solved under these operating arrangements. This baseline retains existing segmentation and sets the additional cost change to zero ($d = 0$); its operating restrictions differ from those of the EU baseline. All four distributions are computed on the same expanded set of candidate portfolios, and every policy effect uses the same baseline reference. Appendix H.3 reports convergence and search gaps; Appendix H.4 reports sensitivity to initial distributions and numerical integration.

Greater mainland core presence can accompany a smaller global network. Higher offshore delivery costs increase Chinese providers' mainland core presence while reducing their worldwide networks. These providers also serve mainland customers from offshore facilities, so the shock raises their own cost of supplying those customers from abroad. Table 7 compares fixed networks, responses to unchanged rivals, and equilibrium responses, as defined in the EU application; all effects use the common baseline reference under the specified operating arrangements. Figure 4 shows which providers account for the loca-

Table 7: China: prices and core networks under easing and tightening

Network response	China price (%)	Other prices (%)	World cores (links)	China cores (links)	Residual share (pp)
<i>Easing: trade cost -9.52%</i>					
Fixed networks	-7.84	0.00	0.00	0.00	-2.72
Responses to unchanged rivals	-8.11	-0.14	+1.43	-0.03	-2.80
Equilibrium responses	-8.05	-0.11	+0.96	-0.04	-2.78
<i>Moderate tightening: trade cost +10.52%</i>					
Fixed networks	+7.89	0.00	0.00	0.00	+3.46
Responses to unchanged rivals	+8.36	+0.24	-2.49	+0.07	+3.69
Equilibrium responses	+8.17	+0.13	-1.21	+0.08	+3.59
<i>Larger tightening: trade cost +28.40%</i>					
Fixed networks	+19.37	0.00	0.00	0.00	+10.02
Responses to unchanged rivals	+20.72	+0.70	-7.36	+0.12	+10.94
Equilibrium responses	+20.25	+0.42	-3.90	+0.16	+10.61

Notes: All scenarios use common candidate sets and the same baseline reference, which retains existing segmentation. The three response comparisons are defined in Section 6.1. Prices are solved jointly for every profile; reported changes use expenditure-weighted expected log indices. Core counts are expected provider-country presences, excluding fixed partner paths. Residual supply is unexposed to the shock ($\xi = 0$).

tion changes.

Under the larger tightening, Alibaba's mainland-core probability rises by 10.18 percentage points, Huawei's by 3.43, and Tencent's by 2.21. Nevertheless, their worldwide networks contract by 0.68, 1.20, and 1.79 core-country links. Across all providers, the expected mainland core count increases by 0.158 links, while the offshore count falls by 4.06 links. The net global contraction is 3.90 core-country links: the response is not a one-for-one relocation to China. The three Chinese providers account for about 94 percent of that net global decline. Their worldwide operating profits and net portfolio payoffs also fall, so the greater mainland presence does not imply that these providers benefit from the restriction.

Easing reverses this pattern for the Chinese providers: their mainland-core probabilities decline, but their worldwide networks expand. Across all eight firms, the expected

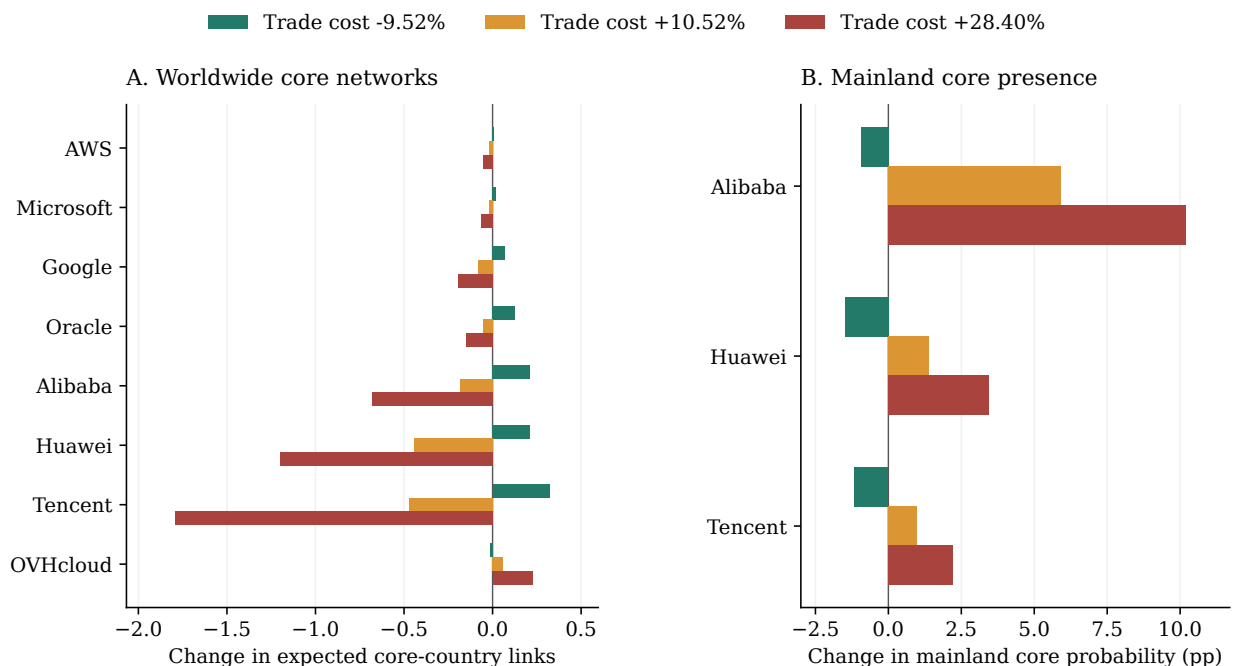


Figure 4: China frictions: mainland core presence and the global portfolio

Notes: Equilibrium responses relative to the common baseline reference. Panel A reports expected worldwide core-country link changes. Panel B reports mainland-core probability changes for the three providers that can select such a core. Foreign direct-core probabilities are fixed at zero under the maintained operating-mode scenario; AWS and Microsoft retain fixed partner paths. All scenarios use the same set of candidate portfolios.

mainland count falls by 0.036 and the offshore count rises by 0.99, giving a worldwide increase of 0.96 core-country links.

Network adjustment transmits the restriction to other markets. The direct friction change accounts for most of the China price effect, while the network response generates its international transmission. With fixed portfolios, easing lowers the China price index by 7.84 percent. Allowing all firms to adjust raises the reduction to 8.05 percent, while the price index outside China falls by 0.11 percent.

The tightening scenarios display the opposite aggregate direction. A 10.52 percent trade-cost increase raises China prices by 7.89 percent at fixed portfolios and 8.17 percent with equilibrium responses. The larger 28.40 percent increase raises them by 19.37 and 20.25 percent, respectively; non-China prices rise by 0.42 percent with network adjustment. As in the EU experiment, the non-China price effect is zero at fixed networks by

construction: the policy changes no direct supply condition or expenditure outside China. The modest difference between fixed-network and equilibrium prices in China therefore coexists with an international price effect that operates entirely through portfolio adjustment.

Competitor feedback moderates the network response. Competitor responses dampen both the global contraction under tightening and the expansion under easing. Holding rivals at their baseline distributions, the combined portfolio responses to easing add 1.43 global core links; with responses to policy-specific competitor portfolios, the increase is 0.96, about 33 percent smaller. For moderate tightening the contraction against rivals at the baseline is 2.49, compared with 1.21 with equilibrium responses. For the larger shock it is 7.36, compared with 3.90, an attenuation of 47 percent. The associated non-China price increase falls from 0.70 to 0.42 percent. As in the EU calculation, a comparison evaluated only against rivals at the baseline would overstate the global network response.

Provider profits. Appendix Figure 6 reports changes in worldwide operating profit and net portfolio payoff. Under the larger tightening, operating profits fall by 7.43, 12.19, and 15.86 percent for Alibaba, Huawei, and Tencent, respectively; their net portfolio payoffs also fall. Easing increases their operating profits by about 2–3 percent. The Chinese providers' greater mainland core presence under tightening therefore does not imply a profit gain. Both the offshore contribution to their China supply and the change in their complete portfolios matter. Effects for some other providers are small or differ across profit measures, so these estimates do not establish a clear ranking of gains across providers. Appendix H.3 reports provider shares and core-count changes in numerical form.

Exposure of other suppliers. With the residual contribution unchanged, its China share rises from about 9.78 to 20.39 percent under the larger tightening, while the Chinese providers' combined total-market share falls by 4.56 percentage points. That comparison assumes the residual remains unexposed to the offshore friction. To assess its importance, let ξ be the fraction of the residual's price-index contribution exposed to the same cost change:

$$b_{\text{CHN}}(d; \xi) = b_{\text{CHN}}(0)[(1 - \xi) + \xi e^{-(\sigma-1)d}]. \quad (6.1)$$

Table 8: China Prices and Provider Incidence Under Alternative Residual Exposure

Trade cost	China price (%)		Chinese provider share (pp)	
	$\xi = 0$	$\xi = 1$	$\xi = 0$	$\xi = 1$
-9.52%	-8.05	-9.04	+1.19	-0.03
+10.52%	+8.17	+9.73	-1.49	+0.09
+28.40%	+20.25	+25.63	-4.56	+0.09

Notes: Effects relative to the baseline at each policy’s central equilibrium portfolio distribution, computed with $\xi = 0$. The $\xi = 1$ columns reprice those same policy portfolios after exposing all residual supply to the offshore friction; portfolios are not reoptimized. Chinese providers are Alibaba, Huawei, and Tencent. Their combined share uses the entire China market, including residual supply, as its denominator. Price effects are percent changes; share effects are percentage points.

The central portfolio calculations use $\xi = 0$. For $\xi = 1$ we re-solve prices holding each regime’s central equilibrium portfolio distribution fixed. The baseline reference is common to both cases because $d = 0$ leaves the residual term unchanged. This is sensitivity to supply origin at given policy portfolios, rather than a new portfolio equilibrium under exposed residual supply.

Table 8 shows that residual exposure preserves the sign of the China price effect in all three scenarios, but reverses the sign of the Chinese providers’ combined share change. For the larger tightening, the price increase is 25.63 rather than 20.25 percent, and their combined share changes by approximately +0.09 rather than -4.56 percentage points. The same distinction appears under easing and moderate tightening. Border frictions act on supply origins, including the origins used by residual suppliers; provider nationality alone does not determine incidence. These repricing comparisons establish neither the robustness of the network response nor the market shares that would result after all firms reoptimize under $\xi = 1$. Table 14 in the appendix reports the corresponding comparison holding baseline networks fixed.

7. CONCLUSION

Geographic restrictions on cloud services change the market reach of multinational infrastructure and the returns to locating in different countries. The return to a location inside a regulated market can rise even as the provider’s global network becomes less valuable. Assessing a policy through domestic entry or the domestic price index alone can therefore

miss its effects on global networks and customers elsewhere.

In the EU application, EU-wide eligibility expands the modeled providers' core presence inside the EU relative to a baseline without geographic eligibility restrictions, whereas national eligibility reduces it. National eligibility also reduces their worldwide core presence by 4.1, about 2.5 percent, relative to EU-wide eligibility. The direct supply restriction accounts for most of the 27.9 percent regulated-price increase; portfolio adjustment generates the price increases outside public procurement. In the China application, higher offshore supply costs increase Chinese providers' probabilities of mainland core presence while contracting their worldwide portfolios and reducing their operating profits. Easing produces the reverse aggregate pattern. In both calculations, competitor responses partly offset the contraction predicted against unchanged rivals under tighter restrictions.

International transmission depends on complete location portfolios and repriced market competition. Our completion bounds support global responses where simpler elimination rules lack their required ordering. The probability-flow argument separately provides sufficient conditions for uniqueness of the distributional portfolio equilibrium. Conditional pricing uniqueness does not replace that argument.

The quantitative results are conditional on baseline geography, residual supply, and operating arrangements. The computations approximate portfolio equilibria on finite candidate sets expanded by global search, with the remaining search gaps reported. Within this framework, domestic presence is not a sufficient measure of geographic restrictions' effects: the same policy can change the scale of multinational networks and the prices faced by customers elsewhere.

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APPENDIX

The appendix contains the proofs, computational procedures, supporting evidence, data conventions, and policy diagnostics.

A. EQUILIBRIUM AND PRICING PROOFS

This appendix proves the main portfolio results and conditional pricing uniqueness. Further sufficient conditions are in Appendix B. Where the appendices factor deterministic costs as $\phi_{fj}w_{fj}$, with $w_{fj} > 0$, that product equals C_{fj}^0 in Section 4.1.

A.1 Proof of Proposition 2

The state space $Q = \prod_f \Delta(\mathcal{X}_f)$ is nonempty, compact, and convex. For each portfolio, $V_f(S_f; Q_{-f})$ is continuous in Q . Two distinct portfolios have payoff difference $V_f(S_f; Q_{-f}) - V_f(T_f; Q_{-f}) - \sigma_\varepsilon \sum_j \ell_{fj} (\mathbf{1}\{j \in S_f\} - \mathbf{1}\{j \in T_f\}) \varepsilon_{fj}$, whose stochastic loading is nonzero, and absolute continuity assigns zero probability to the associated affine indifference hyperplane. If $Q^k \rightarrow Q$, the best-response indicator therefore converges for almost every type. Dominated convergence implies $\mathcal{B}_f(Q_{-f}^k) \rightarrow \mathcal{B}_f(Q_{-f})$. The joint map $\mathcal{B} : Q \rightarrow Q$ is continuous, so Brouwer's theorem supplies a fixed point. The same argument applies when the deterministic payoff tables are any continuous functions of rival distributions; multilinearity is not needed for existence.

A.2 Contingent cutoffs and interaction notation

For Lemma 1, condition on all costs except ε_{fj} . Every include- j portfolio has the same slope $-\sigma_\varepsilon \ell_{fj}$ in that cost; every exclude- j portfolio has slope zero. Maximizing separately within the groups therefore gives $V_{fj}^{\text{in}} - \sigma_\varepsilon \ell_{fj} \varepsilon_{fj}$ and V_{fj}^{out} . Their comparison proves (4.6). Absolute continuity removes ties. Coordinates fixed throughout the feasible family are not decision margins.

For comparison with the pair-cover seminorm, define

$$\|h\|_{\partial f} = \sum_{j \in \mathcal{J}_f} \frac{\bar{g}_{fj}}{\ell_{fj}} \max_{S \ni j, T \not\ni j} |h(S) - h(T)|, \quad (\text{A.1})$$

where $\mathcal{J}_f$ consists of coordinates varying across $\mathcal{X}_f$. The difference of optimized include and ex-

include values obeys

$$|\bar{\varepsilon}_{fj}(v+h) - \bar{\varepsilon}_{fj}(v)| \leq \frac{\max_{S \ni j, T \not\ni j} |h(S) - h(T)|}{\sigma_\varepsilon \ell_{fj}}. \quad (\text{A.2})$$

Indeed, each group's maximum changes between the smallest and largest perturbations in that group. Coupling the old and new choices with the same cost vector, a link changes only between the two cutoffs. Integrating the conditional density bound and taking a union over links gives the include-exclude sensitivity bound $\|h\|_{\partial f}/\sigma_\varepsilon$. For $g \neq f$, let

$$K_{fg} = \max_{A,B,U} \|\Delta r_{A,B,U}^{fg}\|_{\partial f}, \quad K_{ff} = 0. \quad (\text{A.3})$$

The payoff differences $\Delta r_{A,B,U}^{fg}$ and the sharper matrix $K^\sharp$ are defined in Appendix A.3. The argument below proves $K^\sharp \leq K$ and, in particular,

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})) \leq \sigma_\varepsilon^{-1} \sum_{g \neq f} K_{fg} d_{\text{TV}}(Q_g, Q'_g). \quad (\text{A.4})$$

All cost-density weights use the private loading ℓ_{fj} , not the possibly different deterministic loading w_{fj} .

A.3 Constructing the portfolio-response bound

The include-exclude argument gives a valid interaction bound, but it can charge the same portfolio switch to several links. A sharper argument follows probability as it moves between complete-portfolio choices. It changes how we bound sensitivity, while leaving the firm's optimization problem unchanged.

Write $P_f(v)$ for the distribution of the exact optimizer when its deterministic portfolio-value table is v . In the game, $P_f(V_f(\cdot; Q_{-f})) = \mathcal{B}_f(Q_{-f})$: the value table determined by rivals induces the firm's distributional response. Let $\mathcal{J}_f$ contain the links whose inclusion varies across $\mathcal{X}_f$, and put $\gamma_{fj} = \bar{g}_{fj}/\ell_{fj}$ for $j \in \mathcal{J}_f$.

For any nonnegative weight vector γ , define

$$\begin{aligned} C_f(h; \gamma) &= \min_{c \geq 0} \sum_{j \in \mathcal{J}_f} \gamma_j c_j, \\ \text{subject to } \underbrace{|h(S) - h(T)|}_{\text{change in relative payoff}} &\leq \underbrace{\sum_{j \in S \Delta T} c_j}_{\text{cover over changing links}}, \quad S, T \in \mathcal{X}_f. \end{aligned} \quad (\text{A.5})$$

Abbreviate $C_f(h) = C_f(h; \gamma_f)$. Here $S \Delta T$ contains the links on which the portfolios differ. The variables c_j are auxiliary bounds in payoff units, not portfolio choices or realized costs. The constraints cover every change in relative portfolio value, including pairs that differ at many links. Minimizing their density-weighted sum tightens a sensitivity bound; the firm's portfolio optimization is unchanged.

Lemma A.1 (Probability flow across portfolio boundaries). *Under the private-cost and absolute-continuity assumptions of Section 4.1,*

$$d_{\text{TV}}(P_f(v + h), P_f(v)) \leq \frac{C_f(h)}{\sigma_\varepsilon}. \quad (\text{A.6})$$

Moreover, C_f is a seminorm and $C_f(h) \leq \|h\|_{\partial f}$, where $\|h\|_{\partial f}$ is the include-exclude seminorm in (A.1).

Total variation bounds the change in the probability of any collection of portfolios, such as those containing at least one EU location. It need not equal the probability that a type's chosen portfolio changes when the same cost realization is used under both value tables.

To understand the bound, interpolate between the old and new payoff tables. Probability moves across boundaries at which two complete portfolios tie for the highest utility. A boundary's movement depends on the difference between their payoff perturbations, which (A.5) covers using the links that change. For a given link and fixed other costs, however, there is only one include-exclude cutoff. All relevant portfolio boundaries draw on that same density budget, $\bar{g}_{fj}/\ell_{fj}$. The argument therefore accounts for switches involving several links without assuming that those links are chosen separately. Appendix A.4 proves the result, including dependent within-firm costs.

A two-link sensitivity calculation. Return to the four portfolios in Example 1, and suppose $(v_\emptyset, v_a, v_b, v_{ab}) = (0, -2, -2, -1)$. An improvement $h(S) = 1.5 \mathbf{1}\{a \in S\}$ changes the optimal choice from no links to both links for costs near zero. Two inclusion decisions change even though only portfolios containing a receive a higher value. With equal weights $\gamma_a = \gamma_b = \gamma$, the assignment $c_a = 1.5, c_b = 0$ satisfies every constraint in (A.5); pairs differing only in a also require $c_a \geq 1.5$. Hence $C_f(h) = 1.5\gamma$, whereas the include-exclude seminorm gives $\|h\|_{\partial f} = 3\gamma$. The latter permits the same joint switch to be charged at both inclusion margins. The sharper bound halves this sensitivity upper bound, not necessarily the actual change in choice probabilities. Applying it to equilibrium requires bounding the payoff perturbations generated by every relevant rival change.

For rival-g portfolios A, B and remaining rivals' portfolios U , write

$$\Delta r_{A,B,U}^{fg}(S) = r_f(S, A, U) - r_f(S, B, U).$$

Define

$$K_{fg}^\# = \max_{A,B,U} C_f(\Delta r_{A,B,U}^{fg}) \quad (g \neq f), \quad K_{ff}^\# = 0. \quad (\text{A.7})$$

The maximum covers every pure-rival change and every configuration of the remaining rivals. It turns the payoff-sensitivity bound into a bound on strategic responses:

$$\underbrace{d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f}))}_{\text{own response}} \leq \sum_{g \neq f} \underbrace{\frac{K_{fg}^\#}{\sigma_\varepsilon}}_{\text{sensitivity bound}} \underbrace{d_{\text{TV}}(Q_g, Q'_g)}_{\text{rival change}}. \quad (\text{A.8})$$

The left side is the change in the firm's optimal portfolio distribution; the right side weights each rival's distributional change by $K_{fg}^\#/\sigma_\varepsilon$. These coefficients bound response magnitudes, not their signs. Their nonnegativity does not impose strategic complementarity: a rival can raise or lower the relative return to an own portfolio. In particular, $K^\# \leq K$ entrywise and $\rho(K^\#) \leq \rho(K)$. The include-exclude matrix (A.3) remains a valid, potentially more conservative bound.

A.4 Probability flow and rival-distribution sensitivity

We prove Lemma A.1 and (A.8). Fix a firm and suppress its index when no confusion can arise. Remove coordinates constant on the feasible family $\mathcal{X}$, since their realized costs add the same amount to every alternative. The conditional density bounds remain valid after marginalizing these coordinates, by conditional expectation. The remaining incidence vectors $z_S \in \{0, 1\}^m$ are distinct. Write $L = \text{diag}(\ell_j)$ and retain $\sigma_\varepsilon > 0$. A singleton feasible family has constant choice and zero sensitivity; all the statements below then hold with empty sums.

The pair-cover seminorm. For a finite table h , put $\text{osc}(h) = \max_S h(S) - \min_S h(S)$. The pair-cover problem is feasible: $c_j = \text{osc}(h)$ covers every distinct pair. Any feasible vector can be truncated coordinatewise at $\text{osc}(h)$ without violating a constraint. Indeed, if a coordinate in a constraint is truncated, its remaining contribution alone covers that constraint. Hence the minimum in (A.5) is attained, including for nonnegative weights with zero components. Scaling feasible covers proves absolute homogeneity. If c^1 and c^2 cover h_1 and h_2 , their sum covers $h_1 + h_2$ by the triangle inequality. Thus C is a seminorm. With strictly positive weights its kernel consists of constant tables.

Let

$$M_j(h) = \max_{S \ni j, T \not\ni j} |h(S) - h(T)|, \quad N(h) = \sum_j \gamma_j M_j(h) = \|h\|_{\partial f}.$$

For any distinct S, T and every $j \in S \Delta T$, the corresponding $M_j(h)$ is at least $|h(S) - h(T)|$. Therefore

$c_j = M_j(h)$ is a feasible pair cover, proving $C(h) \leq N(h)$.

Proof of the probability-flow lemma. First suppose the joint density p is smooth and satisfies $p(e) \leq \bar{g}_j p_{-j}(e_{-j})$ pointwise for each j . Interpolate $v_t = v + th$, $0 \leq t \leq 1$. For an unordered pair of distinct portfolios S, T , let $\Gamma_{ST}(t)$ denote the portion of their indifference hyperplane on which they both maximize utility. Lower-dimensional intersections are omitted from its surface integral. Set

$$w_{ST}(t) = \int_{\Gamma_{ST}(t)} \frac{p(e)}{\|L(z_S - z_T)\|_2} d\mathcal{H}^{m-1}(e).$$

The indifference equation is $v_t(S) - v_t(T) = \sigma_\varepsilon [L(z_S - z_T)]' e$, so the normal speed of this boundary is $|h(S) - h(T)|/(\sigma_\varepsilon \|L(z_S - z_T)\|_2)$. Probability crossing a shared boundary enters one choice cell and leaves the other. The triangle inequality for these fluxes gives, for almost every t ,

$$\frac{1}{2} \sum_S \left| \frac{d}{dt} P(v_t)(S) \right| \leq \frac{1}{\sigma_\varepsilon} \sum_{\{S,T\}} |h(S) - h(T)| w_{ST}(t). \quad (\text{A.9})$$

The sum on the right counts an unordered boundary once; the factor $1/2$ on the left removes its two appearances in the vector of cell probabilities.

We next bound the total boundary weight assigned to one coordinate. Fix j and e_{-j} . Within the include- j group, every utility has the same slope $-\sigma_\varepsilon \ell_j$ in e_j ; within the exclude- j group, every utility is constant in e_j . The maximizing portfolio inside each group therefore does not depend on e_j . Except on a null set of e_{-j} , the two group maximizers are unique and meet at one finite cutoff $\bar{\varepsilon}_j(t, e_{-j})$. At that cutoff they are also global maximizers. Consequently the projections of the top-choice facets with $j \in S \Delta T$ partition the e_{-j} space up to null sets.

The projection Jacobian of such a facet onto the other coordinates is $\ell_j / \|L(z_S - z_T)\|_2$. It follows that

$$\sum_{\{S,T\}: j \in S \Delta T} w_{ST}(t) = \frac{1}{\ell_j} \int p(\bar{\varepsilon}_j(t, e_{-j}), e_{-j}) de_{-j} \leq \frac{\bar{g}_j}{\ell_j}. \quad (\text{A.10})$$

There is no independence assumption in this step. The last inequality uses the conditional density bound and integrates the marginal density p_{-j} . It is essential to use top-choice facets, rather than all pairwise tie hyperplanes, for the partition statement.

For completeness, degeneracies do not create an additional boundary charge. Three distinct binary vectors cannot be collinear: in a coordinate where two differ, a point on their line can have binary coordinates only at the two endpoints. Since L is invertible, two independent tie equations for three alternatives intersect in codimension at least two. Their projections onto an $(m - 1)$ -dimensional coordinate space have Lebesgue measure zero. Ties within an include or exclude

group similarly lie on nontrivial affine hyperplanes in the other coordinates. Absolute continuity of their marginal distribution makes these exceptional sets null.

If c is any feasible pair cover, combining (A.9) and (A.10) yields

$$\frac{1}{2} \sum_S \left| \frac{d}{dt} P(v_t)(S) \right| \leq \frac{1}{\sigma_\varepsilon} \sum_j \gamma_j c_j.$$

Integrating from zero to one proves the desired inequality for this cover. The transport calculation on an unbounded domain can be justified by first intersecting each cell with a fixed ball. Its exterior boundary does not move with t , while its moving internal facets satisfy the same upper bound in (A.10). Integrate on the ball and let its radius increase. The omitted probability is bounded by the density's tail mass, uniformly in t , and tends to zero. This also justifies passage from the integrated flux inequality to endpoint total variation without a compact-support assumption.

Now let p be an arbitrary absolutely continuous density satisfying the maintained conditional bounds. Convolve it with a nonnegative product smooth approximate identity κ_δ of total mass one, obtaining $p^\delta = p * \kappa_\delta$. The almost-everywhere inequalities $p(e) \leq \bar{g}_j p_{-j}(e_{-j})$ imply

$$p^\delta(e) \leq \bar{g}_j (p_{-j} * \kappa_{\delta, -j})(e_{-j}) = \bar{g}_j p_{-j}^\delta(e_{-j}).$$

Integration of the j th smoothing kernel gives the first equality of marginal convolutions. Smoothness makes the resulting inequalities pointwise, so the preceding argument applies with unchanged density caps. Moreover, $p^\delta \rightarrow p$ in L^1 . Applying any fixed measurable choice map cannot increase total variation; hence the endpoint choice distributions under p^δ converge to those under p . Passing to the limit and minimizing over c proves Lemma A.1.

The proof also gives the local version used in Appendix B: it suffices that the integrated cutoff-density expression in (A.10) be bounded by a common $\tilde{\gamma}_j$ along the entire interpolating segment. The resulting bound is $C(h; \tilde{\gamma})/\sigma_\varepsilon$. A bound at only one endpoint does not supply this conclusion.

A.5 A common equilibrium comparison

The preceding probability-flow bound controls choice probabilities for given payoff tables. The next result converts such a bound into a comparison across rival distributions. It also collects the matrix arguments used in the global, localized, and approximation results.

Proposition A.1 (Equilibrium comparison). *Under the assumptions of Proposition 2, let $\mathcal{U} \subseteq \mathcal{Q}$ be*

nonempty. Suppose a seminorm $\mathfrak{n}_f$ on finite payoff tables satisfies, for every $Q, Q' \in \mathcal{U}$,

$$d_{\text{TV}}(P_f(V_f(\cdot; Q_{-f})), P_f(V_f(\cdot; Q'_{-f}))) \leq \mathfrak{n}_f(V_f(\cdot; Q_{-f}) - V_f(\cdot; Q'_{-f})).$$

Define $M_{ff} = 0$ and $M_{fg} = \max_{A,B,U} \mathfrak{n}_f(\Delta r_{A,B,U}^{fg})$ for $g \neq f$. Then

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})) \leq \sum_g M_{fg} d_{\text{TV}}(Q_g, Q'_g), \quad Q, Q' \in \mathcal{U}. \quad (\text{A.11})$$

For any nonnegative matrix M satisfying this comparison, if $\mathcal{U}$ contains every fixed point and $\rho(M) < 1$, the BNE Q^* is unique. For $\tilde{Q} \in \mathcal{U}$, its residual against the exact response map, $e_f = d_{\text{TV}}(\tilde{Q}_f, \mathcal{B}_f(\tilde{Q}_{-f}))$, gives

$$(d_{\text{TV}}(\tilde{Q}_f, Q_f^*))_f \leq (I - M)^{-1} e. \quad (\text{A.12})$$

Response iteration converges from every point in $\mathcal{U}$ if $\mathcal{B}(\mathcal{U}) \subseteq \mathcal{U}$, and from every point in $\mathcal{Q}$ if $\mathcal{B}^L(\mathcal{Q}) \subseteq \mathcal{U}$ for some finite L . The latter condition also implies containment of every fixed point.

Proof. Telescope the difference of the two product rival distributions one marginal at a time. For the term changing rival g , put $\delta_g = d_{\text{TV}}(Q_g, Q'_g)$. If positive, the signed parts of $Q'_g - Q_g$ are $\delta_g \mu_g^+$ and $\delta_g \mu_g^-$, with both normalized measures of mass one. Couple them by $\pi_g(A, B)$ and let $\omega_g(U)$ be the hybrid product distribution of the other rivals. The corresponding payoff-table change is

$$h_g(S) = \delta_g \sum_{A,B,U} \pi_g(A, B) \omega_g(U) \Delta r_{A,B,U}^{fg}(S).$$

Homogeneity and subadditivity give $\mathfrak{n}_f(h_g) \leq \delta_g M_{fg}$; a term with $\delta_g = 0$ vanishes. Summing these bounds and applying the assumed response inequality to the actual endpoint tables proves (A.11). The hybrid and pure-switch tables enter only this algebraic seminorm bound; no local density assumption is applied to them. Each signed part has mass δ_g , not $2\delta_g$.

For $M \geq 0$ and $\rho(M) < 1$, $M^k \rightarrow 0$ and $(I - M)^{-1} = \sum_{k \geq 0} M^k \geq 0$. The distance vector between two fixed points in $\mathcal{U}$ satisfies $d \leq Md \leq M^k d$, hence is zero. Containment and Proposition 2 give global uniqueness. Inserting $\mathcal{B}(\tilde{Q})$ between $\tilde{Q}$ and Q^* gives $d \leq e + Md$; iteration yields $d \leq \sum_{k=0}^{N-1} M^k e + M^N d$. Letting $N \rightarrow \infty$ proves (A.12).

For iterates in $\mathcal{U}$, their distances from Q^* obey $d^{t+1} \leq Md^t$ and converge to zero. Invariance keeps iterates in that set. Alternatively, $\mathcal{B}^L(\mathcal{Q}) \subseteq \mathcal{U}$ puts every fixed point there and gives $\mathcal{B}^t(Q) = \mathcal{B}^L(\mathcal{B}^{t-L}(Q)) \in \mathcal{U}$ for $t \geq L$. The same distance argument applies after round L . No closedness assumption on $\mathcal{U}$ is needed because the fixed point already exists and lies in it. $\square$

Global uniqueness and firm reweighting. Use $\mathfrak{n}_f = C_f/\sigma_\varepsilon$ and $\mathcal{U} = \mathcal{Q}$ in Proposition A.1. This gives $M = K^\#/\sigma_\varepsilon$ and proves (A.8). Since $C_f(h) \leq \|h\|_{\partial f}$, it also proves $K^\# \leq K$, $\rho(K^\#) \leq \rho(K)$, and (A.4). When $\rho(M) < 1$, set $a = (I - M)^{-1}\mathbf{1} \gg 0$. Then $Ma = a - \mathbf{1}$, so the response map is a contraction in $\max_f d_{\text{TV}}(Q_f, Q'_f)/a_f$, with modulus at most $\max_f(1 - 1/a_f) < 1$. This proves Theorem 1, including for reducible matrices. A componentwise upper bound on $K^\#$ gives the same conclusion whenever its spectral condition holds.

The residual e in (A.12) uses the exact response over the entire feasible family. A residual computed on a finite candidate set or sampled cost types requires an additional error bound before it can be substituted for e .

Proposition A.2 (Perron minimality within firm reweightings). *Let M be any fixed nonnegative response matrix for Theorem 1, such as $K^\#$ or K . For any positive firm weights u , the response comparison inequality shows that simultaneous best response is Lipschitz in the weighted metric $\max_f d_{\text{TV}}(Q_f, Q'_f)/u_f$ with constant $(1/\sigma_\varepsilon) \max_f (Mu)_f/u_f$, hence a contraction whenever this constant is below one. The smallest comparison constant is determined by*

$$\inf_{u \gg 0} \max_f \frac{(Mu)_f}{u_f} = \rho(M),$$

with the infimum attained by the right Perron vector when M is irreducible. Moreover, rescaling the firm-level metrics by positive constants c_f replaces M by DMD^{-1} . Here D is diagonal with entries c_f , so the rescaling leaves $\rho(M)$ unchanged.

Proof. The first claim applies the componentwise response comparison directly with positive weights u . The variational identity is the Collatz–Wielandt characterization of the spectral radius of a nonnegative matrix (Horn and Johnson, 2013, Chapter 8), and irreducibility gives attainment at the right Perron vector. Multiplying each firm’s distance by c_f transforms the comparison matrix into DMD^{-1} . Similar matrices share eigenvalues, proving the rescaling claim. $\square$

A.6 Conditional AB pricing and derivatives

Fix $\sigma > \eta > 1$, an active set, and a market, suppressing its index. Let $y_f = \psi_f p_f^{1-\sigma}$ and $D_f = b + \sum_{g \neq f} y_g$, where $b \geq 0$ is fixed residual price-index contribution. For a given $D_f > 0$, define $\xi_f = \psi_f c_f^{1-\sigma} > 0$. A scalar pricing response solves

$$G_f(y; D_f) = \log y - \log \xi_f - (1 - \sigma) \log \mu \left(\frac{y}{y + D_f} \right) = 0.$$

Writing $s_f = y/(y + D_f)$ and $\beta_f = (\sigma - 1)(\sigma - \eta)\mu(s_f)/(\sigma\eta)$, its derivative with respect to $\log y$ is $1 + s_f(1 - s_f)\beta_f > 0$. Bounded markups make G_f tend to $-\infty$ as $y \downarrow 0$ and to $+\infty$ as $y \uparrow \infty$. Thus there is a unique smooth response $y_f = \mathcal{Y}_f(D_f)$.

Differentiating $\log y_f = \log \xi_f + (1 - \sigma) \log \mu(s_f)$ gives $d \log y_f = -\beta_f ds_f$. Combining this with $ds_f = s_f(1 - s_f)(d \log y_f - d \log D_f)$ yields

$$\tilde{\kappa}_f = \frac{\partial \log y_f}{\partial \log D_f} = \frac{s_f(1 - s_f)\beta_f}{1 + s_f(1 - s_f)\beta_f}. \quad (\text{A.13})$$

Here s_f is evaluated at the solved scalar response $\mathcal{Y}_f(D_f)$. Define $x_f = \log y_f$ and $\Phi_f(x) = \log \mathcal{Y}_f(b + \sum_{g \neq f} e^{x_g})$. Its Jacobian has

$$\Lambda_{fg} = \tilde{\kappa}_f \frac{y_g}{D_f} \quad (g \neq f), \quad \Lambda_{ff} = 0. \quad (\text{A.14})$$

In this Jacobian, $y_g = e^{x_g}$ for rivals, while $\tilde{\kappa}_f$ uses the solved scalar response, not the share implied by the input x_f . Cross-market entries are zero. Since $\sum_{g \neq f} y_g/D_f \leq 1$, $\beta_f \leq \bar{\beta} = (\sigma - 1)(\sigma - \eta)/[\sigma(\eta - 1)]$, and $s_f(1 - s_f) \leq 1/4$,

$$\|\Lambda(x)\|_\infty \leq \bar{\lambda}_{AB} = \frac{\bar{\beta}/4}{1 + \bar{\beta}/4} < 1. \quad (\text{A.15})$$

The map is defined on the Euclidean space of log contributions x ; the mean-value inequality and Banach's theorem give a unique fixed point. If $b = 0$ and only one firm is active, $s_f = 1$ and $p_f = \mu(1)c_f$ directly. With $b > 0$ and one named firm, its scalar response faces the constant $D_f = b$. An empty active set has no named pricing variables; if residual supply is also absent, no aggregate price index is asserted. These cases prove existence and uniqueness for every portfolio profile in Proposition 1.

On a fixed active set, $I - \Lambda$ is invertible and the implicit-function theorem gives smooth dependence on positive costs and accessibility. Moreover $\|(I - \Lambda)^{-1}\|_\infty \leq (1 - \bar{\lambda}_{AB})^{-1}$. At $(\sigma, \eta) = (5, 2)$ the uniform modulus is 0.375. This bounds conditional pricing feedback. Uniqueness of the portfolio equilibrium requires a separate condition.

Lemma A.2 (Equilibrium share and profit derivatives). *On a nonempty fixed active set I , write $F_0 = b \geq 0$ for residual price power and set $a_0 = 1 - 1/\sigma$, $b_0 = 1/\eta - 1/\sigma$. For log pre-markup scores $t_i = \log \xi_i$, the equilibrium system is*

$$Y s_i = \xi_i(a_0 - b_0 s_i)^{\sigma-1}, \quad s_0 = F_0/Y, \quad s_0 + \sum_{i \in I} s_i = 1. \quad (\text{A.16})$$

Define

$$r(s) = \frac{s(a_0 - b_0 s)}{a_0 + (\sigma - 2)b_0 s}, \quad W = s_0 + \sum_i r(s_i), \quad \omega_i = r(s_i)/W.$$

The exact share derivatives are

$$J_{ik} = r(s_i)(\mathbf{1}\{i = k\} - \omega_k), \quad J_{0k} = -s_0 \omega_k, \quad J_{0k} + \sum_i J_{ik} = 0. \quad (\text{A.17})$$

With $\kappa = (a_0 - b_0)/[a_0 + (\sigma - 2)b_0] > 0$,

$$\kappa s_i(1 - s_i) \leq J_{ii} \leq s_i(1 - s_i). \quad (\text{A.18})$$

Hold rival scores and F_0 fixed. For $R \geq 0$ and $G(t_i) = R[s_i/\sigma + b_0 s_i^2]$, if $[s^L, s^U]$ encloses the own share throughout a fixed-active-set score interval, then

$$G'(t_i) \geq \kappa R \min\{u(s^L), u(s^U)\}, \quad u(s) = (1/\sigma + 2b_0 s)s(1 - s). \quad (\text{A.19})$$

Proof. Logarithmic differentiation of (A.16) gives $ds_i = r(s_i)(dt_i - d \log Y)$ and $ds_0 = -s_0 d \log Y$. Adding these equations yields $d \log Y = \sum_i \omega_i dt_i$, proving (A.17). The denominator W is positive on a nonempty active set, including a monopoly without residual supply.

The ratio $r(s)/s = (a_0 - b_0 s)/[a_0 + (\sigma - 2)b_0 s]$, continuously extended at zero, has derivative $-(\sigma - 1)a_0 b_0/[a_0 + (\sigma - 2)b_0 s]^2 < 0$. Its denominator remains positive when $\sigma < 2$, because $a_0 + (\sigma - 2)b_0 = (a_0 - b_0) + (\sigma - 1)b_0 > 0$. Thus $\kappa s \leq r(s) \leq s$. Put $B_i = s_0 + \sum_{g \neq i} r(s_g)$. Then $\kappa(1 - s_i) \leq B_i \leq 1 - s_i$ and

$$J_{ii} = \frac{r(s_i)B_i}{r(s_i) + B_i}. \quad (\text{A.20})$$

Monotonicity of $xy/(x + y)$ in nonnegative x, y gives (A.18); both bounds are zero for a monopoly without residual supply. Finally, $G' = R(1/\sigma + 2b_0 s_i)J_{ii} \geq \kappa R u(s_i)$. The derivative $u'(s) = 1/\sigma + (4b_0 - 2/\sigma)s - 6b_0 s^2$ has one negative and one positive root, is positive at zero and negative at one. Hence u increases then decreases on $[0, 1]$; its minimum on an interval is attained at an endpoint. This proves (A.19), including the zero bounds when $R = 0$ or an endpoint is zero or one. $\square$

Residual price-index contribution is fixed in this calculation, but its share varies. The no-residual formulas follow by setting $s_0 = 0$. Derivatives on a fixed active set do not justify integrating through an entry or exit boundary; those comparisons use fully repriced corner payoffs.

B. FURTHER PORTFOLIO-UNIQUENESS CONDITIONS

The uniform matrix in Theorem 1 covers all rival distributions. The following refinements restrict that domain or retain signed responses. They apply to arbitrary finite feasible portfolio families; none follows from convergence at a finite collection of starting values.

B.1 Containment and localized contraction

Let $\mathcal{Q} = \prod_f \Delta(\mathcal{X}_f)$. At the specified positive noise level, a nonnegative matrix $C^{\mathcal{U}}$ with zero diagonal is a response bound on $\mathcal{U} \subseteq \mathcal{Q}$ if

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})) \leq \sum_{g \neq f} C_{fg}^{\mathcal{U}} d_{\text{TV}}(Q_g, Q'_g), \quad Q, Q' \in \mathcal{U}. \quad (\text{B.1})$$

For the full domain, $C^{\mathcal{Q}} = K^{\#}/\sigma_{\varepsilon}$ is valid. A smaller region may allow smaller weights on the choice boundaries that can be reached there. It must still describe distributions over whole portfolios, not independent link probabilities.

Proposition B.1 (Localized exact-portfolio uniqueness). *Under the assumptions of Proposition 2, suppose $\mathcal{U}$ contains every fixed point of $\mathcal{B}$ and admits the bound (B.1). If*

$$\rho(C^{\mathcal{U}}) < 1, \quad (\text{B.2})$$

the portfolio BNE is unique in $\mathcal{Q}$.

Corollary B.1 (Localized best-response convergence). *Under the proposition's assumptions, iteration converges to the unique BNE from every point in $\mathcal{U}$ if $\mathcal{B}(\mathcal{U}) \subseteq \mathcal{U}$. Alternatively, if for some finite L ,*

$$\mathcal{B}^L(\mathcal{Q}) \subseteq \mathcal{U}, \quad (\text{B.3})$$

containment follows and iteration converges from every point in $\mathcal{Q}$, without a separate invariance assumption.

Proof. Apply Proposition A.1 with $M = C^{\mathcal{U}}$. Two fixed points in $\mathcal{U}$ have distance vector $d \leq Md$, hence $d = 0$. Invariance keeps iterates in the comparison region. Under finite-step trapping, $\mathcal{B}^t(Q) = \mathcal{B}^L(\mathcal{B}^{t-L}(Q)) \in \mathcal{U}$ for every $t \geq L$, and the same distance contraction applies thereafter. $\square$

Containment and contraction must be verified jointly at the chosen noise level. A Jacobian at one computed equilibrium supplies neither requirement. On a nonconvex region, even pointwise derivative bounds require control of connecting paths to imply the pairwise inequality above. Failure of a uniform spectral test is not evidence of multiple equilibria.

B.2 Gaussian containment at a fixed positive noise level

Suppose link shocks are independent standard normal within each firm. Choose any feasible reference profile $\widehat{S}$ and, for $S \neq \widehat{S}_f$, define

$$\begin{aligned} D_{fS}(S_{-f}) &= r_f(\widehat{S}_f, S_{-f}) - r_f(S, S_{-f}), & \Delta_{fS} &= D_{fS}(\widehat{S}_{-f}), \\ A_{fGS} &= \max_{T_g, U} [D_{fS}(\widehat{S}_g, U) - D_{fS}(T_g, U)]_+. \end{aligned} \quad (\text{B.4})$$

Here U ranges over the other firms' pure portfolios. The reduced payoffs include repricing at every profile. If $e_g(Q) = 1 - Q_g(\widehat{S}_g)$, telescoping rivals away from the reference and taking expectations gives

$$V_f(\widehat{S}_f; Q_{-f}) - V_f(S; Q_{-f}) \geq \delta_{fS}(e(Q)), \quad \delta_{fS}(e) = \Delta_{fS} - \sum_{g \neq f} A_{fGS} e_g. \quad (\text{B.5})$$

Let $\overline{\Phi} = 1 - \Phi$ and φ be the standard normal tail and density. Define $v_{fS}^2 = \sum_j \ell_{fj}^2 (\mathbf{1}\{j \in S\} - \mathbf{1}\{j \in \widehat{S}_f\})^2 > 0$ and

$$T_f(e) = \min \left\{ 1, \sum_{S \neq \widehat{S}_f} \overline{\Phi} \left(\frac{\delta_{fS}(e)}{\sigma_\varepsilon v_{fS}} \right) \right\}. \quad (\text{B.6})$$

An alternative must overcome the reference's deterministic advantage with its Gaussian cost advantage. A union bound therefore gives $e(\mathcal{B}(Q)) \leq T(e(Q))$; it requires no independence across portfolio costs. The operator is increasing. Starting at $e^0 = \mathbf{1}$, set $e^{t+1} = T(e^t)$ and

$$\mathcal{U}_t = \{Q : Q_f(\widehat{S}_f) \geq 1 - e_f^t \forall f\}. \quad (\text{B.7})$$

Then $e^{t+1} \leq e^t$ and $\mathcal{B}^t(Q) \subseteq \mathcal{U}_t$, $\mathcal{B}(\mathcal{U}_t) \subseteq \mathcal{U}_{t+1} \subseteq \mathcal{U}_t$. Every equilibrium thus belongs to every $\mathcal{U}_t$. The construction bounds probability mass without removing feasible portfolios.

For each firm with more than one feasible action, suppose a finite L gives $m_f = \min_{S \neq \widehat{S}_f} \delta_{fS}(e^L) > 0$. For each varying coordinate $j \in \mathcal{J}_f$, choose positive $t_{fk}^{(j)}$, $k \in \mathcal{J}_f \setminus \{j\}$,

such that $\sigma_\varepsilon \sum_{k \neq j} \ell_{fk} t_{fk}^{(j)} \leq m_f/2$. Set

$$g_{fj}^{\text{loc}} = \min \left\{ \varphi(0), \varphi\left(\frac{m_f}{2\sigma_\varepsilon \ell_{fj}}\right) + \varphi(0) \sum_{k \neq j} 2\bar{\Phi}(t_{fk}^{(j)}) \right\}. \quad (\text{B.8})$$

Use weights $g_{fj}^{\text{loc}}/\ell_{fj}$ in the pair-cover problem (A.5) to define C_f^{loc} , and put

$$C_{fg}^{\text{loc}} = \sigma_\varepsilon^{-1} \max_{A,B,U} C_f^{\text{loc}}(\Delta r_{A,B,U}^{fg}), \quad g \neq f, \quad C_{ff}^{\text{loc}} = 0. \quad (\text{B.9})$$

The pure-rival switches are the same global comparisons used in Appendix A.2.

Proposition B.2 (Gaussian containment and finite-noise uniqueness). *Under the assumptions of Proposition 2 and independent standard normal link shocks within each firm, suppose some finite L gives $m_f > 0$ for every firm with more than one feasible portfolio and $\rho(C^{\text{loc}}) < 1$. The portfolio BNE is unique, and simultaneous response iteration converges to it from every initial distribution. A firm with only one feasible portfolio has $T_f = 0$ and a zero matrix row, and requires no gap condition.*

Proof. For a pure rival profile, the definition of A_{fgs} implies $D_{fs}(S_{-f}) \geq \Delta_{fs} - \sum_{g \neq f} A_{fgs} \mathbf{1}\{S_g \neq \widehat{S}_g\}$. This proves (B.5). If S is chosen, its private cost advantage over $\widehat{S}_f$ exceeds that gap. The advantage is centered normal with standard deviation $\sigma_\varepsilon v_{fs}$, yielding (B.6), including when the gap is negative. Monotonicity of T , $T(\mathbf{1}) \leq \mathbf{1}$, and induction give the nested sets and finite-step containment just stated.

It remains to justify the smaller density weights. Consider any value table v satisfying

$$v(\widehat{S}_f) - v(S) \geq m_f \quad (S \neq \widehat{S}_f). \quad (\text{B.10})$$

Conditional on shocks other than j , let M_j^{in} and M_j^{out} be the optimized include- j and exclude- j values after removing the j -shock, and let $c_j = (M_j^{\text{in}} - M_j^{\text{out}})/(\sigma_\varepsilon \ell_{fj})$. On $H_j = \bigcap_{k \neq j} \{|\varepsilon_{fk}| \leq t_{fk}^{(j)}\}$, the other shocks change any reference-alternative comparison by at most $m_f/2$. The reference belongs to one of the two groups and exceeds every portfolio in the opposite group by at least $m_f/2$. Hence $|c_j| \geq m_f/(2\sigma_\varepsilon \ell_{fj})$ on H_j . Independence and a union bound give

$$\mathbb{E}_{\varepsilon_{f,-j}}[\varphi(c_j)] \leq \min\{\varphi(0), \varphi(m_f/(2\sigma_\varepsilon \ell_{fj})) + \varphi(0) \Pr(H_j^c)\} \leq g_{fj}^{\text{loc}}. \quad (\text{B.11})$$

This bounds an integrated cutoff density, not the unconditional Gaussian density. The empty-coordinate event has probability one.

If v, v' both satisfy (B.10), so does their straight interpolation. The coordinate boundary budget

in (A.10) is therefore at most $g_{fj}^{\text{loc}}/\ell_{fj}$ throughout that segment. Integrating the probability-flow bound and minimizing over pair-cover weights gives

$$d_{\text{TV}}(P_f(v), P_f(v')) \leq \sigma_\varepsilon^{-1} C_f^{\text{loc}}(v' - v). \quad (\text{B.12})$$

The actual endpoint tables generated by $Q, Q' \in \mathcal{U}_L$ have the required gap by (B.5). Apply Proposition A.1 with $\eta_f = C_f^{\text{loc}}/\sigma_\varepsilon$ to obtain

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})) \leq \sum_{g \neq f} C_{fg}^{\text{loc}} d_{\text{TV}}(Q_g, Q'_g). \quad (\text{B.13})$$

The pure-switch tables enter only the algebraic seminorm comparison; they need not share the reference's strict gap. Finite-step containment and Corollary B.1 complete the proof. $\square$

The reference is unrestricted at the outset, but success implies $\Delta_{fS} \geq m_f > 0$: it must be a strict pure equilibrium of the deterministic game. No uniqueness assumption for that game's mixed equilibria is imposed. Failure of either gap or spectral condition proves no multiplicity. A verified increasing upper operator $\tilde{T} \geq T$ can replace T ; pointwise numerical upper values without monotonicity do not suffice. Every feasible portfolio still has positive probability at positive Gaussian noise: assigning shocks $-M/\ell_{fj}$ on a target and $+M/\ell_{fj}$ off it gives that target a cost advantage $\sigma_\varepsilon M |S \Delta S^0|$ over any alternative. For large finite M , it is strictly optimal on an open set of shocks with positive probability.

B.3 Signed responses and uniqueness

Contraction also ensures iterative convergence. For uniqueness alone, one can retain signed responses in the residual and simplex tangent space

$$\mathcal{F}(Q) = Q - \mathcal{B}(Q), \quad (\text{B.14})$$

$$\mathcal{T} = \{z : \mathbf{1}' z_f = 0 \ \forall f\}. \quad (\text{B.15})$$

Proposition B.3 (One-sided exact-portfolio uniqueness). *Under the assumptions of Proposition 2, let $\mathcal{U}$ contain every fixed point. If a symmetric W , positive definite on $\mathcal{T}$, satisfies*

$$(Q - Q')' W [\mathcal{F}(Q) - \mathcal{F}(Q')] > 0 \quad \text{for all distinct } Q, Q' \in \mathcal{U}, \quad (\text{B.16})$$

the portfolio BNE is unique in $\mathcal{Q}$.

Proof. Two distinct fixed points in $\mathcal{U}$ would make the left side zero. Existence follows from Proposition 2. $\square$

If $\mathcal{U}$ is convex and $\mathcal{B}$ is continuously differentiable on a relative open neighborhood of it in $\text{aff}(\mathcal{Q})$, a sufficient condition is

$$z' \frac{W[I - D\mathcal{B}(Q)] + [I - D\mathcal{B}(Q)]'W}{2} z \geq \mu z'Wz, \quad Q \in \mathcal{U}, z \in \mathcal{T}, \quad \mu > 0. \quad (\text{B.17})$$

Indeed, for $z = Q - Q'$, integration along $Q' + tz$ gives $z'W[\mathcal{F}(Q) - \mathcal{F}(Q')] \geq \mu z'Wz > 0$. Large negative responses can aid this inequality while violating an absolute-response contraction. The conclusion is uniqueness, not convergence of simultaneous iteration. Nonsingularity at one computed fixed point yields at most local isolation under the usual differentiability conditions; the uniform inequality and containment are essential for global uniqueness.

C. VALID PORTFOLIO-COMPLETION BOUNDS

The bounds refer to complete portfolios and fully repriced continuation values. Feasibility restrictions may reduce the completion family, and the number of links may vary. The proof does not impose positive realized costs.

C.1 Validity

Under branch and bound, fix a partial-portfolio subproblem (S_f, U_f) and a feasible completion $T_f = S_f \cup H$, where $H \subseteq U_f$ and $|H| = k$. For every rival state ς and statistic m ,

$$\begin{aligned} z_{\varsigma m}(T_f) &= z_{\varsigma m}(S_f) + \sum_{j \in H} a_{\varsigma m j} \\ &\leq z_{\varsigma m}(S_f) + \sum_{h=1}^k a_{\varsigma m, (h)}^{\downarrow}(U_f) = \bar{z}_{\varsigma m}(S_f, U_f, k). \end{aligned}$$

The inequality holds component by component. Coordinatewise monotonicity of the continuation envelope gives

$$\pi_{f\varsigma}(T_f) \leq \bar{\pi}_{f\varsigma}(z_{\varsigma}(T_f)) \leq \bar{\pi}_{f\varsigma}(\bar{z}_{\varsigma}(S_f, U_f, k)).$$

Since any k -element subset of U_f costs at least the sum of the k smallest link costs, the total cost of T_f satisfies $\sum_{j \in T_f} C_{fj} \geq \underline{C}_f(S_f, U_f, k)$. Multiplying the state inequalities by $q_{\varsigma} \geq 0$, summing over ς ,

and subtracting the cost lower bound proves that the expression inside braces in (4.10) dominates the payoff of T_f . Maximizing over k proves Proposition 3.

If the attainable completion sizes for a subproblem are known, the maximization over k can be restricted to them. Negative link costs also require no separate treatment: the order statistic $\sum_{h=1}^k C_{f,(h)}^\uparrow$ is a lower bound for every k -link completion whether costs are positive or negative. A preprocessing rule that instead fixes every negative-cost link into the portfolio would need additional closure and monotonicity conditions; the implementation does not use that rule.

C.2 Pricing specializations

Corollary C.1 (Constant-markup CES). *Suppose $\sigma > 1$, $\theta > 0$, and the portfolio enters each market through nonnegative additive service and accessibility statistics. Suppose accessibility has a nonnegative loading. Holding a rival-portfolio state fixed, the closed-form CES variable profit satisfies the monotonicity conditions of Proposition 3.*

Proof. Write the service statistic as $H_{nf}(T_f) = H_{nf}^0 + \sum_{j \in T_f} h_{fjn}$ with $h_{fjn} \geq 0$. The Fréchet service cost is proportional to $H_{nf}^{-1/\theta}$, so own competitiveness contains the increasing term $H_{nf}^{(\sigma-1)/\theta}$. The accessibility log-sum is increasing in its nonnegative additive weights, and a nonnegative loading preserves that order. With rivals' price-index contributions fixed, the constant-markup CES share is increasing in own competitiveness, and variable profit is $R_n s_{nf}/\sigma$. Applying the componentwise top- k envelopes therefore gives a valid continuation bound. $\square$

Corollary C.2 (Atkeson–Burstein pricing). *Suppose $\sigma > \eta > 1$ and the primitive service and accessibility conditions of Corollary C.1 hold. If $\bar{s}_{nfs}(z_\varsigma)$ is a valid coordinatewise increasing upper bound on the AB equilibrium share in rival state ς , then*

$$\bar{\pi}_{f\varsigma}(z_\varsigma) = \sum_n R_n \left[\frac{\bar{s}_{nfs}(z_\varsigma)}{\sigma} + \left(\frac{1}{\eta} - \frac{1}{\sigma} \right) \bar{s}_{nfs}(z_\varsigma)^2 \right]$$

satisfies Proposition 3.

Proof. Under AB pricing, variable profit in market n is $R_n g(s_{nf})$ with g the bracketed function in the corollary display; because $\sigma > \eta > 1$, $g'(s) = 1/\sigma + 2(1/\eta - 1/\sigma)s > 0$ on $[0, 1]$. Substituting any valid share upper bound therefore gives a variable-profit upper bound. A core-only implementation obtains $\bar{s}_{nfs}$ from the markup interval $\sigma/(\sigma - 1) \leq \mu \leq \eta/(\eta - 1)$ and the AB equilibrium-share envelope documented in Appendix C.6. $\square$

These corollaries distinguish the mathematical completion upper bound from the payoff evaluation used at complete portfolios. CES prices and shares have a closed-form continuation in the implementation. Complete AB portfolios are evaluated by numerical fixed point iteration. Once every discrete subproblem has been evaluated or ruled out, the AB calculation therefore proves global optimality of the portfolio response conditional on that numerical continuation.

C.3 Relation to the maximal-portfolio envelope

For comparison, construct a pure maximal-portfolio bound using the same continuation envelopes, and write $a_{sj} = (a_{s1j}, \dots, a_{sLj})$:

$$B_f^{\max}(S_f, U_f) = \sum_{\varsigma} q_{\varsigma} \bar{\pi}_{f\varsigma} \left(z_{\varsigma}(S_f) + \sum_{j \in U_f} a_{sj} \right) - \sum_{j \in S_f} C_{fj} - \sum_{j \in U_f} \min\{C_{fj}, 0\}. \quad (\text{C.1})$$

Corollary C.3 (Envelope dominance). *When (4.10) and (C.1) use the same componentwise continuation envelopes, $B_f^{\text{card}}(S_f, U_f) \leq B_f^{\max}(S_f, U_f)$.*

Proof. For every k , each top- k statistic is no larger than the corresponding sum over all undecided links. Moreover $\sum_{h=1}^k C_{f,(h)}^{\uparrow}(U_f) \geq \sum_{j \in U_f} \min\{C_{fj}, 0\}$. Monotonicity of the continuation envelope gives the result for each k , and maximizing over k preserves it. $\square$

The dominance statement compares bounds built from the same component envelopes. Other valid caps need not be dominated by the cardinality bound. Writing their minimum as B_f^{other} , distinguish

$$B_f^{\text{base}} = \min\{B_f^{\max}, B_f^{\text{other}}\}, \quad B_f^{\text{used}} = \min\{B_f^{\text{base}}, B_f^{\text{card}}\}.$$

Without an additional cap, the latter reduces to B_f^{card} . The service-path and joint-affine bounds used in the policy searches are specified in Appendix C.5.

C.4 A common affine ranking of benefits and costs

Tightening the bound with a common set of links. Cardinality conditioning still allows a completion to receive benefits that no single portfolio can deliver together. A further refinement charges for this incompatibility. For example, a facility that serves Korea especially well may leave some of Japan's optimistic benefit unattained. If repriced profit rises by at least a known amount with competitiveness, that shortfall implies a minimum loss relative to the optimistic profit bound. We construct such a loss for each market, add the losses across markets and rival states, and choose

the same k links to trade off these benefits against their actual costs. The resulting *joint-affine bound* requires only a ranking of link scores once its coefficients are available.

Fix a firm and a subproblem, write $c_j = C_{fj}$, suppress f , and write its selected and undecided sets as (S, U) . For a given number k of additions, let $\text{Top}_k(v)$ and $\text{Bot}_k(v)$ denote the sums of the k largest and smallest entries of a vector indexed by U . Both sums are zero at $k = 0$. Index market-rival-state pairs by $a = (n, s)$, with weight $\omega_a = q_s \geq 0$. Let $\xi_a(H) > 0$ be own competitiveness before markups for completion $S \cup H$, and let $G_a(t)$ be its market operating profit after equilibrium repricing when log competitiveness is t , holding the rival state fixed. Thus the completion's objective is $\sum_a \omega_a G_a(\log \xi_a(H)) - C(S) - \sum_{j \in H} c_j$, where $C(S) = \sum_{j \in S} c_j$.

Proposition C.1 (Joint-affine completion bound). *For each channel a , suppose a positive endpoint $\bar{\xi}_a$ and affine coefficients satisfy, for every $H \subseteq U$ with $|H| = k$,*

$$\log \frac{\xi_a(H)}{\bar{\xi}_a} \leq d_a + \sum_{j \in H} A_{aj} \leq 0. \quad (\text{C.2})$$

Suppose $\bar{G}_a \geq G_a(\log \bar{\xi}_a)$ and G_a is continuously differentiable with $G'_a(t) \geq \mu_a \geq 0$ on every interval from $\log \xi_a(H)$ to $\log \bar{\xi}_a$. Define

$$\bar{G} = \sum_a \omega_a \bar{G}_a, \quad D = \sum_a \omega_a \mu_a d_a, \quad \beta_j = \sum_a \omega_a \mu_a A_{aj}.$$

Then an upper bound for all completions with k additions is

$$B^{\text{joint}}(S, U, k) = \bar{G} - C(S) + D + \text{Top}_k((\beta_j - c_j)_{j \in U}). \quad (\text{C.3})$$

Using the same endpoint profit bounds, it satisfies

$$B^{\text{joint}}(S, U, k) \leq B^{\text{card}}(S, U, k) := \bar{G} - C(S) - \text{Bot}_k((c_j)_{j \in U}). \quad (\text{C.4})$$

The result remains valid when the feasible family excludes some of these completions. For unrestricted size, maximize over admissible k ; for a fixed total size K , use $k = K - |S|$.

Proof of Proposition C.1. Let $\mathcal{H}_k = \{H \subseteq U : |H| = k\}$ be the relaxed completion family. The log-envelope condition holds on all of $\mathcal{H}_k$, including any completions excluded by additional constraints. For an active channel, set $t_a(H) = \log \xi_a(H)$ and $\bar{t}_a = \log \bar{\xi}_a$. The log-envelope condition gives $t_a(H) \leq \bar{t}_a$. Since $G'_a \geq \mu_a$ throughout the interval, the mean value theorem makes $G_a(t) - \mu_a t$

nondecreasing there. Hence

$$\begin{aligned} G_a(t_a(H)) &\leq G_a(\bar{t}_a) + \mu_a[t_a(H) - \bar{t}_a] \\ &\leq \bar{G}_a + \mu_a[t_a(H) - \bar{t}_a] \leq \bar{G}_a + \mu_a \left(d_a + \sum_{j \in H} A_{aj} \right). \end{aligned} \quad (\text{C.5})$$

The remaining inequalities use $\bar{G}_a \geq G_a(\bar{t}_a)$ and the log envelope together with $\mu_a \geq 0$. Multiply by $\omega_a \geq 0$, sum over channels, and subtract the actual costs of the same completion. With the definitions of $\bar{G}$, D , β_j , and $C(S)$ in the proposition, its payoff is at most

$$\bar{G} - C(S) + D + \sum_{j \in H} (\beta_j - c_j).$$

Maximizing this expression over $\mathcal{H}_k$ gives (C.3). If the true feasible family is smaller, the maximization remains an upper bound. No sign restriction on c_j is used.

For every $H \in \mathcal{H}_k$, the channel log-envelope conditions also imply $D + \sum_{j \in H} \beta_j \leq 0$. Moreover, $\sum_{j \in H} c_j \geq \text{Bot}_k(c)$, even for signed costs. Thus the displayed affine payoff is at most $\bar{G} - C(S) - \text{Bot}_k(c) = B^{\text{card}}(k)$. Maximization proves (C.4). The further comparison with the maximal-portfolio bound follows separately from Corollary C.3 when the same continuation envelopes are used. $\square$

The amount of improvement has an exact characterization. Define $c(H) = \sum_{j \in H} c_j$, $C_{\min} = \text{Bot}_k(c)$, and $L(H) = -D - \sum_{j \in H} \beta_j \geq 0$. Then

$$B^{\text{card}}(k) - B^{\text{joint}}(k) = \min_{H \in \mathcal{H}_k} \{c(H) - C_{\min} + L(H)\}. \quad (\text{C.6})$$

Indeed, subtracting the top- k expression for $B^{\text{joint}}(k)$ from $B^{\text{card}}(k)$ converts its maximum into this minimum. Both terms are nonnegative. Since $\mathcal{H}_k$ is finite, equality of the two bounds holds if and only if some H simultaneously satisfies $c(H) = C_{\min}$ and $L(H) = 0$. Strict improvement occurs precisely when no cheapest completion also has zero weighted affine shortfall. Ties matter: different representatives of cost or statistic ties do not establish incompatibility between their maximizing set families. The identity describes the relaxation, not an observed economic loss.

The lower derivative bound makes subtraction from the optimistic endpoint safe: every shortfall in log competitiveness reduces profit by at least that slope times the shortfall. No concavity of profit is required. Appendix I.2 constructs the affine coefficients and the share intervals used to obtain the slopes. The profit cap must cover the synthetic endpoint itself, even if no feasible portfolio attains it. Zero slope bounds recover the cardinality bound.

If portfolio size is unrestricted, maximize both bounds over the same admissible completion sizes. Weak dominance is preserved. Strict improvement of the overall maximum occurs if and only if every size attaining the old maximum strictly improves. Separately treated inactive completions must be included in both comparisons. An additional valid cap can be combined by taking a minimum, but strict improvement of the raw cardinality bound need not change that minimum when the other cap is already tighter.

Implemented service paths. The additive-statistic specializations above are sufficient conditions. When installed local coverage depends on parent cores, the exact payoff must use that complete service mapping, and a node envelope must dominate every feasible supported path. The fixed partner paths in the China exercise enter both exact and upper-bound supply; they do not become optional direct cores. Appendices C.5 and C.7 state the algorithm and error accounting, and Appendix I.2 gives the active-branch service and accessibility construction. An envelope valid only on a restricted choice set cannot bound returns over all feasible portfolios.

C.5 Branch search and termination

For a fixed cost type and supplied rival distribution, define the repriced objective

$$v_f(T) = \mathbb{E}_{Q_{-f}}[r_f(T, S_{-f})] - \sigma_\epsilon \sum_{j \in T} \ell_{fj} \epsilon_{fj}.$$

A node fixes selected links S and undecided links U and represents $\{T \in \mathcal{X}_f : S \subseteq T \subseteq S \cup U\}$. Start with any feasible fully repriced incumbent. Process a node by discarding it if infeasible or if its valid completion upper bound does not exceed the incumbent; evaluate it if complete; otherwise split on one undecided link. The two children impose inclusion and exclusion. They partition the node's completion set, so neither branch requires an ordered marginal value. Best-bound selection processes the largest remaining upper bound first; it changes scheduling, not validity.

At interruption let L be the best attained value and U^* the maximum of L and all unresolved node bounds. Every unevaluated feasible portfolio is in one unresolved node, hence

$$L \leq \max_{T \in \mathcal{X}_f} v_f(T) \leq U^*.$$

A response is solved to tolerance ϵ when $U^* - L \leq \epsilon$. Finite exact enumeration terminates because there are finitely many nodes. Time limits do not ensure a small gap. Both the maximal-portfolio bound and the cardinality-conditioned bound must dominate the feasible completion

family even with negative realized costs. The joint-affine construction and its assumptions are proved in Appendix C. For parent-dependent service, maximal and active-branch envelopes must cover the same paths as exact payoffs.

Bounds used in the cloud policy searches. Both applications search across admissible portfolio sizes, applying the bounds within each size and retaining the largest unresolved upper bound. For a node and size, a route-feasibility envelope includes every service path that some completion can support. A second, additive relaxation attributes local service to its possible parent cores, allowing duplicate contributions and relaxing joint parent requirements. Componentwise top- k sums bound that relaxation. The smaller of these two competitiveness envelopes is repriced against the supplied rival states; monotonicity of AB profit gives the gross-profit cap, enlarged for numerical error. Subtracting the cheapest feasible number of remaining link costs gives the cardinality cap. The joint-affine cap is also computed on the additive relaxation, using a common ranking of benefits and costs. The search retains the smaller cap.

The EU search permits every subset of 42 core countries for each provider. China excludes direct mainland core choice for the five non-Chinese providers. Its fixed AWS and Microsoft partner paths enter both exact service and each upper envelope as fixed supply, without a selectable core or private core cost. Thus joint-affine bounds are used in both policy searches. The separate fixed-count comparison in Appendix E.3 measures bound performance under that diagnostic's design; it does not measure the speed gain in the endogenous-count policy searches.

C.6 An AB share envelope for a search node

Fix a rival state r . Suppose $\bar{\psi}_{nf}$ and $\underline{c}_{nf}$ bound own accessibility from above and delivered cost from below for every completion. In an additive core-only model, the portfolio containing all undecided locations supplies such bounds; in a parent-dependent model, domination of every feasible supported path must be checked. With fixed rival accessibility and costs, put

$$\bar{y}_{nf} = \bar{\psi}_{nf}(\mu_{\min}\underline{c}_{nf})^{1-\sigma}, \quad \underline{y}_{ng} = \psi_{ng}(\mu_{\max}c_{ng})^{1-\sigma},$$

where $\mu_{\min} = \sigma/(\sigma - 1)$ and $\mu_{\max} = \eta/(\eta - 1)$. The markup interval yields

$$\bar{s}_{nfr} = \frac{\bar{y}_{nf}}{b_n + \bar{y}_{nf} + \sum_{g \neq f} \underline{y}_{ng}}. \quad (\text{C.7})$$

Use zero share when no own completion can supply the market, and omit inactive rivals from the denominator. The fixed residual price-index contribution b_n remains in the bound. Since $h(s) = s/\sigma + (1/\eta - 1/\sigma)s^2$ is increasing, a valid maximal completion bound is

$$\sum_r q_r \sum_n R_n h(\bar{s}_{nfr}) - \sum_{j \in S} C_j - \sum_{j \in U} \min\{C_j, 0\}, \quad (\text{C.8})$$

where $C_j = \phi_{fj} w_{fj} + \sigma_\varepsilon \ell_{fj} \varepsilon_{fj}$. Any completion's cost is at least the two cost terms, including with signed shocks. Cardinality conditioning tightens those terms as in Appendix C. The bound relaxes own and rival markups; complete portfolios are still evaluated at their joint pricing equilibrium.

C.7 Pricing, integration, and equilibrium error

Write $x = (\log y_i)_i$ for the active price-index contributions in logs. If a computed x has fixed-point residual r_x , the conditional pricing contraction gives error at most $r_x/(1 - \bar{\lambda}_{AB})$ in the sup norm. This statement uses the undamped residual of the map in log contributions; a stopping criterion in shares must first be converted to a bound in the required units. Verified share and profit allowances can then enlarge every node bound and lower the incumbent to give an upper bound on the remaining payoff error. Floating-point residuals alone do not account for rounding error. That error must also be bounded, for example by outward-rounded interval arithmetic.

A global optimality bound applies to a specified own type and the supplied rival values. Continuous cost types and rival distributions require a separate integration argument. Optimality within a finite candidate set does not rule out profitable deviations outside it. We add discovered portfolios to common candidate sets, recompute every policy distribution, and repeat the search. Checks using independent cost draws assess type coverage; repeating searches at the same types does not provide that assessment. The F8 records in Appendix H report those limits explicitly.

A uniform contraction also converts a response-distribution residual into a distance bound through Proposition A.1; without a verified contraction bound, small regret, small total variation, and agreement across starts remain separate diagnostics. For quasi-Monte Carlo blocks, differences across scrambles are numerical stability measures and are not independent-sample binomial confidence bounds. A small payoff error need not imply a small policy-price error when near-tied portfolios have different market consequences.

D. THE BOUNDARY OF INDEPENDENT COUNTRY RULES

We characterize when independent country cutoffs recover complete-portfolio choice, and when an additive approximation preserves the optimal portfolio. Appendix J develops the separate fixed-point and numerical-error implications.

D.1 Link-by-link cutoffs and the modularity boundary

Let z index the policy or counterfactual environment, and fix a rival portfolio distribution Q_{-f} . We suppress z when no confusion arises. For this subsection, let the feasible family be the full power set $\mathcal{X}_f = 2^{\mathcal{A}_f}$. Write $W_f(S_f; Q_{-f}, z)$ for firm f 's interim operating value before deterministic and private link costs. For a deterministic link-cost vector ϕ_f , define

$$V_f^\phi(S_f; Q_{-f}, z) = W_f(S_f; Q_{-f}, z) - \sum_{j \in S_f} \phi_{fj} w_{fj}. \quad (\text{D.1})$$

Thus $V_f^\phi(\cdot; Q_{-f}, z)$ is the Section 4 interim value in environment z . The corresponding type payoff subtracts the private-cost term in (4.2).

Definition D.1 (Separable link-by-link cutoff representation). At (Q_{-f}, z, ϕ_f) , a separable link-by-link cutoff representation is a vector $c_f = (c_{fj})_{j \in \mathcal{A}_f}$ such that, outside a null tie set, the exact best response is

$$b_f(\varepsilon_f; Q_{-f}, z) = \{j \in \mathcal{A}_f : \varepsilon_{fj} < c_{fj}\}. \quad (\text{D.2})$$

The cutoffs may depend on beliefs, policy, and deterministic costs, but not on the firm's other realized private costs.

Here “separable” describes the coordinatewise payoff representation, not the joint distribution of link choices. The coordinates of ε_f may be correlated, in which case the induced choices and portfolio distribution are also generally correlated.

Assumption D.1 (Rich within-firm cost support). For each firm, the joint distribution of ε_f is absolutely continuous and assigns positive probability to every nonempty open subset of $\mathbb{R}^{|\mathcal{A}_f|}$.

Theorem D.1 (Modularity is the exact separability boundary). Fix (f, Q_{-f}, z) , suppose $\mathcal{X}_f = 2^{\mathcal{A}_f}$, and impose Assumption D.1. The following are equivalent:

(i) $W_f(\cdot; Q_{-f}, z)$ is modular, so there are numbers a_f and $(v_{fj})_{j \in \mathcal{A}_f}$ such that

$$W_f(S_f; Q_{-f}, z) = a_f + \sum_{j \in S_f} v_{fj} \quad \text{for every } S_f \subseteq \mathcal{A}_f.$$

(ii) For every deterministic link-cost vector ϕ_f , the exact best response admits a separable link-by-link cutoff representation almost surely.

A representation for even one fixed deterministic cost vector already implies modularity. The result permits correlated private costs within a firm.

Proof. For a modular value, maximizing one coordinate at a time includes j when $\varepsilon_{fj} < (v_{fj} - \phi_{fj}w_{fj})/(\sigma_\varepsilon \ell_{fj})$, proving sufficiency. For necessity, fix any ϕ_f admitting a cutoff representation, suppress f, Q_{-f}, z , and write

$$x_j = \phi_j w_j + \sigma_\varepsilon \ell_j \varepsilon_j, \quad q_j = \phi_j w_j + \sigma_\varepsilon \ell_j c_j.$$

This invertible affine transformation preserves absolute continuity and positive probability on every nonempty open set. The empty candidate set is immediate; otherwise put $B = \max_{k, S \subseteq \mathcal{A} \setminus \{k\}} |W(S \cup \{k\}) - W(S)| < \infty$.

Fix j and $T \subseteq \mathcal{A} \setminus \{j\}$. On an open box of other costs, choose $x_k < -B - 1$ for $k \in T$ and $x_k > B + 1$ for $k \notin T \cup \{j\}$. Every maximizer contains T and excludes those other links: adding an omitted link in T gains at least $-B - x_k > 0$, and deleting a wrongly included link gains at least $x_k - B > 0$. Full-power-set feasibility permits both changes. The remaining choice is between T and $T \cup \{j\}$, so exact inclusion occurs when $x_j < W(T \cup \{j\}) - W(T)$. If this threshold differed from q_j , an open interval between them, together with the box of other costs, would have positive probability of disagreement. Thus $W(T \cup \{j\}) - W(T) = q_j$ for every T . Telescoping from the empty portfolio gives $W(S) = W(\emptyset) + \sum_{j \in S} q_j$. No independence within a firm is used. $\square$

D.2 Approximation error and portfolio choice

Now allow any nonempty finite feasible family $\mathcal{X}_f$. Fix beliefs and policy, and let V_f be the interim value net of deterministic link costs. Let M_f be any approximating table on the same family. Define

$$R_f^M = V_f - M_f, \quad \Omega_f^M = \text{osc}_{\mathcal{X}_f} R_f^M, \quad \varrho_f^M = \max_{S \in \mathcal{X}_f} |R_f^M(S)|. \quad (\text{D.3})$$

Both tables subtract the same private costs,

$$U_V(S) = V_f(S) - \sigma_\varepsilon \sum_{j \in S} \ell_{fj} \varepsilon_{fj}, \quad U_M(S) = M_f(S) - \sigma_\varepsilon \sum_{j \in S} \ell_{fj} \varepsilon_{fj}.$$

Let S_V, S_M maximize these respective objectives over $\mathcal{X}_f$. For a unique exact maximizer define its decision margin by

$$\Delta_f = \min_{T \neq S_V} \{U_V(S_V) - U_V(T)\}, \quad (\text{D.4})$$

with $\Delta_f = +\infty$ for a singleton feasible family.

Proposition D.1 (Payoff loss, portfolio preservation, and distribution error). *For every cost type,*

$$0 \leq U_V(S_V) - U_V(S_M) \leq \Omega_f^M \leq 2\varrho_f^M. \quad (\text{D.5})$$

The two maximizers coincide whenever

$$\Delta_f > \Omega_f^M. \quad (\text{D.6})$$

Under the conditional-density assumptions of Lemma A.1,

$$d_{\text{TV}}(P_f(V_f), P_f(M_f)) \leq \min \left\{ 1, \frac{C_f(R_f^M)}{\sigma_\varepsilon} \right\} \leq \min \left\{ 1, \frac{\Omega_f^M}{\sigma_\varepsilon} \sum_{j \in \mathcal{J}_f} \frac{\bar{g}_{fj}}{\ell_{fj}} \right\}. \quad (\text{D.7})$$

Here $\mathcal{J}_f$ contains the coordinates varying across $\mathcal{X}_f$. Neither additivity of M_f nor a full power set is required.

Proof. Optimality under M_f and cancellation of the common private costs give

$$U_V(S_V) - U_V(S_M) = U_M(S_V) - U_M(S_M) + R_f^M(S_V) - R_f^M(S_M) \leq \Omega_f^M \leq 2\varrho_f^M.$$

For every $T \neq S_V$, $U_M(S_V) - U_M(T) \geq \Delta_f - \Omega_f^M$; positivity proves preservation. Lemma A.1 applied to M_f and $M_f + R_f^M = V_f$ gives the first distribution bound. For any distinct feasible S, T , $|R_f^M(S) - R_f^M(T)| \leq \Omega_f^M$ and their symmetric difference contains a varying coordinate. Thus $c_j = \Omega_f^M$ is a feasible pair cover and proves the second bound. Total variation is at most one. $\square$

Only the variation of the approximation error across portfolios matters. A common error in every portfolio's value has no effect on choice. Conversely, a small value error can change investment when alternative networks are nearly tied. The probability-flow bound controls distance between the two choice distributions; it is not a bound on disagreement under a particular coupling that uses the same cost draws.

Independent country rules as an application. On the full power set, an additive table

$$M_f(S; Q_{-f}, z) = \alpha_f(Q_{-f}, z) + \sum_{j \in S} v_{fj}(Q_{-f}, z) \quad (\text{D.8})$$

is maximized by

$$\tilde{b}_f^M(\varepsilon_f; Q_{-f}, z) = \{j : \varepsilon_{fj} < v_{fj}(Q_{-f}, z)/(\sigma_\varepsilon \ell_{fj})\}. \quad (\text{D.9})$$

The best sup-norm additive fit is

$$\varrho_f^*(Q_{-f}, z) = \inf_{\alpha, v} \max_{S \subseteq \mathcal{A}_f} |V_f(S; Q_{-f}, z) - \alpha - \sum_{j \in S} v_j|. \quad (\text{D.10})$$

On a constrained feasible family, S_M must instead solve the constrained additive portfolio problem. Independent cutoffs need not be feasible or optimal there. Throughout, the deterministic loading w_{fj} and private loading ℓ_{fj} remain distinct.

D.3 Comparisons with Limited Portfolio Adjustment

An eight-provider exercise solves 24 firm–regime problems allowing adjustment in ten countries, with 256 private-cost types per problem. Rival portfolios and own choices outside those countries are fixed at their observed values. Under AB pricing, Shapley and minimax country rules disagree with the optimum within that restricted choice set for 19.38 and 19.50 percent of types. Mean losses are $5.47 \times 10^{-4} G_{\text{ref}}$ and $5.43 \times 10^{-4} G_{\text{ref}}$; the largest tested threshold loss is $0.0219 G_{\text{ref}}$. Optimal AB and CES choices within those countries differ for 80.66, 79.54, and 77.93 percent of types in the baseline, EU-wide, and National. Both pricing specifications use the same AB-calibrated parameters; CES is not recalibrated. The comparison therefore measures sensitivity to pricing assumptions, not their relative empirical fit. Choice disagreement is distinct from the profit losses above, and neither measures the error in a policy equilibrium. They do not test AES applicability, whose distinct order requirements are examined in Appendix E.2.

E. METHOD DIAGNOSTICS

E.1 Comparisons in the eight-provider specification

Appendix D.3 reports the AB–CES and independent-country comparisons using the eight-provider inputs. Each problem enumerates all 2^{10} portfolios within the ten-country choice set,

holding the other 32 country choices and rival portfolios fixed at observed values. Enumeration establishes the optimum for this conditional problem. It does not solve the full 42-country game or establish uniqueness of its equilibrium.

E.2 Why order-based portfolio elimination need not apply

The contingent cutoff in Lemma 1 exists for every finite portfolio-value table. Order-based elimination requires more. The lattice bounds in Jia (2008) use supermodularity of the firm’s own portfolio payoff. The procedure in Arkolakis et al. (2026) permits weaker conditions: the single-crossing differences condition in choices (SCD-C, Definition 3 of their March 2026 revision) requires non-negative marginal values to remain nonnegative under either expansion or contraction of the other choices.

Proposition E.1 (AB repricing need not preserve portfolio order). *There is a two-firm AB market with a unique pricing equilibrium at every portfolio and a three-link portfolio problem whose fully repriced payoff is neither supermodular nor submodular and satisfies neither direction of the AES SCD-C condition.*

Proof. Use the primitives in Example 2. Solving the unique pricing system before and after each addition and subtracting the stated link cost gives the marginal values of link 1 along $\emptyset \subset \{2\} \subset \{2, 3\}$ as -0.000337 , 0.000897 , and -0.001047 . The first change is positive and the second negative, so the payoff is neither submodular nor supermodular. The positive marginal value at $\{2\}$ becomes negative under both contraction to $\emptyset$ and expansion to $\{2, 3\}$, violating the two respective AES ordering directions. $\square$

The example separates pricing uniqueness from portfolio order. Jia or AES can still be used as an exact inner procedure when its ordering condition is verified over all relevant portfolios, including those not chosen in equilibrium. The calculation is reproduced by `scripts/verify_ab_aes_counterexample.py`. The F8 comparisons in Appendix D.3 evaluate exact and additive choices directly and do not assume that either ordering holds.

E.3 A paired completion-bound diagnostic

An eight-provider fixed-count diagnostic uses 42 countries, two antithetic own types per provider, and a 300-second limit for each method. The 16 paired problems use the same inputs and rule for selecting the next search node. Both methods finish for two pairs: AWS takes 280.67 seconds with the cardinality bound and 152.94 with the joint bound; Microsoft takes 180.76 and 135.14 seconds. For the 14 pairs that reach the time limit under both methods, the median remaining upper gap

falls from 20.28 to 19.75 percent of G_{ref} , a median paired reduction of 2.65 percent. The calculation using the joint bound runs second, so these times may also reflect a benefit from running it after the first calculation. The diagnostic records 356 bound checks with no violation, 72 uses of the stronger bound, and 32 independent repricing checks. It illustrates the distinction between a tighter upper bound and shorter total computation time. It does not establish numerical accuracy for the policy calculations with endogenous portfolio size.

F. DATA AND CALIBRATION UNITS

The current inputs distinguish core locations from the countries in which they support service. The matching equations and parameter values are in Section 5.2; the conventions below determine their units.

F.1 Core presence and installed local service

A core-country link is a selectable location. Multiple named core regions in one country are combined at the country choice unit. Installed local paths supply a weaker service weight and do not each count as a new core location. Their availability is evaluated by checking the presence of required parent cores in each complete portfolio, before averaging across the distribution. A local path with a confirmed parent abroad therefore differs from one whose parent is in the same country. Neither path implies unrestricted access to the parent's entire product set. Unknown parent links remain available through unmodeled support; no parent assignment is imputed. The empirical extent of this uncertainty is reported in Appendix G.

The 40 composition statistics concern core locations: three regional fractions, construction cost, and host-market expenditure for each provider. Installed service coverage is a separate geographic diagnostic. Treating a local-service country as an additional core in those moments would combine a different location unit with the calibrated core cost.

The EU economy allows all 8×42 core choices. The China economy retains the same parameters but removes direct mainland core choice for the five non-Chinese providers, while retaining fixed branded partner paths for AWS and Microsoft. The baseline distribution is re-solved after that change. These partner paths contribute service and branded profit under the assumption that the provider receives all branded operating profit; they are excluded from selectable core counts and core private shocks.

F.2 Expenditure, delivery, and accessibility inputs

Country expenditure is the dimensionless proxy

$$\bar{R}_n = \left[\frac{\text{GDP}_{n,2023}(I_{n,2023}/100)}{\text{GDP}_{\text{US},2023}(I_{\text{US},2023}/100)} \right]^{0.7}, \quad \bar{R}_{\text{US}} = 1.$$

GDP is nominal GDP in current U.S. dollars; I is the percentage of the population using the internet, not the number of users. The inputs use 2023 World Bank indicators, with an IMF GDP estimate and a maintained 93 percent internet-use rate for Taiwan. The exponent 0.7 compresses cross-country differences and is held fixed. The proxy is not observed cloud revenue. For EU countries the customer segments have expenditure $R_{n,\text{reg}} = 0.18\bar{R}_n$ and $R_{n,\text{private}} = 0.82\bar{R}_n$; other countries retain $\bar{R}_n$. These expenditure weights are unchanged across policy regimes.

The baseline delivery wedges take the values in (5.1). The starting accessibility score is

$$q_{in}^0 = -0.25 \log(1 + d_{in}/1000) + 0.201\{i = n\} + 0.151\{\tau_{in} \leq 1.15\} + 0.20(\omega_n - 1), \quad (\text{F.1})$$

where d_{in} is distance in kilometers between the input country's reference coordinates. These include facility-based geographic averages and city fallbacks. The final term is a destination-specific index from the 2025 OECD digital-policy inputs, held fixed across counterfactuals, with neutral value $\omega_n = 1$ when unavailable. Its coefficient is assumed; the index is neither a provider revenue share nor a measured willingness to pay. It is held fixed across the policy experiments. The proximity indicator uses the baseline delivery wedge, so a counterfactual change in τ does not mechanically change q .

For each EU destination, we reorder the eleven foreign-EU scores in (F.1) using RIPE round-trip-time rankings. The country-pair statistic is the median of anchor-pair minimum ping measurements from the August 18, 2026 archive. The sample retains anchor pairs observed in both directions, while computing each direction's statistic separately. Shorter round-trip times receive higher scores. Domestic scores, non-EU destinations, and non-EU origins retain q^0 . This replaces the assignment of scores within 132 directional EU country pairs; it does not estimate a latency coefficient or use latency differences elsewhere to change supply costs.

Reordering preserves the sum of $e^{\lambda_A q_{in}}$ over origins at each destination. We use the fixed all-origin frontier

$$A_n^{\text{frontier}} = \lambda_A^{-1} \log \left(1 + \sum_{i=1}^{42} e^{\lambda_A q_{in}} \right),$$

computed before applying policy eligibility. Both the ranking replacement and later policy com-

Table 9: Profit and Cost Normalizations

Provider	ℓ_f^{ref}	K_f^{ref}	G_f^{ref}
AWS	0.02383	28	0.66711
Microsoft	0.01113	33	0.36740
Google	0.00398	29	0.11553
Oracle	0.00178	27	0.04796
Alibaba	0.00604	15	0.09054
Huawei	0.00494	10	0.04938
Tencent	0.00212	11	0.02332
OVHcloud	0.00006	18	0.00113

$G_f^{\text{ref}} = \ell_f^{\text{ref}} K_f^{\text{ref}}$. These constants define the units of private costs and payoff gains. They are held fixed across calculations and differ from the current observed core counts. Unrounded values are included in the replication data.

parisons therefore leave this reference unchanged. Otherwise an eligibility restriction could lower a provider’s access score and its normalization simultaneously, obscuring the service loss. This convention does not imply that any provider operates the frontier network.

The central residual term is common across markets: $b_n = b^0 \exp(\zeta)$, with $b^0 = 1.77817$ and calibrated log multiplier $\zeta = 0.1396$ (rounded). EU public and private segments inherit the same term. Its realized share is endogenous because the contributions of the eight providers change. Residual sensitivities vary this term or its exposure to the China shock without treating it as a separate strategic firm. The common profile is a maintained simplification of unmodeled supply, not a fitted map of regional supplier capacity.

F.3 Profit and Cost Normalizations

The reference cardinalities used to normalize private costs and gains in Table 9 differ from the current observed core counts, especially for Alibaba, Huawei, Tencent, and OVHcloud. Rescaling these reference units with the current counts would change the meaning of noise and regret tolerances. The numerical implementation keeps them fixed. The maintained missing-construction-data penalty is $\delta = 0.614$. It applies to Belgium, Bahrain, Hong Kong, Israel, Morocco, the Philippines, Qatar, Serbia, Thailand, and Taiwan. Belgium is the only model-EU country in this list. Thus the penalty is not an EU-wide entry-cost adjustment; it also changes the attractiveness of non-EU alternatives. Its empirical validity requires a separate sensitivity of the portfolio problem, not merely repricing fixed networks.

F.4 Location score and interpretation of fit

For the composition moments in (5.2), the summary score is

$$Q_{\text{loc}} = \sum_{f,j} v_{fj} \left[\frac{m_{fj}(Q_f) - m_{fj}^{\text{obs}}}{a_{fj}^{\text{ref}}} \right]^2. \quad (\text{F.2})$$

Each provider has equal weight. Geography, construction cost, and expenditure receive equal block weights, with the three regional fractions sharing the geography block. The scales a_{fj}^{ref} are fixed feasible moment ranges under reference portfolio sizes, rather than estimated standard errors or current variable-count ranges. Missing-country fractions receive zero weight and are outside the 40 moments. This score informed specification assessment; the reported count and share calibration does not minimize it over additional spatial parameters.

The matched worldwide counts discipline provider cost intercepts, while global share ratios discipline relative competitiveness and residual intensity. Location moments are correlated summaries used to assess fit conditional on the maintained spatial inputs; country-level provider revenue shares are not targets or independent validation data. Table 2 distinguishes the five public 2024 IaaS share targets from the assumptions for the three smaller providers.

The specification contains no regional cost intercept. Its remaining EU core overprediction matters for an origin-eligibility experiment even though global shares and worldwide location counts fit closely. The observed-core comparison in Appendix H addresses one part of this sensitivity without recalibrating the model.

G. CLOUD-NETWORK MEASUREMENT AND INSTITUTIONAL SOURCES

This appendix records the scope of the descriptive evidence in Section 2. It is separate from the 2026 F8 calibration targets. Historical launch dates, current core-country presences, local service paths, and regulatory qualifications are different objects and are not forced to have identical totals.

G.1 Historical country-entry samples

Table 10 reconstructs the original five-provider launch sample and then changes provider coverage and the definition of a documented public-compute presence. Only entry dates documented to a

Table 10: Historical Entry Counts under Alternative Measurement Scopes

Notes: The window is 2007–2024. A provider-year is included when it has at least one first country entry. A multi-country provider-year has at least two. The final two columns report multi-country provider-years and entries occurring in those years. Calendar-year clustering is not evidence of simultaneous decisions. Rows C and E add Huawei, Tencent, and OVHcloud, whose historical product and date documentation is less uniform than for the original five providers.

Sample	Entries	Provider-years with entry	Multi years	Entries in multi years	Share
A. Original five-provider launch data	115	52	28	91	0.791
B. Five providers, documented entry year	115	52	28	91	0.791
C. F8, documented core entry year	139	65	35	109	0.784
E. F8, documented public-compute entry year	145	67	36	114	0.786

specific year enter these counts. The three additional providers have less uniform legacy documentation. We therefore report the eight-provider histories as a measurement sensitivity alongside the more consistently documented five-provider historical series.

In the 42-country frame, the historical eight-provider panel contains 336 provider-country pairs. It records a specific documented presence year for 159, an upper bound on the presence year for 11, and no documented presence for 166. The resulting 170 documented-presence pairs are not the 169 current core presences used in the calibration: they differ in date, product scope, and the treatment of local and partner-operated supply. An upper bound from an archived roster is not coded as a launch year.

The original five-provider launch panel starts each provider after generally available paid public-cloud service and records its first documented region in each country. Restricting eligibility to compute-IaaS general availability leaves the multi-country pattern similar: 53 percent of provider-years with at least one entry and 78 percent of entries are in multi-country years. These eligibility changes concern historical descriptive counts, not the current core-presence targets.

G.2 Local service and parent regions

The data on local regions and their parents begins from 82 source records. Sixty local records pass the 42-country and product screens: 24 have a confirmed parent in the same country, nine have a confirmed parent abroad, and 27 have an unknown parent. All 33 confirmed parent assignments in this screened sample are AWS records; 25 of the unknown assignments are OVHcloud records.

The counts therefore describe documentation coverage and cannot be interpreted as provider-wide or capacity-weighted frequencies. Unknown parentage is not evidence that a local service is independent.

AWS's official roster assigns Copenhagen and Helsinki to eu-north-1 and Warsaw and Hamburg to eu-central-1. Its architecture guide calls a Local Zone an extension of a region, while documenting some direct network paths from the Local Zone. Figure 1 consequently depicts a product relationship and makes no claim about the route of every workload, legal data location, or realized traffic.

G.3 Case chronologies and limits

In France, the national cloud strategy was announced on May 17, 2021, Bleu was announced ten days later, circular 6282-SG followed on July 5, and SecNumCloud v3.2 was issued on March 8, 2022. S3NS entered general availability on October 16, 2025 and qualified on December 17. Bleu remained in the qualification process in the ANSSI record consulted on September 8, 2026. These intervals are chronology, not review durations or causal effects: application dates and product scopes are not held constant.

In mainland China, the Azure–21Vianet arrangement predates the 2016 catalogue: the parties signed their memorandum in 2012 and Azure China became generally available in 2014. AWS stated that Sinnet would become the counterparty for the Beijing region on August 1, 2016 and referred to compliance with evolving Chinese telecommunications regulation. The statement does not attribute the change to one named instrument and says that the change did not alter prices. The record therefore supports the operator and account distinction, not a price or cost effect of the 2016 catalogue.

In Korea, notice 2023-4 introduced the tiered CSAP framework on January 31, 2023. Microsoft dates its low-tier IaaS certification to November 20, 2024 and AWS dates its certification to March 26, 2025. Both refer to services supplied from infrastructure operating before the notice. Google confirms that its certification covers IaaS and PaaS resources in the Seoul region but publishes no certification date on the page used here. This evidence does not rule out internal infrastructure, isolation, or cost changes made for certification.

Germany provides a useful boundary case. Microsoft Cloud Germany became generally available in September 2016 with a data-trustee arrangement, before Microsoft's ordinary German regions opened in 2019. The company later closed the trustee-based product to new customers and ultimately retired it. Supply form can therefore coincide with country entry and can also be with-

Table 11: Regulatory Access and Cloud Supply Arrangements

Notes: The table reports documented institutional and supply distinctions. It does not assign a causal effect to a regulation. Qualification dates refer to the specified offer, not to a provider’s first country entry. “Held fixed” means that the feature is not an endogenous choice in the current quantitative model. Sources are listed below; Section G discusses their interpretation.

Case and covered demand	Supply boundary	Documented arrangement	Implication for the model
France: specified sensitive public-sector workloads	SecNumCloud qualification attaches to an offer and legal entity. Version 3.2 includes EU location and provider-control requirements.	S3NS, which uses Google Cloud technology under a Thales-controlled entity, qualified on December 17, 2025. Bleu uses Microsoft technology and remained in the qualification process in the September 2026 record. OVHcloud qualifies named offers.	Motivates buyer-specific eligibility. The EU experiment varies geographic recognition conditional on assurance; ownership and joint-venture choice are held fixed.
Mainland China: customers of locally supplied cloud services	Licensed local operators and counterparties supply the China regions in a separate contracting and account environment.	Sinnet and NWCD operate the two AWS China regions; 21Vianet operates Azure China. AWS documented a change in the Beijing-region counterparty on August 1, 2016 while referring generally to Chinese telecommunications regulation.	The baseline distinguishes brand, operator, and directly controlled investment. The counterfactual changes cross-border costs relative to this segmented baseline.
Korea: public-sector workloads covered by the CSAP tier	Specified services require certification to serve public buyers, separately from the opening of a commercial region.	Microsoft certified services in Korea Central and Korea South on November 20, 2024; AWS certified services using its Seoul region and edge location on March 26, 2025. Google’s page confirms covered Seoul services but gives no certification date.	Motivates customer-specific eligibility distinct from core-country entry. Internal adaptation and certification costs are not observed and are held fixed.

Primary records: ANSSI SecNumCloud v3.2 and the ANSSI qualification catalogue; AWS–Sinnet operating relationship and Microsoft–21Vianet; Korean notice 2023-4, Microsoft CSAP, AWS CSAP, and Google Cloud CSAP.

drawn; country-entry counts alone cannot establish that regulation affects form but not entry.

The source archive preserves regulator documents, provider pages, the event and qualification ledgers, retrieval dates, and page hashes. Public records do not identify rival responses, capacity changes, certification expenditure, or cross-market price effects. Those outcomes remain conditional predictions of the quantitative model.

H. POLICY ACCOUNTING AND NUMERICAL CHECKS

H.1 Convergence on the candidate sets and unrestricted search

The EU results use the calibrated parameters throughout. When a search finds a profitable deviation, we add the portfolio to every regime’s candidate set and recompute all distributions. The reported results use the resulting common candidate sets. The replication files record the parameter estimates, candidate sets, and numerical samples for each result. The China application below imposes additional restrictions on local operations.

For the EU application, each firm’s candidate set contains between 9,850 and 12,284 complete portfolios, and distributions use 1,536 simulated own-cost types. At the maintained elasticities, the conditional AB pricing contraction modulus is $0.375 < 1$, giving unique prices for each fixed portfolio profile.

The distributional residual is measured before damping:

$$\text{TV} = \max_f \frac{1}{2} \sum_{S_f} \left| \widehat{\mathcal{B}}_f(Q_{-f})(S_f) - Q_f(S_f) \right|.$$

It compares full portfolio probabilities rather than country-inclusion marginals. The stopping rule requires each firm’s mean candidate-set regret, including the pricing-error allowance, to be at most $10^{-4} G_f^{\text{ref}}$ and the undamped TV residual to be at most 0.01 at two consecutive checks. The same criteria apply to the China calculations below. All three EU distributions meet the mean-regret and TV stopping criteria on their candidate sets, as reported in Table 12.

What each numerical check establishes. Candidate-set regret compares each simulated type’s current assignment with its best candidate in the maintained set. Full-choice-set search instead asks whether an omitted portfolio improves on the incumbent. An open search gap means that the time-limited search has not proved the absence of such an improvement. Checks at selected cost types and convergence on a simulated type sample therefore do not establish optimality for every type in the continuous population.

Table 12 reports the final search round separately from convergence on the candidate sets. It finds no further improvement beyond numerical error at the examined cost types, although most global search bounds remain open. Because the same cost types are reused across rounds, the exercise is a check on the reported candidate sets rather than an independent validation sample.

Table 12: Convergence and Full-Portfolio Search Diagnostics

Panel A. Convergence on the candidate sets

	Maximum firm mean regret / G_f^{ref}	Maximum sampled type regret / G_f^{ref}	Undamped portfolio TV
Baseline	2.80×10^{-6}	2.16×10^{-4}	0.00731
EU-wide	3.38×10^{-6}	4.22×10^{-4}	0.00605
National	6.50×10^{-6}	3.54×10^{-3}	0.00584

Panel B. Search over all feasible country portfolios

Regime	Problems	Improving portfolios found	Maximum open gap / G_f^{ref}	Closed bounds
Baseline	32	0	1.070	1
EU-wide	32	0	1.071	0
National	32	0	0.989	0

Notes: Panel A reports candidate-set diagnostics for simulated cost types; all three regimes meet the prespecified stopping rules. Panel B reports 96 full-choice-set search problems evaluated at four cost types per firm. No improvement beyond numerical error is found, but time limits leave most global upper bounds open. The cost types are reused across search rounds. These checks do not prove optimality for every type in the continuous distribution or uniqueness of the portfolio equilibrium.

H.2 EU accounting and additional comparisons

Holding the currently observed cores fixed and retaining the same installed local topology and parameters yields a National-minus-EU-wide regulated price effect of 22.02 percent. Residual shares are 40.92 percent under EU-wide and 90.80 percent under National. This calculation is distinct from the fixed baseline portfolio result of 27.49 percent and from the earlier fixed-count specification. It shows that starting geography affects the magnitude even without any portfolio response.

For F8 concentration we normalize named shares within each priced profile: $\hat{s}_{nf} = s_{nf} / \sum_g s_{ng}$, and then compute $10,000 \sum_f \hat{s}_{nf}^2$ before averaging over profiles and regulated expenditure. The implementation assigns zero to a profile with no eligible modeled provider; it does not renormalize the integration after excluding that profile. Under the National comparison, the expenditure-weighted probability of this event is below 0.01 percent at both the baseline and policy portfolio distributions. The much larger residual share therefore primarily reflects low supply from modeled providers, not profiles in which all modeled firms are absent. Residual concentration itself remains unspecified: treating all residual sales as one firm would not give a supported total-market HHI.

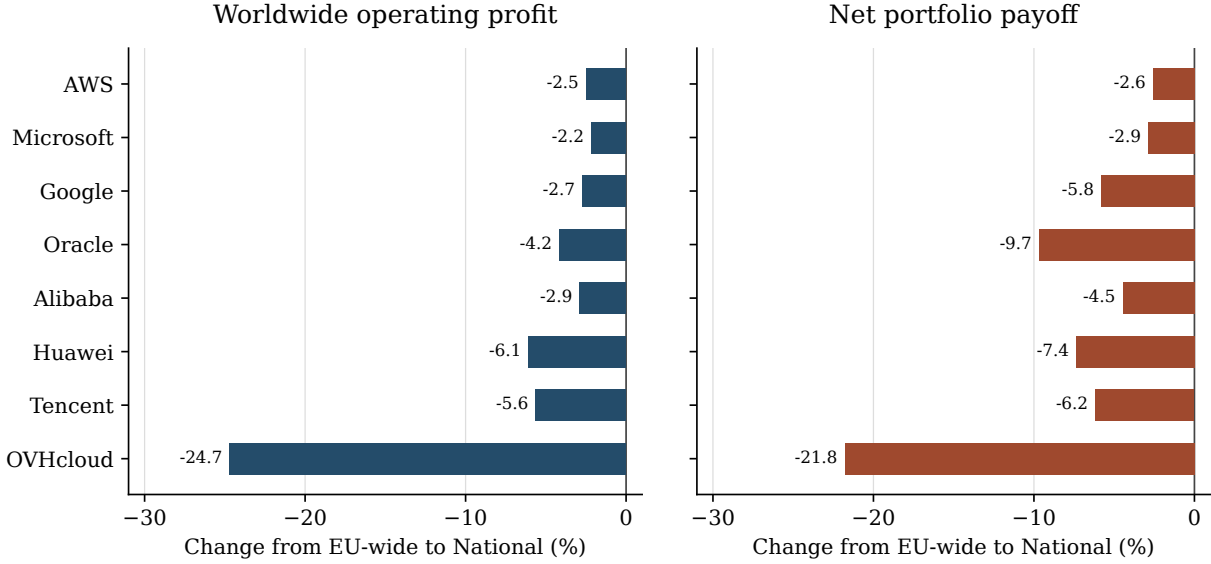


Figure 5: Provider incidence: operating profit and net portfolio payoff

Notes: Percentage changes from EU-wide to National in each firm’s worldwide expected profit. Operating profit precedes costs of core presence. Net portfolio payoff subtracts the model’s signed net link costs, including private Gaussian draws; it is not accounting net income. Ratios use mean levels across two integration blocks, and all EU-wide denominators are positive.

One-firm and joint response comparisons. Holding rivals at their baseline distributions, a one-firm response has distribution

$$Q^{r,\{f\}} = \widehat{\mathcal{B}}_f^r(Q_{-f}^O) \otimes Q_{-f}^O.$$

Combining all eight responses to rivals at the baseline instead gives $Q^{r,u} = \bigotimes_f \widehat{\mathcal{B}}_f^r(Q_{-f}^O)$ and requires joint repricing. Comparing these responses with the policy-specific equilibrium distribution measures the additional effects of responding to changed rivals. This experiment is different from adding the eight one-firm price effects. The exact response operator is the computational target; the hat denotes the approximation using common candidate sets. The full-choice-set search in Table 12 targets the recomputed policy distributions, not unrestricted optimality of each response to rivals at the baseline.

Table 5 reports the resulting worldwide counts and non-EU price effects.

Profit and margin accounting. Operating profit is revenue less variable production costs, summed over customer markets; net portfolio payoff also subtracts modeled link costs, including signed private cost draws. Consequently the latter is not financial statement net income. Each

operating margin is the ratio of expected operating profit to expected revenue after averaging the two integration blocks. All eight margins change by less than 0.07 percentage points, and all EU-wide profit denominators are positive.

The regional location accounting does not by itself attribute profit changes to a particular customer market: prices, sales, and link costs change jointly across the global portfolio.

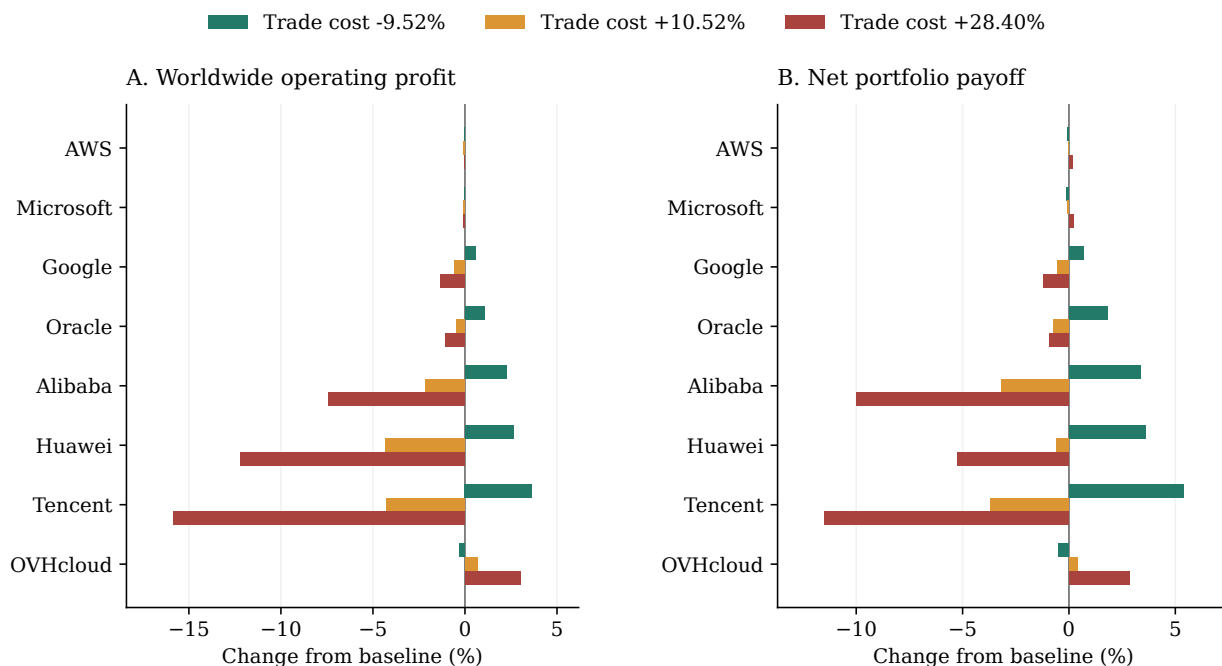


Figure 6: China frictions: worldwide provider profits

Notes: Percentage changes under equilibrium responses, with residual supply unexposed. Ratios use mean profit levels across two integration samples, with positive baseline denominators. Operating profit precedes costs of core presence; net portfolio payoff subtracts modeled signed link costs, including private draws, and is not accounting net income. AWS and Microsoft include branded partner-path profit under the assumption that the provider receives all branded operating profit. All scenarios use the same candidate portfolios and baseline reference.

H.3 China policy accounting and additional results

The baseline, easing, and both tightening scenarios use one common set of candidate portfolios. An initial full-choice-set search under easing found one additional Alibaba portfolio. After adding it to every scenario and recomputing all four distributions, the repeat search found no further improvement beyond numerical error in the examined problems. Most global search bounds remained open, with the largest remaining gap approximately equal to the reference payoff. The search covers four specified cost types per firm and finite rival quadrature. Establishing equilibrium for the continuous type distribution or its uniqueness requires further analysis.

Relative to the earlier calculations, the largest revision across the three central equilibrium scenarios is below 0.03 percentage points for prices and 0.03 expected worldwide core links. The directions of the reported aggregate effects are unchanged. This comparison establishes stability with respect to the completed candidate-set expansion, rather than a bound on all remaining numerical error.

Table 13: China provider shares and changes

Provider	Baseline (%)	-9.52% (pp)	+10.52% (pp)	+28.40% (pp)
AWS	29.44	+0.62	-0.84	-2.47
Microsoft	17.50	+0.45	-0.63	-1.84
Google	5.23	+0.35	-0.43	-1.21
Oracle	2.21	+0.17	-0.19	-0.52
Alibaba	17.90	+0.59	-0.63	-2.02
Huawei	11.36	+0.31	-0.50	-1.37
Tencent	6.57	+0.29	-0.35	-1.17
OVHcloud	0.03	0.00	0.00	-0.01
Residual	9.78	-2.78	+3.59	+10.61

Notes: All shocks use the same candidate portfolios and baseline reference. Equilibrium responses use portfolio distributions computed on the common candidate sets. Changes use equilibrium responses and $\xi = 0$. Shares include residual supply in the denominator; they are model predictions, not observed China market shares. AWS and Microsoft include branded partner-operated supply. Rounded columns may not sum exactly to 100 or zero.

All four distributions meet the prespecified candidate-set stopping rules. Reported price effects average two numerical integration samples; the estimates from the two samples differ by less than 0.01 percentage points in the central comparisons. This range is a numerical integration diagnostic rather than a sampling confidence interval, and it does not resolve candidate-set or empirical uncertainty.

The baseline shares and location changes are predictions of the model under the specified operating arrangements. Chinese market shares are not separate calibration targets. Direct-core probabilities are zero for the five non-Chinese providers by the scenario's feasible-set restriction. AWS and Microsoft's fixed mainland partner paths remain available and earn branded profit without a new direct core presence or private core-cost shock. Chinese providers retain both mainland and offshore location choices. The definition excludes fixed partner paths from worldwide core counts; those totals should not be compared mechanically with the EU totals or observed physical facilities.

Profit effects are ratios of expected levels, with positive baseline denominators. The replication generator checks share adding-up, location totals, and the identity between operating profit, signed link costs, and net payoff. These aggregation checks are not an independent equilibrium verification. AWS and Microsoft's profits depend on the assumed attribution of branded partner profit. Operating margins are not reported for the China application.

Table 14: Residual Exposure at Fixed Baseline Portfolios

Trade cost	China price (%)		Chinese provider share (pp)	
	$\xi = 0$	$\xi = 1$	$\xi = 0$	$\xi = 1$
-9.52%	-7.84	-8.84	+0.81	-0.41
+10.52%	+7.89	+9.43	-1.06	+0.52
+28.40%	+19.37	+24.53	-3.13	+1.56

Notes: Effects relative to the baseline with its portfolio distribution held fixed in every policy. At $\xi = 1$ the entire residual price-index contribution receives the offshore friction. All firms reprice; no investment is reoptimized. Chinese providers are Alibaba, Huawei, and Tencent, measured as a share of the entire China market. The main-text comparison in Table 8 instead holds each policy’s central equilibrium portfolios fixed.

Residual supply and the interpretation of concentration. Section 6.2 defines the residual exposure share ξ . Table 14 supplements the main-text comparison at central equilibrium portfolios with repricing at the common baseline portfolios. Neither comparison reoptimizes portfolios under $\xi = 1$ or estimates where residual supply originates.

In the central equilibrium calculations, HHI conditional on sales by the eight providers changes by roughly -19 , $+26$, and $+81$ points for easing, moderate tightening, and larger tightening. These are concentration measures within the eight providers’ sales, computed for each priced profile before averaging. They do not treat the residual as a single firm and do not identify total-market concentration. The Chinese providers’ combined total-market share rises by 1.19 percentage points under easing and falls by 1.49 and 4.56 points under tightening. Exposing residual supply reverses those share-change signs at the same central networks. The provider-nationality ranking is therefore a conditional incidence result, rather than evidence that a border restriction inherently favors or harms domestic brands.

H.4 Sensitivity to initialization and numerical integration

For each of the four China regimes, we repeat portfolio iteration from three initial distributions: concentration on a smallest-cardinality candidate, concentration on a largest-cardinality candidate, and a uniform distribution over each firm’s candidates. Eleven of twelve runs meet the prespecified stopping rules. The remaining baseline run ends just above the distributional criterion and is retained in every comparison.

The final distributions are close in economically relevant summaries. Relative to the corresponding reported distribution, the largest difference in a country-inclusion probability is 0.26 percentage points and the largest difference in worldwide expected core counts is 0.11. The signs

of the reported China and non-China price effects are unchanged across the restart comparisons. The largest departure from a central policy-price effect is 0.08 percentage points in China and 0.04 percentage points outside China. This comparison assesses the sensitivity of these statistics to the tested initial distributions on the common candidate sets; it is not a bound across all possible equilibria.

At the original integration resolution, fresh rival-integration samples give small payoff regret, but the distributional stopping criterion is not met by any of the twelve restarts or the four reference distributions. Nearly tied portfolios can generate this pattern, although the check does not establish its cause. A separate exercise increases integration accuracy at the fixed baseline reference and one easing restart. Disagreement between best responses across integration samples falls sharply for the baseline and more modestly for easing. This exercise holds rival distributions fixed; it does not recompute policy equilibria or bound the remaining error in policy effects.

A subsequent calculation increases rival integration to 4,096 points and re-solves all three EU and four China distributions on the original cost-type sample and common candidate sets. All seven meet the stopping criteria and pass a separate rival-integration check on that sample. Relative to the source distributions evaluated on the same outcome samples, changes in worldwide policy core-count differences are below 0.10 presences, and the regulated EU and China price effects change by less than 0.02 percentage points. These checks support numerical stability conditional on the calibration's own-cost sample.

Own-cost sampling presents a separate issue. Applying each frozen decision rule to 1,536 new cost vectors per provider produces worldwide core counts about five to eleven presences below those implied by the calibration sample. The test holds rival distributions and the decision rule's expected profits fixed; it does not compute a new equilibrium on the new types. Small payoff regret under independent rival integration therefore does not establish stability of the calibrated location moments across own-cost samples. The reported counts and policy responses retain the calibration sample, and their generalization beyond it requires further assessment.

Finally, unique conditional prices and stable policy statistics do not establish portfolio uniqueness. A proof based on a smaller region would require showing that it contains all fixed points and bounding responses uniformly within it. Neither repeated convergence nor the conditional pricing result supplies that argument.

I. AB INTERACTIONS AND SERVICE ENVELOPES

The market envelopes below bound K from Appendix A.2, and hence $K^\sharp$. Each envelope must cover its stated domain, including off-equilibrium states.

We bound interactions by finite market-level payoff differences, including entry and exit. Let $a = 1/\sigma$, $b = 1/\eta - 1/\sigma$, and $h(s) = as + bs^2$. Write $v_{fn}(S) = R_n h(s_{fn}(S))$. Because h is increasing on $[0, 1]$, $0 \leq v_{fn}(S) \leq R_n/\eta$. For own portfolios S, T , rival-g portfolios A, B , and fixed portfolios $U = S_{-(f,g)}$, define

$$\begin{aligned} \Delta_{fg}^n(S, T; A, B, U) &= v_{fn}(S, A, U) - v_{fn}(T, A, U) \\ &\quad - v_{fn}(S, B, U) + v_{fn}(T, B, U), \end{aligned}$$

and

$$\Omega_{fgn}^j = \max_{\substack{S \ni j, T \not\ni j \\ A, B, U}} |\Delta_{fg}^n(S, T; A, B, U)|. \quad (\text{I.1})$$

Proposition I.1 (Market-by-market AB bound). *In the fixed-expenditure AB model,*

$$K_{fg} \leq \bar{K}_{fg}^{FD} \equiv \sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{\ell_{fj}} \sum_n \Omega_{fgn}^j, \quad g \neq f. \quad (\text{I.2})$$

Set $\bar{K}_{ff}^{FD} = 0$. Moreover, $\Omega_{fgn}^j \leq R_n/\eta$, so

$$K_{fg} \leq \left(\sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{\ell_{fj}} \right) \sum_n \frac{R_n}{\eta}. \quad (\text{I.3})$$

The finite-difference spectral condition $\sigma_\varepsilon > \rho(\bar{K}^{FD})$ is therefore sufficient for uniqueness of the exact portfolio BNE.

Proof. Apply the triangle inequality across markets inside (A.3) and then maximize each market term separately to obtain (I.2). In a given market, order each pair of own competitiveness levels and each pair of rival competitiveness levels. AB share and $h(s)$ are increasing in own competitiveness and decreasing in rival competitiveness. Each one-direction payoff difference therefore lies in $[0, R_n/\eta]$, so the difference between two such increments has absolute value at most R_n/η . This proves (I.3). The spectral conclusion follows from Theorem 1 and entrywise matrix domination. $\square$

The finite-difference bound covers entry and exit but discards the calibrated share response. A hybrid bound uses exact derivatives where the active set is fixed and fully repriced corner payoffs where it changes. Define log effective competitiveness by

$$t_{hn}(S_h) = \log \psi_{hn}(S_h) + (1 - \sigma) \log c_{hn}(S_h).$$

Let H_{fgn} be a verified upper bound on the absolute mixed derivative of firm f 's market- n variable profit with respect to t_{fn} and t_{gn} on the relevant fixed-active-set share region. Let $\mathcal{X}_{hn}^{\text{act}}$ contain the portfolios that give provider h positive eligible supply in market n . Define the active-face diameters

$$d_{fn}^j = \max_{\substack{S_f \ni j, T_f \not\ni j \\ S_f, T_f \in \mathcal{X}_{fn}^{\text{act}}}} |t_{fn}(S_f) - t_{fn}(T_f)|,$$

$$d_{gn} = \max_{A, B \in \mathcal{X}_{gn}^{\text{act}}} |t_{gn}(A) - t_{gn}(B)|.$$

Either diameter is set to zero when its maximizing class is empty. Finally, let $\Omega_{fgn}^{j, \partial}$ be the fully repriced four-corner maximum over portfolio rectangles that reach an entry or exit face, with value zero when no such rectangle exists.

Corollary I.1 (Hybrid AB interaction bound). *In the fixed-expenditure AB model,*

$$K_{fg} \leq \bar{K}_{fg}^{\text{HYB}} \equiv \sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{\ell_{fj}} \sum_n \max \left\{ \Omega_{fgn}^{j, \partial}, \min \left(\frac{R_n}{\eta}, H_{fgn} d_{fn}^j d_{gn} \right) \right\}. \quad (\text{I.4})$$

Set $\bar{K}_{ff}^{\text{HYB}} = 0$. Hence $\sigma_\varepsilon > \rho(\bar{K}^{\text{HYB}})$ is sufficient for uniqueness of the exact portfolio BNE.

Proof. Partition the portfolio rectangles in (I.1) into those that remain on a fixed active face and those that reach entry or exit. On a fixed active face, the fundamental theorem of calculus gives the product $H_{fgn} d_{fn}^j d_{gn}$, capped by the global market envelope R_n/η . The second class is bounded directly by its fully repriced four-corner maximum. Taking the larger class bound and substituting it into (I.2) proves (I.4). The spectral conclusion follows from Theorem 1. $\square$

I.1 Mixed profit derivatives and the hybrid bound

Lemma A.2 supplies the equilibrium share Jacobian with fixed residual price-index contribution. Use its notation $r_i = r(s_i)$, $W = s_0 + \sum_h r_h$, and $\omega_i = r_i/W$, with $\beta_s(s) = (\sigma - 1)b_0/(a_0 - b_0s)$.

Differentiation gives

$$\begin{aligned}
r'(s) &= \frac{1 - s^2(\sigma - 1)b_0^2/(a_0 - b_0s)^2}{[1 + s\beta_s(s)]^2}, \\
W_{,g} &= J_{0g} + \sum_h r'(s_h)J_{hg}, \\
P_{fg} &:= \partial_{t_g}\omega_f = \frac{r'(s_f)J_{fg} - \omega_f W_{,g}}{W}, \\
Q_{ffg} &:= \partial_{t_g}J_{ff} = r'(s_f)J_{fg}(1 - \omega_f) - r_f P_{fg}.
\end{aligned}$$

Differentiating $\partial_{t_f}VP_{fn} = R_n(1/\sigma + 2b_0s_f)J_{ff}$ then gives

$$\mathcal{H}_{fgn}(s) = R_n\{2b_0J_{ff}J_{fg} + (1/\sigma + 2b_0s_f)Q_{ffg}\}. \quad (\text{I.5})$$

The J_{0g} term is required when residual price-index contribution is positive. Setting $s_0 = J_{0g} = 0$ recovers the no-residual benchmark.

A verified upper bound H_{fgn} on $\sup_s |\mathcal{H}_{fgn}(s)|$ throughout the relevant fixed-active-set region therefore gives

$$\Omega_{fgn}^{j,\text{act}} \leq \min\{R_n/\eta, H_{fgn}d_{fn}^j d_{gn}^j\}. \quad (\text{I.6})$$

This derivative bound must hold throughout every interpolating score rectangle, including unattainable intermediate scores, to justify the active-face term in (I.4). Entry and exit use the separate corner bound.

I.2 Service and accessibility envelopes

A common portfolio score has the form

$$\xi_a(H) = \zeta_a V_a(H)^{p_v} W_a(H)^{p_w}, \quad V_a(H) = V_a^0 + \sum_{j \in H} v_{aj}, \quad W_a(H) = W_a^0 + \sum_{j \in H} u_{aj}, \quad (\text{I.7})$$

where $\zeta_a > 0$, $v_{aj}, u_{aj} \geq 0$, and $p_v, p_w \geq 0$. The exponents may vary by channel; their channel subscripts are suppressed. Selected and mandatory facilities enter the bases. In a log-sum accessibility specification, W_a^0 includes the constant 1 as well as their contributions. Fixed normalizations enter ζ_a or the exponents; they do not change as H varies within the conditional problem. Every statistic with a positive exponent is positive on the active domain. The cloud service aggregator gives $p_v = (\sigma - 1)/\theta$. For accessibility $\lambda_A^{-1} \log W_a$ entering log competitiveness with loading χ , $p_w = \chi/\lambda_A$; subtracting a fixed accessibility normalizer changes ζ_a .

Let $\bar{V}_a = V_a^0 + \text{Top}_k(v_{a\cdot})$ and $\bar{W}_a = W_a^0 + \text{Top}_k(u_{a\cdot})$, and set $\bar{\xi}_a = \zeta_a \bar{V}_a^{p_v} \bar{W}_a^{p_w}$. The inequality $\log z \leq z - 1$ gives

$$\begin{aligned} \log \frac{\xi_a(H)}{\bar{\xi}_a} &\leq p_v \frac{V_a(H) - \bar{V}_a}{\bar{V}_a} + p_w \frac{W_a(H) - \bar{W}_a}{\bar{W}_a} = d_a + \sum_{j \in H} A_{aj} \leq 0, \\ d_a &= p_v \frac{V_a^0 - \bar{V}_a}{\bar{V}_a} + p_w \frac{W_a^0 - \bar{W}_a}{\bar{W}_a}, \quad A_{aj} = \frac{p_v v_{aj}}{\bar{V}_a} + \frac{p_w u_{aj}}{\bar{W}_a}. \end{aligned} \quad (\text{I.8})$$

Each separate top- k statistic bounds every relaxed completion, proving the last inequality on all of $\mathcal{H}_k$. A zero exponent removes its channel before any logarithm or ratio is evaluated. For this construction, $L(H) = 0$ means that every statistic with positive exponent and positive weight $\omega_a \mu_a$ attains its top- k upper value. Channels with zero weight or zero slope impose no equality requirement.

Lower slopes. For each fixed market and rival state, hold rival pre-markup scores and residual price-index contribution F_0 fixed. Lemma A.2 gives the nonnegative lower slope

$$\mu = \kappa R \min\{u(s^L), u(s^U)\}$$

whenever $[s^L, s^U]$ contains the own share along every path from a completion's score to its synthetic upper endpoint $\bar{\xi}$. This is a lower derivative bound, not a concavity assumption.

The markup interval gives readily computed share intervals. Let $\xi^L \leq \xi_f \leq \bar{\xi}$, where a lower score can be formed using bottom- k service and accessibility sums in (I.7). Put $m_- = (1 - 1/\eta)^{\sigma-1}$ and $m_+ = (1 - 1/\sigma)^{\sigma-1}$. Then

$$s^L = \frac{\xi^L m_-}{\xi^L m_- + \sum_{g \neq f} \xi_g m_+ + F_0}, \quad s^U = \frac{\bar{\xi} m_+}{\bar{\xi} m_+ + \sum_{g \neq f} \xi_g m_- + F_0} \quad (\text{I.9})$$

enclose every equilibrium share on that score interval whenever the denominators are positive. Each inverse markup lies between $1 - 1/\eta$ and $1 - 1/\sigma$, so the numerator and competing price-index bounds give the two inequalities directly. The boxes cover the synthetic upper score as well as attainable portfolios; a sharper verified enclosure may replace them. If a displayed lower denominator is zero, the trivial interval $[0, 1]$ and $\mu = 0$ avoid an undefined ratio. An economically inactive completion is handled separately, without taking its log score.

Validity in numerical implementation. The required upper bound at the reference score is $\bar{G}_a \geq G_a(\log \bar{\xi}_a)$. A cap known only to exceed profits at feasible portfolios does not satisfy this

requirement when $\bar{\xi}_a$ is an unattainable combination of statistics. Subtracting the slope correction from such a cap can make the result invalid. In contrast, the existing AB share envelope can be evaluated at the synthetic score itself and reused in (C.5). The construction then needs no additional AB pricing solve: bottom- k sums, share intervals, coefficient aggregation, and a common top- k sort suffice. Coefficients generally depend on k , so one sorting need not serve every completion size.

These are exact-arithmetic statements. A numerical implementation must enclose the profit bound at the synthetic score and the shares, use a nonnegative lower slope, and control the complete affine intercept, link scores, and final sum, including cancellation and signed costs. Pricing residual allowances alone do not establish those additional bounds. An uncertain candidate cannot be made valid by taking its minimum with an existing valid bound; it must first be discarded or independently enclosed. Forced completions, zero-loaded channels, and inactive alternatives can retain their original valid bounds.

Avoiding an additional pricing solve does not guarantee faster search: coefficient calculations take time and can change when incumbents are found.

J. EQUILIBRIUM ERROR UNDER ADDITIVE APPROXIMATION

We propagate the response errors in Appendix D to equilibrium outcomes. The approximate table must be optimized on the original feasible family; independent country cutoffs additionally require full-power-set feasibility. Search errors are treated in Appendix C.7.

J.1 Fixed points under additive approximation

A continuous choice of additive coefficients gives an approximate response map. One example is the Shapley allocation, which averages each link's marginal contribution over all insertion orders:

$$\alpha_f^{\text{Sh}}(Q_{-f}, z) = V_f(\emptyset; Q_{-f}, z), \quad (\text{J.1})$$

$$v_{fj}^{\text{Sh}}(Q_{-f}, z) = \sum_{T \subseteq \mathcal{A}_f \setminus \{j\}} \frac{|T|!(|\mathcal{A}_f| - |T| - 1)!}{|\mathcal{A}_f|!} [V_f(T \cup \{j\}; Q_{-f}, z) - V_f(T; Q_{-f}, z)]. \quad (\text{J.2})$$

Proposition J.1 (Existence under additive approximation). *Suppose $\mathcal{X}_f = 2^{\mathcal{A}_f}$, approximation coefficients are continuous in rival distributions, and within-firm type distributions are absolutely continuous. The response map $\tilde{\mathcal{B}}^M$ induced by (D.9) then has a fixed point. Continuity of interim values ensures that the Shapley coefficients satisfy the requirement.*

Proof. Continuous coefficients and null ties give a continuous response map, as in Appendix A.1. Apply Brouwer's theorem. $\square$

At such a fixed point, the chosen portfolio loses at most $2\gamma_f^M$ relative to its exact best response for every type, by Appendix D.

J.2 Response error and equilibrium distance

For a fixed policy state and an approximate fixed point $\tilde{Q}^M$, set

$$e_f^M = d_{\text{TV}}\left(\mathcal{B}_f(\tilde{Q}_{-f}^M), \tilde{\mathcal{B}}_f^M(\tilde{Q}_{-f}^M)\right). \quad (\text{J.3})$$

Theorem J.1 (Equilibrium propagation of one-step portfolio error). *Under the assumptions of Theorem 1, let $\bar{K} \geq K^\sharp$ be verified and nonnegative, and let $A = \bar{K}/\sigma_\varepsilon$ satisfy $\rho(A) < 1$. Then*

$$d_f \equiv d_{\text{TV}}(Q_f^*, \tilde{Q}_f^M) \leq [(I - A)^{-1}e^M]_f. \quad (\text{J.4})$$

Here Q^* is the unique exact BNE; the approximate fixed point need not be unique.

Proof. Apply Proposition A.1 with $M = A$, $\mathcal{U} = \mathcal{Q}$, and exact-response residual e^M . $\square$

Under the conditional-density assumptions, the probability-flow bound (D.7), evaluated at $\tilde{Q}_{-f}^M$, gives

$$e_f^M \leq \delta_f^M \equiv \min\{1, C_f(R_f^M)/\sigma_\varepsilon\}. \quad (\text{J.5})$$

The oscillation and sup-norm bounds from Appendix D apply to the same residual. Substitution into (J.4) yields $d \leq (I - A)^{-1}\delta^M$. If $\gamma_f^M = 0$ for every firm and relevant rival distribution, $\tilde{\mathcal{B}}^M = \mathcal{B}$ and the error is zero.

Corollary J.1 (Counterfactual outcome error). *For policy states z_0, z_1 , let $r(z) = (I - A(z))^{-1}e^M(z)$ be valid endpoint distance bounds. For bounded outcomes H_z , the absolute error in the predicted counterfactual change is at most*

$$\text{osc}(H_{z_1}) \min\{1, \sum_f r_f(z_1)\} + \text{osc}(H_{z_0}) \min\{1, \sum_f r_f(z_0)\}, \quad (\text{J.6})$$

where $\text{osc}(H_z) = \max_S H_z(S) - \min_S H_z(S)$.

Proof. At each endpoint, product-distribution total variation is at most $\min\{1, \sum_f d_f\}$, bounding the expectation error by that distance times outcome oscillation. Add the two endpoint bounds. $\square$